Lecture Notes on Kinetic Theory and Statistical Physics

(Oxford Physics Paper A1)

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These are the notes for my lectures on Kinetic Theory and Statistical Physics, being part of the 2nd-year course (Paper A1) at Oxford. I have taught the course since 2011, jointly with Professor Andrew Boothroyd (2011-15) and Professor Julien Devriendt (2015-). Only my part of the course is covered in these notes. I will be grateful for any feedback from students, tutors or sympathisers.

New in this version: Minor edits in §6; more figures added. Where figures are missing, look for them in my handwritten notes.

PART I

Basic Thermodynamics

This part of the course was taught by Professor Julien Devriendt.

PART II

Kinetic Theory

1. Statistical Description of a Gas

1.1. *Introduction*

You have so far encountered two basic types of physics:

1) Physics of single objects (or of groups of just a few such objects). For classical (*macroscopic*) objects, we had a completely deterministic description based on Newton's 2nd Law: given initial positions and velocities of all participating objects and the forces acting on them (or between them), we could predict their behaviour forever. In the case of *microscopic* objects, this failed and had to be replaced by Quantum Mechanics—where, however, we again typically deal with single (or not very numerous) objects and can solve differential equations that determine, eventually, probabilities of quantum states (generalising the classical-mechanical notions of momentum, energy, angular momentum etc.)

2) Physics of "systems"—understood to be large collections of objects (e.g., gas is a large collection of particles). This was introduced in Part I—and the description of such

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systems seemed to be cast in completely different terms, the key notions being *internal energy*, *heat*, *temperature*, *entropy*, *volume*, *pressure*, etc. All these quantities were introduced largely without reference to the microscopic composition of the systems considered.

It is clear that a link between the two must exist—and we would like to understand how it works both for our general peace of mind and for the purposes of practical calculation: for example, whereas the relationship between energy, heat, pressure and volume could be established and then the notions of temperature and entropy introduced without specifying what the system under consideration was made of, we had, in order to make practical quantitative predictions, to rely on experimentally determined empirical relations between

$$P, V, \text{ and } T \quad \text{(equation of state)}$$

and, U being internal energy, between

$$U, V, \text{ and } T \quad \text{(often via heat capacity, } C_V(T, V)\text{)}.$$

Statistical Mechanics (which we will study from Part III onwards) will deal with the question of how, given some basic microphysical information about properties of a system under consideration and some very general principles that a system *in equilibrium* must respect, we can derive the thermodynamics of the system (including, typically, $U(V, T)$, equation of state $P(V, T)$, entropy $S(V, T)$, and hence heat capacities, etc.).

Kinetic Theory (which we are about to study for the simple case of *classical monatomic ideal gas*) is concerned not just with the properties of systems in equilibrium but also—indeed, primarily—with *how the equilibrium is reached* and so how the collective properties of a system evolve with time. This will require both a workable model of the *constituent particles* of the system and of their *interaction (collisions)*. Equilibrium properties will also be derived, but with less generality than in Statistical Mechanics. We study Kinetic Theory first because it is somewhat less abstract and more intuitive than Statistical Mechanics (and we will recover all our *equilibrium* results later on in Statistical Mechanics). Also, it is convenient, in formulating Statistical Mechanics, to refer to some basic knowledge of Quantum Mechanics, whereas our treatment of Kinetic Theory will be completely classical.

So how do we derive the behaviour of a macroscopic system from basic knowledge of the physics of its microscopic constituents?

Let us consider the simplest case: a classical gas. The simplest model is to assume that particles are hard spheres (billiard balls) and that their collisions are elastic (energy- and momentum-conserving). We will forget about Quantum Mechanics for now.

Suppose we know all of these particles' positions and velocities precisely at some time $t = 0$ (in fact, this is quite impossible even in principle, but we are ignoring Quantum Mechanics). Let us solve

$$m\ddot{\boldsymbol{r}} = \boldsymbol{F} \tag{1.1}$$

for each particle and thus obtain $\boldsymbol{r} = \boldsymbol{r}(t)$ and $\boldsymbol{v} = \dot{\boldsymbol{r}}(t)$ for all times $t > 0$. Problem solved? In fact, two difficulties arise:

1) There is too much information. A typical macroscopic gas system will have perhaps 10^{23} particles (**Exercise**: convince yourself that this is a reasonable estimate). Then one data dump ($\boldsymbol{r}$, $\boldsymbol{v}$ for each particle at one time t) needs about $\sim 10^{12}$ Tb. For comparison,

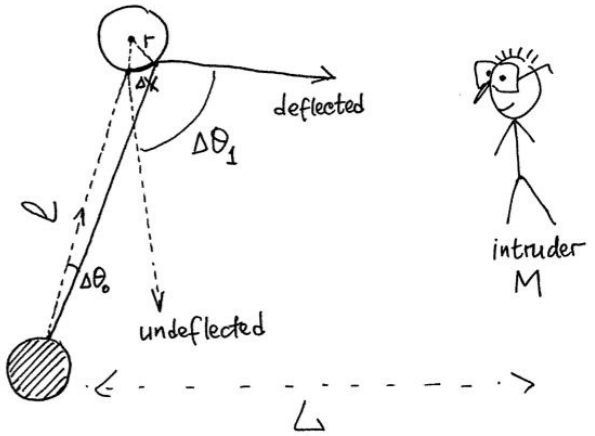


FIGURE 1. Billiard ball gravitationally deflected by an intruder.

the entire contents of the World Wide Web in 2013 were $\sim 4 \times 10^9$ Tb, so we need about 1000 times WWW to save the state of air in a small box.

2) Sensitivity to initial conditions and tiny perturbations. This means that even if we could solve the equations of motions for all these particles, the tiniest errors or imprecisions† would quickly change the solution, because any error in the initial conditions grows exponentially fast with time.

Let me give you an example to illustrate the last point.‡ Imagine we have a set of billiard balls on a frictionless table, we set them in motion (at $t = 0$) and want to observe them as time goes on. We could, in principle, solve their equations of motion and predict where they will all be and how fast they will be moving at any time $t > 0$. It turns out that if someone enters the room during this experiment, the small deflections of the balls due to the intruder's gravitational pull will accumulate to alter their trajectories completely after only ~ 10 collisions!

Proof (Fig. 1). For simplicity of this very rough estimate, let us consider all the balls to be fixed in space, except for one, which moves and collides with them. Assume:

distance between balls $l \sim 20$ cm;
radius of a ball $r \sim 3$ cm;
size of the room $L \sim 5$ m;
mass of intruder $M \sim 80$ kg;
time between collisions $\Delta t \sim 1$ sec.

Then the deflection due to gravity after one collision is

$$\Delta x \sim \frac{MG}{L^2} \Delta t^2 \sim 10^{-8} \text{ cm}. \tag{1.2}$$

So the initial angular deflection is

$$\Delta\theta_0 \sim \frac{\Delta x}{l} \lll\lll\lll 1. \tag{1.3}$$

Angular deflection after the first collision:

$$\Delta\theta_1 \sim \frac{\Delta x}{r} \sim \Delta\theta_0 \, \frac{l}{r}. \tag{1.4}$$

† And of course any saved data will always have finite precision!
‡ I am grateful to G. Hammett for pointing out this example to me.

We see that after each collision, the angular deflection will be amplified by a factor of l/r. Therefore, after n collisions, it will be

$$\Delta\theta_n \sim \Delta\theta_0 \left(\frac{l}{r}\right)^n \sim \frac{\Delta x}{l}\left(\frac{l}{r}\right)^n. \tag{1.5}$$

In order to estimate the number of collisions after which the trajectory changes significantly, we calculate n such that $\Delta\theta_n \sim 1$:

$$n \sim \frac{\ln(l/\Delta x)}{\ln(l/r)} \sim 10, \quad \text{q.e.d.} \tag{1.6}$$

The basic idea is that if errors grow exponentially with the number of collisions that a particle undergoes, you do not need very many collisions to amplify to order unity even very tiny initial perturbations (this is sometimes referred to as the "butterfly effect," after the butterfly that flaps its wings in India, precipitating eventually a change of weather in Britain). A particle of air at 1 atm at room temperature has $\sim 10^9$ collisions per second (we will derive this in §4). Therefore, particle motion becomes essentially *random*—meaning chaotic, deterministically unpredictable in practice even for a classical system.

Thus, particle-by-particle deterministic description [Eq. (1.1)] is useless. Is this a setback? In fact, this is fine because we really are only interested in *bulk properties* of our system, not the motion of individual particles.† *If we can relate those bulk properties to averages over particle motion, we will determine everything we wish to know.*

Let us see how this is done.

1.2. *Energy*

So, we model our gas as a collection of moving point particles of mass m, *whose positions* $\boldsymbol{r}$ *and velocities* $\boldsymbol{v}$ *are random variables.* If we consider a volume of such a gas with no spatial inhomogeneities, then all positions $\boldsymbol{r}$ are equiprobable.

The mean energy of the N particles comprising this system is

$$\langle E\rangle = N\left\langle \frac{mv^2}{2}\right\rangle, \tag{1.7}$$

where $\langle mv^2/2\rangle$ is the mean energy of a particle and we assume that all particles have the same statistical distribution of velocities. In general, particles may have a *mean velocity*, i.e., the whole system may be moving at some speed in some direction:

$$\langle \boldsymbol{v}\rangle = \boldsymbol{u}. \tag{1.8}$$

Let $\boldsymbol{v} = \boldsymbol{u} + \boldsymbol{w}$, where $\boldsymbol{w}$ is *peculiar velocity*, for which $\langle \boldsymbol{w}\rangle = 0$ by definition. Then

$$\langle E\rangle = N\frac{m}{2}\left\langle |\boldsymbol{u}+\boldsymbol{w}|^2\right\rangle = \underbrace{\frac{Mu^2}{2}}_{\equiv K} + \underbrace{N\left\langle \frac{mw^2}{2}\right\rangle}_{\equiv U}, \tag{1.9}$$

where $M = Nm$. The energy consists of the kinetic energy of the system as a whole, K, and the *internal energy*, U. It is U that appears in thermodynamics ("heat")—the *mean energy of the disordered motion of the particles* ("invisible motion," as they called it in the 19th century). The motion is disordered in the sense that it is random and has zero

† We will learn in §11.7 that, in fact, talking about the behaviour of individual particles in a gas is often meaningless anyway, because particles can be indistinguishable.

mean: $\langle \boldsymbol{w} \rangle = 0$. For now, we will assume $\boldsymbol{u} = 0$‡ and so

$$U = \langle E \rangle = N \left\langle \frac{mv^2}{2} \right\rangle . \tag{1.10}$$

1.3. *Thermodynamic Limit*

Is it really enough to know the *average* energy of the system? How is this average energy U related to the *exact* energy E of the system? If they could be very different at any given instance, then, clearly, knowing only U would leave us fairly ignorant about the actual state of the system. Here we are greatly helped by what we previously thought made the description of the system difficult—the very large N. It turns out that for $N \gg 1$, the typical difference between the average energy and the exact energy is very small (and the same will be true about all the other relevant bulk quantities that can be referred to some microscopically exact values).

Let us estimate this difference. The mean energy is $U = \langle E \rangle$ and the exact energy is

$$E = \sum_i \frac{mv_i^2}{2}, \tag{1.11}$$

where the index i runs through all N particles in the system and $\boldsymbol{v}_i$ is the velocity of the ith particle. Then the mean square energy fluctuation is

$$\begin{aligned}
\left\langle (E-U)^2 \right\rangle = \langle E^2 \rangle - U^2 &= \sum_{i,j} \left\langle \frac{mv_i^2}{2} \frac{mv_j^2}{2} \right\rangle - \left(\sum_i \left\langle \frac{mv_i^2}{2} \right\rangle \right)^2 \\
&= \sum_i \left\langle \frac{m^2 v_i^4}{4} \right\rangle + \sum_{i \neq j} \left\langle \frac{mv_i^2}{2} \right\rangle \left\langle \frac{mv_j^2}{2} \right\rangle - \left(\sum_i \left\langle \frac{mv_i^2}{2} \right\rangle \right)^2 \\
&= N \left\langle \frac{m^2 v^4}{4} \right\rangle + N(N-1) \left\langle \frac{mv^2}{2} \right\rangle^2 - \left(N \left\langle \frac{mv^2}{2} \right\rangle \right)^2 \\
&= N \frac{m^2}{4} \left(\langle v^4 \rangle - \langle v^2 \rangle^2 \right) .
\end{aligned} \tag{1.12}$$

Note that, in the second line of this calculation, we are only allowed to write $\langle v_i^2 v_j^2 \rangle = \langle v_i^2 \rangle \langle v_j^2 \rangle$ for $i \neq j$ if we assume that *velocities of different particles are independent random variables*, an important caveat. From Eq. (1.12), we find that the relative root-mean-square fluctuation of energy is

$$\frac{\Delta E_{\mathrm{rms}}}{U} \equiv \frac{\left\langle (E-U)^2 \right\rangle^{1/2}}{U} = \frac{\left[N(m^2/4) \left(\langle v^4 \rangle - \langle v^2 \rangle^2 \right) \right]^{1/2}}{N \langle mv^2/2 \rangle} = \left(\frac{\langle v^4 \rangle}{\langle v^2 \rangle^2} - 1 \right)^{1/2} \frac{1}{\sqrt{N}} \ll 1. \tag{1.13}$$

This is very small for $N \gg 1$ because the prefactor in the above formula is clearly independent of N, as it depends only on single-particle properties, viz., the moments $\langle v^2 \rangle$ and $\langle v^4 \rangle$ of a particle's velocity.

Exercise 1.1. If you like mathematical exercises, figure out how to prove that $\langle v^4 \rangle \geqslant \langle v^2 \rangle^2$, whatever the distribution of v—so we are not taking square root of a negative number!

The result (1.13) implies that the distribution of the system's total energy E (which

‡ Since we have already assumed that the system is homogeneous, we must have $\boldsymbol{u} = \text{const}$ across the system and so, if $\boldsymbol{u}$ is also constant in time, we can just go to a frame moving with velocity $\boldsymbol{u}$.

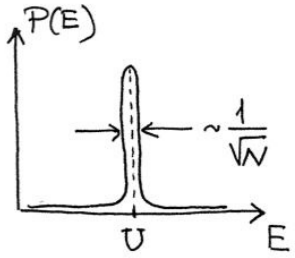


FIGURE 2. Thermodynamic limit, $N \gg 1$: probability distribution of of the exact quantity (here energy E) is sharply peaked around its mean.

is a random variable† because particle velocities are random variables) is very sharply peaked around its mean $U = \langle E \rangle$: the width of this peak is $\sim \Delta E_{\rm rms}/U \sim 1/\sqrt{N} \ll 1$ for $N \ll 1$ (Fig. 2). This is called the *thermodynamic limit*—the statement that mean quantities for systems of very many particles approximate extremely well the exact properties of the system.‡

I hope to have convinced you that *averages* do give us a good representation of the actual state of the system, at least when the number of constituent particles is large.

Exercise 1.2. Consider a large system of volume $\mathcal{V}$ containing $\mathcal{N}$ non-interacting particles. Take some fixed subvolume $V \ll \mathcal{V}$. Calculate the probability to find N particles in volume V. Then assume that both $\mathcal{N}$ and $\mathcal{V}$ tend to ∞, but in such a way that the particle number density is fixed: $\mathcal{N}/\mathcal{V} \to n = \text{const}$.

a) Show that in this limit, the probability p_N to find N particles in volume V (both N and V are fixed, $N \ll \mathcal{N}$) tends to the Poisson distribution whose average is $\langle N \rangle = nV$.

Hint. This involves proving Poisson's limit theorem. You will find inspiration or possibly even the solution in standard probability texts, e.g., Sinai (1992).

b) Prove that

$$\frac{\left\langle (N - \langle N \rangle)^2 \right\rangle^{1/2}}{\langle N \rangle} = \frac{1}{\sqrt{\langle N \rangle}} \tag{1.14}$$

(so fluctuations around the average are very small if $\langle N \rangle \gg 1$).

c) Show that, if $\langle N \rangle \gg 1$, p_N has its maximum at $N \approx \langle N \rangle = nV$; then show that in the vicinity of this maximum, the distribution of N is Gaussian:

$$p_N \approx \frac{1}{\sqrt{2\pi nV}}\, e^{-(N-nV)^2/2nV}.$$

Hint. Use Stirling's formula for $N!$, Taylor-expand $\ln p_N$ around $N = nV$.

The result of (a) is, of course, intuitively obvious, but it is nice to be able to prove it mathematically and even to know with what precision it holds, as you have done in (b)—another demonstration that the world is constructed in a sensible way.

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† Unless the system is completely isolated, in which case $E = \text{const}$ (see §12.2.2). However, completely isolated systems do not really exist (or are, at any rate, inaccessible to observation, on account of being completely isolated) and it tends to be more interesting and more useful to think of systems in which some exchange with outside world is permitted and only mean quantities, in particular mean energy, are fixed (§9).

‡ This may break down if there are strong correlations between particles, i.e., $\langle v_i^2 v_j^2 \rangle \neq \langle v_i^2 \rangle \langle v_j^2 \rangle$: indeed, as I noted after Eq. (1.12), our result is only valid if the averages can be split. Fluctuations in *strongly coupled systems*, where they cannot be split, can be very strong. This is why we focus on "ideal gas" (non-interacting particles; see §2).

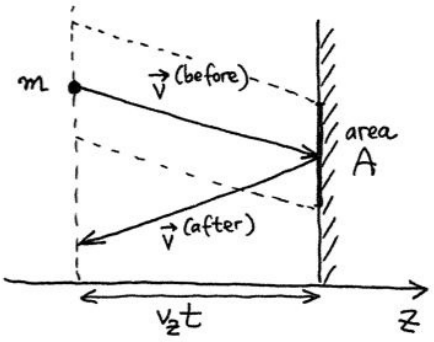


FIGURE 3. Kinetic calculation of pressure. Particles within the volume $Av_z t$ will hit area A during time t and bounce, each delivering momentum $2mv_z$ to the wall.

1.4. *Kinetic Calculation of Pressure*

[Literature: Pauli (2003), §24]

Our objective now is to work out how an important bulk property of a volume of gas—pressure P felt by the walls of a container (or by a body immersed in the gas, or by an imaginary surface separating one part of the gas from another)—is related to average properties of the velocity distribution of the moving particles.

Particles hit a surface (wall) and bounce off; we assume that they do it *elastically*. Recall that

pressure = force per unit area,

force = momentum per unit time.

Therefore, pressure on the wall is the momentum delivered to the wall by the bouncing particles per unit time per unit area ("*momentum flux*").

Let z be the direction perpendicular to the wall (Fig. 3). When a particle bounces off the wall, the projection of its velocity on the z axis changes sign,

$$v_z^{(\text{after})} = -v_z^{(\text{before})}, \tag{1.15}$$

while the two other components of the velocity (v_x and v_y) are unchanged. Therefore, the momentum delivered by the particle to the wall is

$$\Delta p = 2mv_z. \tag{1.16}$$

Consider the particles the z component of whose velocity lies in a small interval $[v_z, v_z + \mathrm{d}v_z]$, where $\mathrm{d}v_z \ll v_z$. Then the contribution of these particles to pressure is

$$\mathrm{d}P(v_z) = \Delta p\, \mathrm{d}\Phi(v_z) = 2mv_z \mathrm{d}\Phi(v_z), \tag{1.17}$$

where $\mathrm{d}\Phi(v_z)$ is the differential *particle flux*, i.e., the number of particles with velocities in the interval $[v_z, v_z + \mathrm{d}v_z]$ hitting the wall per unit time per unit area. In other words, if we consider a wall area A and time t, then

$$\mathrm{d}\Phi(v_z) = \frac{\mathrm{d}N(v_z)}{At}. \tag{1.18}$$

Here $\mathrm{d}N(v_z)$ is the number of particles with velocity in the interval $[v_z, vz + \mathrm{d}v_z]$ that

hit area A over time t:

$$\mathrm{d}N(v_z) = Av_z t \cdot n \cdot f(v_z)\mathrm{d}v_z, \tag{1.19}$$

where $Av_z t$ is the volume where a particle with velocity v_z must be to hit the wall during time t, $n = N/V$ is the number density of particles in the gas and $f(v_z)\mathrm{d}v_z$ is, by definition, the fraction of particles whose velocities are in the interval $[v_z, v_z + \mathrm{d}v_z]$. The differential particle flux is, therefore,

$$\mathrm{d}\Phi(v_z) = nv_z f(v_z)\mathrm{d}v_z \tag{1.20}$$

(perhaps this is just obvious without this lengthy explanation).

We have found that we need to know the *particle distribution function* ("pdf") $f(v_z)$, which is the *probability density function* (also "pdf") of the velocity distribution for a single particle—i.e., the fraction of particles in our infinitesimal interval, $f(v_z)\mathrm{d}v_z$, is the *probability* for a single particle to have its velocity in this interval.† As always in probability theory, the normalisation of the pdf is

$$\int_{-\infty}^{+\infty} f(v_z)\mathrm{d}v_z = 1 \tag{1.21}$$

(the probability for a particle to have some velocity between $-\infty$ and $+\infty$ is 1). We assume that all particles have the same velocity pdf: there is nothing special, statistically, about any given particle or subset of particles and they are all in equilibrium with each other.

From Eqs. (1.17) and (1.20), we have

$$\mathrm{d}P(v_z) = 2mnv_z^2 f(v_z)\mathrm{d}v_z. \tag{1.22}$$

To get the total pressure, we integrate this over all particles with $v_z > 0$ (those that are moving towards the wall rather than away from it):

$$P = \int_0^{\infty} 2mnv_z^2 f(v_z)\mathrm{d}v_z. \tag{1.23}$$

Let us further assume that $f(v_z) = f(-v_z)$, i.e., there is no preference for motion in any particular direction (e.g., the wall is not attractive). Then

$$P = mn\int_{-\infty}^{+\infty} v_z^2 f(v_z)\mathrm{d}v_z = mn\langle v_z^2\rangle. \tag{1.24}$$

The pdf we introduced was in 1D, describing particle velocities in one direction only. It is easily generalised to 3D: we introduce $f(v_x, v_y, v_z)$, which we will abbreviate as $f(\boldsymbol{v})$, such that $f(\boldsymbol{v})\mathrm{d}v_x\mathrm{d}v_y\mathrm{d}v_z$ is the probability for the particle velocity to be in the "cube" $\boldsymbol{v} \in [v_x, v_x + \mathrm{d}v_x] \times [v_y, v_y + \mathrm{d}v_y] \times [v_z, v_z + \mathrm{d}v_z]$ (mathematically speaking, this a *joint probability* for three random variables v_x, v_y and v_z). Then the 1D pdf of v_z is simply

$$f(v_z) = \int_{-\infty}^{+\infty}\mathrm{d}v_x\int_{-\infty}^{+\infty}\mathrm{d}v_y f(v_x, v_y, v_z). \tag{1.25}$$

Therefore, the pressure is

$$\boxed{P = mn\int \mathrm{d}^3\boldsymbol{v}\, v_z^2 f(\boldsymbol{v}) = mn\langle v_z^2\rangle}\,. \tag{1.26}$$

† In §12.3, we will examine somewhat more critically this "frequentist" interpretation of probabilities. A more precise statistical-mechanical definition of $f(v_z)$ will be given in §11.10.

So the pressure on a wall is simply proportional to the mean square z component of the velocity of the particles, where z, by definition, is the direction perpendicular to the wall on which we are calculating the pressure.‡

1.5. *Isotropic Distributions*

Let us now make a further assumption: all directions are statistically the same, the system is *isotropic* (there are no special directions). Then

$$\langle v_x^2\rangle = \langle v_y^2\rangle = \langle v_z^2\rangle = \frac{1}{3}\langle v^2\rangle \tag{1.27}$$

because $v^2 = v_x^2 + v_y^2 + v_z^2$. Therefore, from Eq. (1.26),

$$\boxed{P = \frac{1}{3}\,mn\langle v^2\rangle = \frac{2}{3}\frac{U}{V}}\,, \tag{1.28}$$

where V is the volume of the system and U is its mean internal energy (defined in §1.3). We have discovered the interesting result that in isotropic, 3D systems, pressure is equal to 2/3 of the mean internal energy density (**Exercise**: what is it in an isotropic 2D system?). This relationship between pressure and the energy of the particles makes physical sense: pressure is to do with how vigorously particles bombard the wall and that depends on how fast they are, on average.

How large are the particle velocities? In view of Eq. (1.28) for pressure, we can relate them to a macroscopic quantity that you might have encountered before: the sound speed in a medium of pressure P and mass density $\rho = mn$ is (omitting constants of order unity) $c_s \sim \sqrt{P/\rho} \sim \langle v^2\rangle^{1/2} \sim 300$ m/s.

For future use, let us see what isotropy implies for the pdf. Obviously, f in an isotropic system must be independent of the direction of $\boldsymbol{v}$, it is a function of the *speed* $v = |\boldsymbol{v}|$ alone:

$$f(\boldsymbol{v}) = f(v). \tag{1.29}$$

This amounts to the system being spherically symmetric in $\boldsymbol{v}$ space, so it is convenient to change the $\boldsymbol{v}$-space variables to polars (Fig. 4):

$$(v_x, v_y, v_z) \to (v, \theta, \phi). \tag{1.30}$$

If we know $f(v_x, v_y, v_z)$, what is the joint pdf of v, θ, ϕ, which we will denote $\tilde{f}(v, \theta, \phi)$? Here is how pdfs transform under change of variables:

$$f(\boldsymbol{v})\mathrm{d}v_x\mathrm{d}v_y\mathrm{d}v_z = \underbrace{f(\boldsymbol{v})\underbrace{\left|\frac{\partial(v_x, v_y, v_z)}{\partial(v, \theta, \phi)}\right|}_{\text{Jacobian}}}_{\tilde{f}(v,\theta,\phi)}\mathrm{d}v\mathrm{d}\theta\mathrm{d}\phi = f(\boldsymbol{v})v^2\sin\theta\,\mathrm{d}v\mathrm{d}\theta\mathrm{d}\phi. \tag{1.31}$$

Thus,

$$\tilde{f}(v, \theta, \phi) = f(\boldsymbol{v})v^2\sin\theta = f(v)v^2\sin\theta. \tag{1.32}$$

The last equality is a consequence of isotropy [Eq. (1.29)]. It implies that an isotropic distribution of particle velocities is uniform in ϕ, but not in θ and the pdf of particle

‡ This raises the interesting possibility that pressure need not, in general, be the same in all directions—a possibility that we will eliminate under the additional assumptions of §1.5, but resurrect in Exercise 1.4.

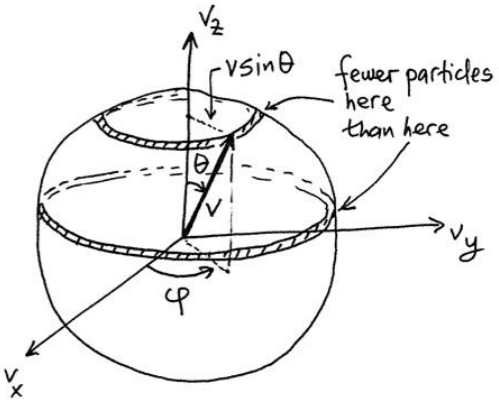


FIGURE 4. Polar coordinates in velocity space. The factor of $\sin\theta$ in Eq. (1.33) accounts for the fact that, if the particles are uniformly distributed over a sphere $|\boldsymbol{v}| = v$, there will be fewer of them in azimuthal bands at low θ than at high θ (the radius of an azimuthal band is $v\sin\theta$).

speeds is†

$$\boxed{\tilde{f}(v) = \int_0^\pi \mathrm{d}\theta \int_0^{2\pi} \mathrm{d}\phi\, \tilde{f}(v,\theta,\phi) = 4\pi v^2 f(v)}\,. \tag{1.33}$$

As an exercise in ascertaining the consistency of our formalism, let us calculate pressure again using polar coordinates in the velocity space (such calculations will prove useful later): as $v_z = v\cos\theta$,

$$\begin{aligned} P = mn\langle v_z^2\rangle &= mn\int \mathrm{d}^3\boldsymbol{v}\, v_z^2 f(\boldsymbol{v}) = mn\int_0^{2\pi}\mathrm{d}\phi\int_0^\pi \mathrm{d}\theta\sin\theta\cos^2\theta\int_0^\infty \mathrm{d}v\, v^4 f(v)\\ &= \frac{4\pi}{3}\,mn\int_0^\infty \mathrm{d}v\, v^4 f(v) = \frac{1}{3}\,mn\int_0^\infty \mathrm{d}v\, v^2\tilde{f}(v) = \frac{1}{3}\,mn\langle v^2\rangle, \end{aligned} \tag{1.34}$$

same as Eq. (1.28).

Exercise 1.3. a) Prove, using $v_x = v\cos\phi\sin\theta$, $v_y = v\sin\phi\sin\theta$ and directly calculating integrals in $\boldsymbol{v}$-space polars, that

$$\langle v_x^2\rangle = \langle v_y^2\rangle = \frac{1}{3}\,\langle v^2\rangle. \tag{1.35}$$

b) Calculate also $\langle v_x v_y\rangle$, $\langle v_x v_z\rangle$, $\langle v_y v_z\rangle$. Could you have worked out the outcome of this last calculation from symmetry arguments?

The answer to the last question is yes. Here is a smart way of computing $\langle v_i v_j\rangle$, where $i,j = x,y,z$ (in fact, you can do all this not just in 3D, but in any number of dimensions, $i,j = 1,2,\ldots,d$). Clearly, $\langle v_i v_j\rangle$ is a symmetric rank-2 tensor (i.e., a tensor, or matrix, with two indices, that remains the same if these indices are swapped). Since the velocity distribution is isotropic, this tensor must be rotationally invariant (i.e., not change under rotations of the coordinate frame). The only symmetric rank-2 tensor that has this property is a constant times Kronecker delta δ_{ij}. So it must be the case that

$$\langle v_i v_j\rangle = C\delta_{ij}, \tag{1.36}$$

where C can only depend on the distribution of *speeds* v (not vectors $\boldsymbol{v}$). Can you figure out what C is? Is it the same in 2D and in 3D? This is a much simpler derivation than doing velocity

† Blundell & Blundell (2009) call the distribution of speeds f and the distribution of vector velocities g, so my f is their g and my $\tilde{f}$ is their f. This variation in notation should help you keep alert and avoid mechanical copying of formulae from textbooks.

integrals directly, but it was worth checking the result by direct integration, as you did above, to convince yourself that the symmetry magic works.

c*) OK, now that you know that it works, calculate $\langle v_i v_j v_k v_l \rangle$ in terms of averages of moments of v (i.e., averages of powers of v such as $\langle v^2 \rangle$ or $\langle v^4 \rangle$).

Hint. Doing this by direct integration is *a lot* of work. Instead, generalise the symmetry argument given above: see what symmetric rotationally invariant rank-4 tensors (i.e., tensors with 4 indices) you can cook up: it turns out that they have to be products of Kronecker deltas, e.g., $\delta_{ij}\delta_{kl}$; what other combinations are there? Then $\langle v_i v_j v_k v_l \rangle$ must be a linear combination of these tensors, with coefficients that depend on moments of v. By examining the symmetry properties of $\langle v_i v_j v_k v_l \rangle$, work out what these coefficients are. How does the answer depend on the dimensionality of the world (2D, 3D, dD)?

Exercise 1.4. Consider an *anisotropic* system, where there exists one (and only one) special direction in space (call it z), which affects the distribution of particle velocities (an example of such a situation is a gas of charged particles—plasma—in a straight magnetic field).

a) How many variables does the distribution function now depend on? (Recall that in the isotropic case, it depended only on one, v.) Write down the most general form of the distribution function under these symmetries—what is the appropriate transformation of variables from (v_x, v_y, v_z)?

b) In terms of averages of those new velocity variables, what is the expression for pressure $P_\parallel$ that the gas will exert on a wall perpendicular to the z axis? (It is called $P_\parallel$ because it is due to particles whose velocities have non-zero projections onto the special direction z.) What is $P_\perp$, pressure on any wall parallel to z?

c) Now consider a wall the normal to which, $\hat{\boldsymbol{n}}$, is at an angle θ to z. What is the pressure on this wall in terms of $P_\parallel$ and $P_\perp$?

Exercise 1.5. Consider an insulated cylindrical vessel filled with monatomic ideal gas. The cylinder is closed on one side and plugged by a piston on the other side. The piston is very slowly pulled out (its velocity u is much smaller than the typical velocities of the gas molecules). Show *using kinetic theory, not thermodynamics,* that during this process the pressure P and volume V of the gas inside the vessel are related by $PV^{5/3} = \text{const}$.

Hint. Consider how the energy of a gas particle changes after each collision with the piston and hence calculate the rate of change of the internal energy of the gas inside the vessel.

[Ginzburg *et al.* 2006, #307]

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2. Classical Ideal Gas in Equilibrium

We now introduce the simplest possible model of a gas and construct its pdf. The assumptions of the model are

- *Particles do not interact* (e.g., they do not attract or repel each other), except for having *elastic* binary collisions, during which they conserve total momentum and energy, do not fracture or stick.

- They are *point particles*, i.e., they do not occupy a significant fraction of the system's volume, however many of them there are. This assumption is necessary to ensure that a particle's ability to be anywhere in space is not restricted by being crowded out by other particles. We will relax this assumption for *"real gases"* in Part VII.

- They are *classical particles*, so there are *no quantum correlations* (which would jeopardise a particle's ability to have a particular momentum if the corresponding quantum state(s) is(are) already occupied by other particles). We will relax this assumption for *"quantum gases"* in Part VI.

• They are *non-relativistic particles*, i.e., their speeds are $v \ll c$. You will have an opportunity to play around with relativistic gases later on (e.g., Exercise 11.3).

In practice, all this is satisfied if the gas is sufficiently *dilute* (low enough number density n) and sufficiently *hot* (high enough temperature T) to avoid Quantum Mechanics, but not so hot as to run into Relativity. I will make these constraints quantitative after I define T (see §2.3).

2.1. *Maxwell's Distribution*

[Literature: Pauli (2003), §25]

Consider our model gas in a container of volume V and assume that there are no changes to external (boundary) conditions or fields—everything is *homogeneous in time and space*.

Let us wait long enough for a sufficient number (=a few) of *collisions* to occur so *all memory of initial conditions is lost* (recall the discussion in §1.1 of how that happens; roughly how long we must wait we will be able to estimate after we discuss collisions in §4).

We will call the resulting state *equilibrium* in the sense that it will be *statistically stationary*, i.e., the particles in the gas will settle into some *velocity distribution independent of time, position or initial conditions* (NB: it is essential to have collisions to achieve this!). How the gas attains such a state will be the subject of §§5–6.

Since the distribution function $f(\boldsymbol{v})$ does not depend on anything, we must be able to work our what it is from some general principles.

First of all, if there are *no special directions* in the system, the pdf must be isotropic [Eq. (1.29)]:

$$f(\boldsymbol{v}) = f(v) = g(v^2), \tag{2.1}$$

where g is some function of v^2 (introduced for the convenience of the upcoming derivation).

Exercise 2.1. In the real world, you might object, there are always special directions. For example, gravity (particles have mass!). After we have finished deriving $f(v)$, think under what condition gravity can be ignored.

Also, the Earth is rotating in a definite direction, so particles in the atmosphere are subject to Coriolis and centrifugal forces. Under what condition can these forces be ignored?

Maxwell (1860) argued (or conjectured) that *the three components of the velocity vector must be independent random variables.*† Then

$$f(\boldsymbol{v}) = h(v_x^2)h(v_y^2)h(v_z^2), \tag{2.2}$$

where all three distributions are the same because of isotropy and depend only on squares of velocity components assuming *mirror symmetry* of the distribution (invariance with respect to the transformation $\boldsymbol{v} \to -\boldsymbol{v}$; this means there are no flows or fluxes in the system).

But in view of isotropy [Eq. (2.1)], Eq. (2.2) implies

$$h(v_x^2)h(v_y^2)h(v_z^2) = g(v^2) = g(v_x^2 + v_y^2 + v_z^2). \tag{2.3}$$

† It is possible to prove this for classical ideal gas either from Statistical Mechanics (see §11.10) or by analysing elastic binary collisions (Boltzmann 1995; Chapman & Cowling 1991), but here we will simply *assume* that this is true.

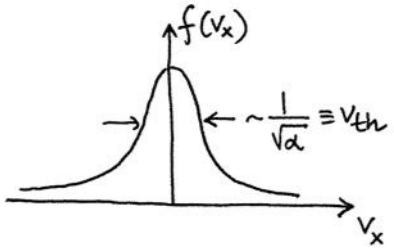


FIGURE 5. The Maxwellian distribution, Eq. (2.14).

Denoting further

$$\varphi(v_x^2) \equiv \ln h(v_x^2) \quad \text{and} \quad \psi(v^2) \equiv \ln g(v^2), \tag{2.4}$$

we find

$$\varphi(v_x^2) + \varphi(v_y^2) + \varphi(v_z^2) = \psi(v_x^2 + v_y^2 + v_z^2). \tag{2.5}$$

Such a functional relationship can only be satisfied if φ and ψ are *linear functions* of their arguments:

$$\varphi(v_x^2) = -\alpha v_x^2 + \beta \quad \text{and} \quad \psi(v^2) = -\alpha v^2 + 3\beta. \tag{2.6}$$

Here α and β are as yet undetermined constants and the minus sign is purely a matter convention (α will turn out to be positive).

Proof. Differentiate Eq. (2.5) with respect to v_x^2 keeping v_y^2 and v_z^2 constant:

$$\psi'(v_x^2 + v_y^2 + v_z^2) = \varphi'(v_x^2). \tag{2.7}$$

Differentiate this again with respect to v_y^2 keeping v_x^2 and v_z^2 constant:

$$\psi''(v_x^2 + v_y^2 + v_z^2) = 0, \quad \text{or} \quad \psi''(v^2) = 0. \tag{2.8}$$

Therefore,

$$\psi(v^2) = -\alpha v^2 + 3\beta, \tag{2.9}$$

where $-\alpha$ and 3β are constants of integration. This is the desired solution (2.6) for ψ.

Now substitute this form of ψ into Eq. (2.5) and let $v_y^2 = v_z^2 = 0$:

$$\varphi(v_x^2) + 2\varphi(0) = \psi(v_x^2) = -\alpha v_x^2 + 3\beta. \tag{2.10}$$

Let $v_x^2 = 0$ in the above: this gives $\varphi(0) = \beta$. Eq. (2.10) then becomes

$$\varphi(v_x^2) = -\alpha v_x^2 + \beta, \quad \text{q.e.d.} \tag{2.11}$$

Eq. (2.6) gives us

$$f(\boldsymbol{v}) = g(v^2) = e^{\psi(v^2)} = C\, e^{-\alpha v^2}, \quad \text{where} \quad C \equiv e^{3\beta}, \tag{2.12}$$

so the velocity distribution has a Gaussian ("bell-curve") shape. It remains to determine the constants α and C. One of them is easy: we know that $\int \mathrm{d}^3\boldsymbol{v}\, f(\boldsymbol{v}) = 1$, so

$$C \int \mathrm{d}^3\boldsymbol{v}\, e^{-\alpha v^2} = C \int \mathrm{d}v_x e^{-\alpha v_x^2} \int \mathrm{d}v_y e^{-\alpha v_y^2} \int \mathrm{d}v_z e^{-\alpha v_z^2} = C \left(\sqrt{\frac{\pi}{\alpha}}\right)^3 = 1. \tag{2.13}$$

Therefore,

$$C = \left(\frac{\alpha}{\pi}\right)^{3/2} \quad \Rightarrow \quad f(\boldsymbol{v}) = \left(\frac{\alpha}{\pi}\right)^{3/2} e^{-\alpha v^2}. \tag{2.14}$$

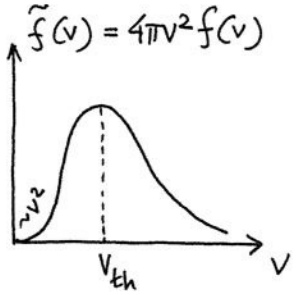


FIGURE 6. The pdf of speeds for a Maxwellian distribution, Eq. (2.16).

Thus, we have expressed $f(\boldsymbol{v})$ in terms of only one scalar parameter α! Have we derived something from nothing? Not quite: the functional form of the pdf followed from a set of assumptions about (statistical) *symmetries* of the equilibrium state.

What is the meaning of α? This parameter tells us about the *width* of the velocity distribution (Fig. 5). Dimensionally, $1/\sqrt{\alpha} = v_{\rm th}$ is some characteristic speed, which we call the *thermal speed* (formally, this is just a renaming, but it helps interpretation). It characterises the typical values particle velocities can take (having $v \gg v_{\rm th}$ is highly improbable because of the strong decay of the Gaussian function). With this new notation, we have

$$\boxed{f(\boldsymbol{v}) = \frac{1}{(\sqrt{\pi}v_{\rm th})^3}\, e^{-v^2/v_{\rm th}^2}}\,, \tag{2.15}$$

an easy-to-remember functional form. This will be called *Maxwell's distribution*, also known as a *Maxwellian*, once we manage to give thermodynamical interpretation to $v_{\rm th}$ (see §2.2).

To complete the formalism, the pdf of speeds is [see Eq. (1.33)]

$$\tilde{f}(v) = \frac{4\pi v^2}{(\sqrt{\pi}v_{\rm th})^3}\, e^{-v^2/v_{\rm th}^2}. \tag{2.16}$$

Note that $v_{\rm th}$ is the most probable speed (Fig. 6; **Exercise**: prove this).

All these manipulations are well and good, but to relate $v_{\rm th}$ to something physical, we need to relate it to something *measurable*. What is measurable about a gas in a box? The two most obviously measurable quantities are

—pressure (we can measure force on a wall),

—temperature (we can stick in a thermometer, as it is defined in Thermodynamics).

We will see in the next section how to relate $v_{\rm th}$, T and P.

Exercise 2.2. a) Work out a general formula for $\langle v^n \rangle$ (n is an arbitrary positive integer) in terms of $v_{\rm th}$, for a Maxwellian gas (*Hint:* it is useful to consider separately odd and even n). If $n < m$, what is larger, $\langle v^n \rangle^{1/n}$ or $\langle v^m \rangle^{1/m}$? Why is this, qualitatively?

b*) What is the distribution of speeds $\tilde{f}(v)$ in a Maxwellian d-dimensional gas?

Hint. This involves calculating the area of a d-dimensional unit sphere in velocity space.

c) Obtain the exact formula for the rms energy fluctuation in a Maxwellian gas (see §1.3).

2.2. *Equation of State and Temperature*

In §1, we learned how, given $f(\boldsymbol{v})$, to compute pressure: the Maxwellian (2.15) is isotropic, so, using Eq. (1.34),

$$P = \frac{1}{3}\, mn\langle v^2\rangle = \frac{1}{3}\, mn \int \mathrm{d}^3\boldsymbol{v}\, v^2 \frac{e^{-v^2/v_{\rm th}^2}}{(\sqrt{\pi}v_{\rm th})^3} = \frac{nmv_{\rm th}^2}{2}. \tag{2.17}$$

This provides us with a clear relationship between $v_{\rm th}$ and the thermodynamic quantities P, and $n = N/V$. Furthermore, we know empirically† that, for 1 mole of ideal gas ($N = N_{\rm A}$, the Avogadro number of particles),

$$PV = RT, \quad \text{where} \quad R = 8.31447 \text{ J/K} \quad (\text{the } \textit{gas constant}), \tag{2.18}$$

and T here is the absolute temperature as defined in Thermodynamics (via Zeroth Law etc.; see Part I). Another, equivalent, form of this *equation of state* is

$$P = nk_{\rm B}T, \quad \text{where} \quad k_{\rm B} = \frac{R}{N_{\rm A}} = 1.3807 \times 10^{-23} \text{ J/K} \quad (\text{the } \textit{Boltzmann constant}). \tag{2.19}$$

Comparing Eqs. (2.19) and (2.17), we can extract the relationship between $v_{\rm th}$ and the thermodynamic temperature:

$$\boxed{\frac{mv_{\rm th}^2}{2} = k_{\rm B}T}\,. \tag{2.20}$$

Thus, *temperature in Kinetic Theory is simply the kinetic energy of a particle moving at the most probable speed in the Maxwellian velocity distribution*,‡ or, vice versa, the width of the Maxwellian is related to temperature via

$$v_{\rm th} = \sqrt{\frac{2k_{\rm B}T}{m}}. \tag{2.21}$$

Two other, equivalent, statements of this sort are that (**Exercise**: prove them)

$$\frac{1}{2}k_{\rm B}T = \frac{m\langle v_x^2\rangle}{2}, \tag{2.22}$$

the mean energy per particle per degree of freedom, and, recalling the definition of U, Eq. (1.10), that

$$\frac{3}{2}\, k_{\rm B}T = \frac{U}{N}, \tag{2.23}$$

the mean energy per particle.¶ From Eq. (2.23), *the heat capacity* of the monatomic classical ideal gas is

$$C_V = \frac{3}{2}\, k_{\rm B}N. \tag{2.24}$$

† From the thermodynamic experiments of Boyle 1662, Mariotte 1676 ($P \propto 1/V$ at constant T), Charles 1787 ($V \propto T$ at constant P), Gay-Lussac 1809 ($P \propto T$ at constant V) and Amontons 1699 (who anticipated the latter two by about a century). To be precise, what we know empirically is that Eq. (2.18) holds for the thermodynamically defined quantities P and T in most gases as long as they are measured in parameter regimes in which we expect the ideal gas approximation to hold.

‡ The Boltzmann constant $k_{\rm B}$ is just a dimensional conversion coefficient owing its existence to the fact that historically T is measured in K rather than (as it should have been) in units of energy.

¶ Note that one sometimes *defines* temperature in Kinetic Theory via Eqs. (2.23), (2.22) or (2.20) and then proves the equivalence of this "kinetic temperature" and the thermodynamic temperature (see, e.g., Chapman & Cowling 1991).

Finally, using our expression (2.21) for $v_{\rm th}$, we arrive at the traditional formula for the Maxwellian: Eq. (2.15) becomes

$$\boxed{f(\boldsymbol{v}) = \left(\frac{m}{2\pi k_{\rm B} T}\right)^{3/2} \exp\left(-\frac{m v^2}{2 k_{\rm B} T}\right)}\,. \tag{2.25}$$

This is a particular case (which we have here derived for our model gas) of a much more general statistical-mechanical result known as the Gibbs distribution—exactly how to recover Maxwell from Gibbs will be explained in §11.10.

The above treatment has not just given us the particle-velocity pdf in equilibrium—we have also learned something new and important about the physical meaning of temperature, *which has turned out to measure how energetic, on average, microscopic particles are*. This is progress compared to Thermodynamics, where T was a purely macroscopic and rather mysterious (if indispensable) quantity: recall that the defining property of T was that it was some quantity that would equalise across a system in equilibrium (e.g., if two systems with initially different temperatures were brought into contact); in Thermodynamics, we were able to prove that such a quantity must exist, but we could not explain exactly what it was or how the equalisation happened. It is now clear how it happens for two volumes of gas when they are mixed together: particles collide and eventually attain a global Maxwellian distribution with a single parameter $\alpha \Leftrightarrow v_{\rm th} \Leftrightarrow T$. When a gas touches a hot or cold wall, particles of the gas collide with the vibrating molecules of the wall—the energy of this vibration is also proportional to T, as we will see in Statistical Mechanics—and again attain a Maxwellian with the same T.

To summarise, we now have the full *thermodynamics of classical monatomic ideal gas:* specific formulae for energy $U = U(N,T)$, Eq. (2.23), heat capacity $C_V = C_V(N)$, Eq. (2.24), equation of state $P = P(N,V,T)$, Eq. (2.19), etc. In addition, we know the full velocity distribution, Eq. (2.25), and so can calculate other interesting things, which thermodynamics is ignorant of (effusion, §3, will be the first example of that, followed by the great and glorious theory of heat and momentum transport, §§5–6).

2.3. *Validity of the Classical Limit*

Here are two very quick estimates for the range of temperatures in which the classical results derived above should hold.

2.3.1. *Nonrelativistic Limit*

Particles must be much slower than light:

$$k_{\rm B} T = \frac{m v_{\rm th}^2}{2} \ll m c^2 \quad \Rightarrow \quad T \ll \frac{m c^2}{k_{\rm B}} \equiv T_{\rm rel}. \tag{2.26}$$

If we formally substitute into this formula the typical molecular mass for air, we get $T_{\rm rel} \sim 10^{14}$ K (but of course molecules will have dissociated and atoms have become ionised at much lower temperatures than this). This being a huge number tells us that the non-relativistic assumption is very safe.

2.3.2. *No Quantum Correlations*

We are thinking of particles as hard point spheres whizzing about with certain velocities and occasionally colliding. But in quantum mechanics, if a particle has a definite velocity (momentum), it cannot have a definite position, so certainly cannot be thought of as a

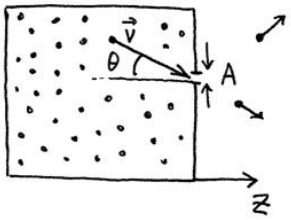


FIGURE 7. Effusion: gas escapes from a container through a small hole.

"point." The relationship between the uncertainties in the particle momentum and its position is

$$\delta r \delta p \sim \hbar. \tag{2.27}$$

Let us estimate the momentum uncertainty as the thermal spread in the particle velocity distribution:

$$\delta p \sim m v_{\rm th} \sim \sqrt{m k_{\rm B} T}. \tag{2.28}$$

Then we can continue thinking of particles as points if the typical extent of the volume of space per particle $(1/n)$ is much larger than the uncertainty in the particle's position:†

$$\frac{1}{n^{1/3}} \gg \delta r \sim \frac{\hbar}{\delta p} \sim \frac{\hbar}{m v_{\rm th}} \sim \frac{\hbar}{\sqrt{m k_{\rm B} T}} \quad \Rightarrow \quad T \gg \frac{\hbar^2 n^{2/3}}{m k_{\rm B}} \equiv T_{\rm deg}. \tag{2.29}$$

The "degeneration temperature" $T_{\rm deg}$ for air at $P = 1$ atm is a few K, but of course most gases will liquefy or even solidify at such temperatures. Again, we see that the classical approximation appears quite safe for our current, mundane purposes. Note, however, that $T_{\rm deg}$ depends on density (or pressure, $P = n k_{\rm B} T$) and when this is very high, gas can become quantum even at quite high temperatures (a famous example is electrons in metals)—then particles get "smeared" over each other and one has to worry about quantum correlations. We will do this in Part VI.

3. Effusion

Let us practice our newly acquired knowledge of particle distributions (§2) and calculations of fluxes (§1.4) on a simple, but interesting problem.

Consider a container containing ideal gas and make a small hole in it (Fig. 7). Suppose the hole is so small that its diameter

$$d \ll \lambda_{\rm mfp}, \tag{3.1}$$

where $\lambda_{\rm mfp}$ is the particle mean free path (the typical distance particles travel between collisions—we will calculate it in §4). Then macroscopically the gas does not "know" about the hole—this is a way to abduct particles without changing their distribution.‡ This can be a way to find out, non-invasively, what the velocity distribution is inside the container, provided we have a way of measuring the velocities of the escaping particles. On an even more applied note, we might be interested in what happens in this set up because we are concerned about gas leaks through small holes in some industrially important walls or partitions.

† Another way to get this is to demand that the volume per particle should contain many de Broglie wave lengths $\lambda_{\rm dB} = h/p$ associated with the thermal motion: $n\lambda_{\rm dB}^3 \sim n(h/m v_{\rm th})^3 \ll 1$.

‡ In §5, we will learn what happens when the gas does "know" and why the hole has to be larger than $\lambda_{\rm mfp}$ for that.

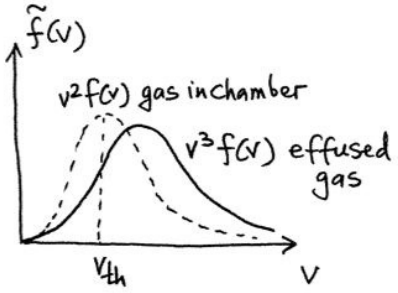


FIGURE 8. Speed distribution of effusing particles: favours faster particles more than the Maxwellian; see Eq. (3.3).

There are two obviously interesting quantitative questions we can ask:

(i) Given some distribution of particles inside the container, $f(\boldsymbol{v})$, what will be the distribution of the particles emerging from the hole?

(ii) Given the area A of the hole, how many particles escape through it per unit time? (i.e., what is the particle *flux* through the hole?)

The answers are quite easy to obtain. Indeed, this is just like the calculation of pressure (§1.4): there we needed to calculate the flux of momentum carried by the particles hitting an area of the wall; here we need the flux of particles themselves that hit an area of the wall (hole of area A)—these particles will obviously be the ones that escape through the hole. Taking, as in §1.4, z to be the direction perpendicular to the wall, we find that the (differential) particle flux, i.e., the number per unit time per unit area of particles with velocities in the 3D cube $[\boldsymbol{v}, \boldsymbol{v}+\mathrm{d}^3\boldsymbol{v}]$, is [see Eq. (1.20)]

$$\mathrm{d}\Phi(\boldsymbol{v}) = nv_z f(\boldsymbol{v})\,\mathrm{d}^3\boldsymbol{v} = n\,\underbrace{v^3 f(v)\mathrm{d}v}_{\substack{\text{speed dis-}\\ \text{tribution}}}\,\underbrace{\cos\theta\sin\theta\,\mathrm{d}\theta\,\mathrm{d}\phi}_{\substack{\text{angular}\\ \text{distribution}}}, \tag{3.2}$$

where, in the second expression, we assumed that the distribution is isotropic, $f(\boldsymbol{v}) = f(v)$, and used $v_z = v\cos\theta$ and $\mathrm{d}^3\boldsymbol{v} = v^2\sin\theta\,\mathrm{d}v\,\mathrm{d}\theta\,\mathrm{d}\phi$.

Thus, we have the answer to our question (i) and conclude that *the distribution of the emerging particles is neither isotropic nor Maxwellian* (even if the gas inside the container is Maxwellian). The angle distribution is not isotropic (has an extra $\cos\theta$ factor) because particles travelling nearly perpendicularly to the wall (small θ) escape with greater probability.† The speed distribution is not Maxwellian (has an extra factor of v; Fig. 8) because faster particles get out with greater probability (somewhat like the smarter students passing with greater probability through the narrow admissions filter into Oxford—not an entirely deterministic process though, just like effusion).

Exercise 3.1. a) Consider a gas effusing out through a small hole into an evacuated sphere, with the particles sticking to the internal surface of the sphere once they hit it. Show that this would produce a uniform coating of the surface.

b) Show that the distribution of the speeds of the particles that might be found in transit between the effuson hole and the surface at any given time is the same as for a Maxwellian gas.

† However, there are fewer of these particles in the original isotropic angle distribution $\propto \sin\theta\mathrm{d}\theta\mathrm{d}\phi$, so statistically, it is $\theta = 45^\circ$ that is the most probable angle for the effusing particles.

If we are only interested in the distribution of speeds, we can integrate out the angular dependence in Eq. (3.2): the flux through the hole of particles with *speeds* in the interval $[v, v + \mathrm{d}v]$ is

$$\mathrm{d}\tilde{\Phi}(v) = nv^3 f(v)\mathrm{d}v \int_0^{\pi/2} \mathrm{d}\theta\, \cos\theta \sin\theta \int_0^{2\pi} \mathrm{d}\phi = \pi n v^3 f(v)\mathrm{d}v = \frac{1}{4}\, nv\tilde{f}(v)\mathrm{d}v, \qquad (3.3)$$

where $\tilde{f}(v)$ is the distribution of speeds inside the container, related to $f(v)$ via Eq. (1.33). Note the upper limit of integration with respect to θ: it is $\pi/2$ and not π because only particles moving *toward* the hole ($v_z = v\cos\theta > 0$) will escape through it.

Finally, the total flux of effusing particles (number of particles per unit time per unit area escaping through the hole, no matter what their speed) is

$$\Phi = \int_0^\infty \mathrm{d}v\, \frac{1}{4}\, nv\tilde{f}(v) = \frac{1}{4}\, n\langle v\rangle, \qquad (3.4)$$

where $\langle v\rangle$ is the average particle speed inside the container. For a Maxwellian distribution, $\tilde{f}(v)$ is given by Eq. (2.16) and so $\langle v\rangle$ can be readily computed:

$$\boxed{\Phi = \frac{1}{4}\, n\sqrt{\frac{8k_\mathrm{B}T}{\pi m}} = \frac{P}{\sqrt{2\pi m k_\mathrm{B} T}}} \qquad (3.5)$$

(**Exercise**: check this result; use the ideal-gas equation of state).

Thus, we have the answer to our question (ii): the number of particles effusing per unit time through a hole of area A is ΦA, where Φ can be calculated via Eq. (3.5) in terms of macroscopic measurable quantities, P (or n) and T, if we know the mass m of the particles.

The fact that, given P and T, the effusion flux $\Phi \propto m^{-1/2}$, implies that if we put a mixture of two particle species into a box with a small hole and let them effuse, the lighter species will effuse at a larger rate than the heavier one, so the composition of the blend emerging on the other side of the hole will favour the lighter particles. This has applications to separation of isotopes which are strictly on a need-to-know basis.

Exercise 3.2. Show that the condition of no mass flow between two insulated chambers containing ideal gas at pressures $P_{1,2}$ and temperatures $T_{1,2}$ and connected by a tiny hole is

$$\frac{P_1}{\sqrt{T_1}} = \frac{P_2}{\sqrt{T_2}}. \qquad (3.6)$$

What would be the condition for no flow if the hole between the chambers were large ($d \gg \lambda_\mathrm{mfp}$)?

Exercise 3.3. What is the *energy flux* through the hole? (i.e., what is the energy lost by the gas in the container per unit time, as particles leave by a hole of area A?)

Exercise 3.4. Consider a thermally insulated container of volume V with a small hole of area A, containing a gas with molecular mass m. At time $t = 0$, the density is n_0 and the temperature is T_0. As gas effuses out through a small hole, both density and temperature inside the container will drop. Work out their time dependence, $n(t)$ and $T(t)$, in terms of the quantities given above. What is the characteristic time over which they will change significantly?

Hint. Temperature is related to the total energy of the particles in the container. The flux of energy of the effusing particles will determine the rate of change of energy inside the container in the same way as the particle flux determines the rate of change of the particle number (and, therefore, their density). Based on this principle, you should be able to derive two differential (with respect to time) equations for two unknowns, n and T. Having derived them, solve them.

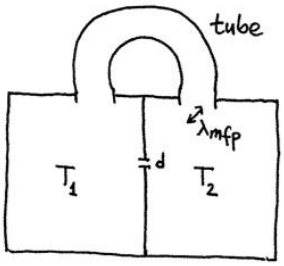


FIGURE 9. Two chambers connected by a tube or an effusion hole (Exrercise 3.6).

Exercise 3.5. A festive helium balloon of radius $R = 20$ cm made of a soft but unstretchable material is tied to a lamppost. The material is not perfect and can have microholes of approximate radius $r = 10^{-5}$ cm, through which helium will be leaking out. As this happens, the balloon shrinks under atmospheric pressure.

a) Assuming the balloon material is a good thermal conductor, calculate how many microholes per cm^2 the balloon can have if it is to lose no more than 10% of its initial volume over one festive week.

b) Now suppose the balloon material is a perfect thermal insulator. Repeat the calculation.

Exercise 3.6. Consider two chambers of equal volume separated by an insulating wall and containing an ideal gas maintained at two distinct temperatures $T_1 < T_2$. Initially the chambers are connected by a long tube (Fig. 9) whose diameter is much larger than the mean free path in either chamber and equilibrium is established (while maintaining T_1 and T_2). Then the tube is removed, the chambers are sealed, but a small hole is opened in the insulating wall, with diameter $d \ll \lambda_{\rm mfp}$ (for either gas).

a) In what direction will the gas flow through the hole, from cold to hot or from hot to cold?

b) If the total mass of the gas in both chambers is M, show that the mass ΔM that will be transferred through the hole from one chamber to the other before a new equilibrium is established is

$$\Delta M = \frac{\sqrt{T_1 T_2}}{T_1 + T_2} \frac{\sqrt{T_2} - \sqrt{T_1}}{\sqrt{T_1} + \sqrt{T_2}} M. \tag{3.7}$$

[Ginzburg *et al.* 2006, #427]

4. Collisions

We argued (on plausible symmetry grounds) that in equilibrium, we should expect the pdf to be Maxwellian for an ideal gas. "In equilibrium" meant that initial conditions were forgotten, i.e., that particles had collided a sufficient number of times. There are certain constraints on the time scales on which we can believe that the gas is in equilibrium (how long do we wait for the gas to "Maxwellianise"?) and on the spatial scales of the system if we are to describe it in these terms. Namely,

- $t \gg \tau_{\rm c}$, the *collision time*, or the typical time that a particle spends in free flight between collisions (it is also convenient to define the *collision rate* $\nu_{\rm c} = 1/\tau_{\rm c}$, the typical number of collisions a particle has per unit time);
- $l \gg \lambda_{\rm mfp}$, the *mean free path*, or the typical distance a particle travels between collisions.

In order to estimate $\tau_{\rm c}$ and $\lambda_{\rm mfp}$, we will have to bring in some information and some assumptions about the microscopic properties of the gas and the nature of collisions.

4.1. *Cross-section*

Assume that particles are *hard spheres* of diameter d. Then they can be considered to collide if their centres approach each other within the distance d. Think of a particle with velocity $\boldsymbol{v}$ moving through a cylinder (Fig. 10) whose axis is $\boldsymbol{v}$ and whose cross section is

$$\sigma = \pi d^2. \tag{4.1}$$

As the particle will necessarily collide with any other particle whose centre is within this cylinder, σ is called the *collisional cross section*.

A useful way of parametrising the more general situation in which particles are *not* hard spheres but instead interact with each other via some smooth potential (e.g., charged particles feeling each other's Coulomb potential), is to introduce the "effective cross section," in which case d tells you how close they have to get to have a "collision," i.e., to be deflected from a straight path.

Exercise 4.1. Coulomb Collisions. For particles with charge e, mass m and temperature T, estimate d.

4.2. *Collision Time*

Moving through the imaginary cylinder of cross section σ, a particle sweeps the volume $\sigma v t$ over time t. The average number of other particles in this volume is $\sigma v t n$. If this is > 1, then there will be at least one collision during the time t. Thus, we define the *collision time* $t = \tau_\mathrm{c}$ so that

$$\sigma v \tau_\mathrm{c} n = 1 \quad \Rightarrow \quad \tau_\mathrm{c} = \frac{1}{\sigma n v}, \quad \nu_\mathrm{c} = \frac{1}{\tau_\mathrm{c}} = \sigma n v. \tag{4.2}$$

As we are interested in a "typical" particle, v here is some typical speed. For a Maxwellian distribution, we may pick any of these:

$$v \sim \langle v \rangle \sim v_\mathrm{rms} \sim v_\mathrm{th}. \tag{4.3}$$

All these speeds have different numerical coefficients (viz., $\langle v \rangle = 2v_\mathrm{th}/\sqrt{\pi}$, $v_\mathrm{rms} = \sqrt{3/2}\, v_\mathrm{th}$), but we are in the realm of order-of-magnitude estimates here, so it does not really matter which we choose. To fix the notation, let us define

$$\boxed{\tau_\mathrm{c} = \frac{1}{\nu_\mathrm{c}} = \frac{1}{\sigma n v_\mathrm{th}} = \frac{1}{\sigma n}\sqrt{\frac{m}{2k_\mathrm{B}T}}}\,. \tag{4.4}$$

4.3. *Mean Free Path*

Then the typical distance a particle travels between collisions is

$$\boxed{\lambda_\mathrm{mfp} = v_\mathrm{th}\tau_\mathrm{c} = \frac{1}{\sigma n}} \tag{4.5}$$

(or we could have said that this is the length of the cylinder of cross section σ such that it contains at least one particle: $\sigma \lambda_\mathrm{mfp} n = 1$).

Note that, given the gas density, λ_mfp is independent of temperature. At constant T, it is $\lambda_\mathrm{mfp} = k_\mathrm{B}T/\sigma P \propto P^{-1}$, inversely proportional to pressure.

4.4. *Relative Speed*

[Literature: Pauli (2003), §26]

If you have a suspicious mind, you might worry that the arguments above are somewhat

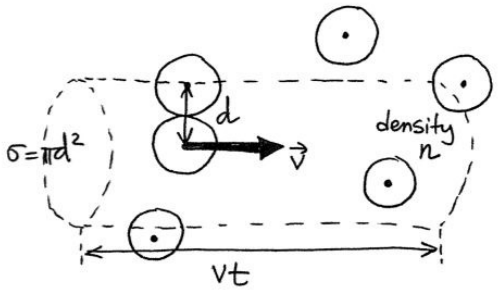


FIGURE 10. Two chambers connected by a tube or an effusion hole (Exrercise 3.6).

dodgy: indeed, we effectively assumed that while our chosen particle moved through its σvt cylinder, all other particles just sat there waiting to be collided with. Surely what matters is, in fact, the *relative* speed of colliding particles? This might prompt us to introduce the following definition for the mean collision rate, which is conventional:

$$\nu_{\rm c} = \sigma n \langle v_{\rm r} \rangle, \tag{4.6}$$

where $v_{\rm r} = |\boldsymbol{v}_1 - \boldsymbol{v}_2|$ is the mean relative speed of a pair of particles. It is more or less obvious that $\langle v_{\rm r} \rangle \sim v_{\rm th}$ just like any other speed in a Maxwellian distribution (what else could it possibly be?!), but let us convince ourselves of this anyway (it is also an instructive exercise to calculate $\langle v_{\rm r} \rangle$).

By definition,

$$\langle v_{\rm r} \rangle = \int {\rm d}^3 \boldsymbol{v}_1 \int {\rm d}^3 \boldsymbol{v}_2 |\boldsymbol{v}_1 - \boldsymbol{v}_2| f(\boldsymbol{v}_1, \boldsymbol{v}_2), \tag{4.7}$$

where $f(\boldsymbol{v}_1, \boldsymbol{v}_2)$ is the joint two-particle distribution function (i.e., the pdf that the first velocity is in a ${\rm d}^3\boldsymbol{v}_1$ interval around $\boldsymbol{v}_1$ and the second in ${\rm d}^3\boldsymbol{v}_2$ around $\boldsymbol{v}_2$). Now we make a key assumption:

$$f(\boldsymbol{v}_1, \boldsymbol{v}_2) = f(\boldsymbol{v}_1) f(\boldsymbol{v}_2), \tag{4.8}$$

i.e., the two particles' velocities are *independent*. This makes sense as long as we are considering them *before* they have undergone a collision—remember that particles are non-interacting in an ideal gas, except for collisions.† Taking the single-particle pdfs f to be Maxwellian, we get

$$\begin{aligned}
\langle v_{\rm r} \rangle &= \int\int {\rm d}^3 \boldsymbol{v}_1 {\rm d}^3 \boldsymbol{v}_2 |\boldsymbol{v}_1 - \boldsymbol{v}_2| \frac{1}{(\pi v_{\rm th}^2)^3} \exp\left(-\frac{v_1^2}{v_{\rm th}^2} - \frac{v_2^2}{v_{\rm th}^2} \right) \\
&= \int {\rm d}^3 \boldsymbol{v}_{\rm r} v_{\rm r} \int {\rm d}^3 \boldsymbol{V} \frac{1}{(\pi v_{\rm th}^2)^3} \exp\left(-\frac{2V^2}{v_{\rm th}^2} - \frac{v_{\rm r}^2}{2v_{\rm th}^2} \right) \\
&= \int {\rm d}^3 \boldsymbol{v}_{\rm r} v_{\rm r} \frac{1}{(\sqrt{2\pi} v_{\rm th})^3} \exp\left(-\frac{v_{\rm r}^2}{2v_{\rm th}^2} \right) = \sqrt{2} \langle v \rangle = \frac{2\sqrt{2}\, v_{\rm th}}{\sqrt{\pi}} = 4\sqrt{\frac{k_{\rm B} T}{\pi m}}.
\end{aligned} \tag{4.9}$$

In the calculation of the double integral, we changed variables $(\boldsymbol{v}_1, \boldsymbol{v}_2) \to (\boldsymbol{V}, \boldsymbol{v}_{\rm r})$, where $\boldsymbol{V} = (\boldsymbol{v}_1 + \boldsymbol{v}_2)/2$ is the centre of mass and $\boldsymbol{v}_{\rm r} = \boldsymbol{v}_1 - \boldsymbol{v}_2$ the relative velocity; then $v_1^2 + v_2^2 = 2V^2 + v_{\rm r}^2/2$ and ${\rm d}^3\boldsymbol{v}_1 {\rm d}^3\boldsymbol{v}_2 = {\rm d}^3\boldsymbol{V} {\rm d}^3\boldsymbol{v}_{\rm r}$.

It is in view of this result that some books define the mean collision rate

$$\nu_{\rm c} = \sqrt{2}\, \sigma n \langle v \rangle = \frac{1}{\tau_{\rm c}} \tag{4.10}$$

and the mean free path

$$\lambda_{\rm mfp} = \langle v \rangle \tau_{\rm c} = \frac{1}{\sqrt{2}\, \sigma n} \tag{4.11}$$

† It certainly would not be sensible to assume that they are independent right *after* a collision. The assumption of independence of particle velocities before a collision is a key one in the derivation of Boltzmann's collision integral (Boltzmann 1995; Chapman & Cowling 1991) and is known as Boltzmann's *Stosszahlansatz*. Boltzmann's derivation would be a central topic in a more advanced course on Kinetic Theory (e.g., Dellar 2015).

(here $\langle v\rangle$, *not* $\langle v_{\rm r}\rangle$, is used because $\lambda_{\rm mfp}$ is just the distance one particle travels at the average speed over time $\tau_{\rm c}$). Of course, these formulae in a sense represent a precision overkill: $\nu_{\rm c}$ and $\lambda_{\rm mfp}$ are quantities whose purpose is order-of-magnitude estimate of the collisional time and spatial scales, so factors of order unity are irrelevant.

Exercise 4.2. Show that $\langle v_{\rm r}^2\rangle = 2\langle v^2\rangle$.

Hint. This is much easier to show than $\langle v_{\rm r}\rangle = \sqrt{2}\langle v\rangle$. You should not need more than one line of trivial algebra to prove it.

Exercise 4.3. Consider a gas that is a mixture of two species of molecules: type-1 with diameter d_1, mass m_1 and mean number density n_1 and type-2 with diameter d_2, mass m_2 and mean number density n_2. If we let them collide with each other for a while, they will eventually settle into a Maxwellian equilibrium and the temperatures of the two species will be the same.

a) What will be the rms speeds of each of the two species?

b) Show that the combined pressure of the mixture will be $P = P_1 + P_2$ (Dalton's law).

c) What is the cross-section for the collisions between type-1 and type-2 molecules?

d) What is the mean collision rate of type-1 molecules with type-2 molecules? Is it the same as the collision rate of type-2 molecules with type-1 molecules? (Think carefully about what exactly you mean when you define these rates.)

Hint. In (d), you will need to find the mean relative speed of the two types of particles, a calculation analogous to the one in §4.4. Note however, that as the masses of the particles of the two different types can be very different, the distinction between $\langle v_{\rm r}\rangle$ and $\langle v_1\rangle$ or $\langle v_2\rangle$ can now be much more important than in the case of like-particle collisions.

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5. From Local to Global Equilibrium (Transport Equations)

5.1. *Inhomogeneous Distributions*

We have so far discussed a very simple situation in which the gas was homogeneous, so the velocity pdf $f(\boldsymbol{v})$ described the state of affairs at any point in space and quantities such as n, P, T were constants in space. This also meant that we could assume that there were no flows (if there was a constant mean flow $\boldsymbol{u}$, we could always go to the frame moving with it). This is obviously not the most general situation: thus, we know from experience that if we open a window from a warm room onto a cold Oxford autumn, it will be colder near the window than far away from it (so T will be a function of space), a draft may develop (mean flow $\boldsymbol{u}$ of air, with some gradients across the room), etc. Clearly such systems will have a particle velocity distribution that is different in different places. Let us therefore generalise our notion of the velocity pdf and introduce *the particle distribution function in the position and velocity space* ("phase space"):

$$F(t,\boldsymbol{r},\boldsymbol{v})\mathrm{d}^3\boldsymbol{r}\mathrm{d}^3\boldsymbol{v} = \text{average number of particles with velocities in the 3D } \boldsymbol{v}\text{-space volume } [v_x, v_x+\mathrm{d}v_x]\times[v_y, v_y+\mathrm{d}v_y]\times[v_z, v_z+\mathrm{d}v_z] \text{ finding themselves in the spatial cube } [x, x+\mathrm{d}x]\times[y, y+\mathrm{d}y]\times[z, z+\mathrm{d}z] \text{ at time } t.$$

We have followed convention in choosing the normalisation

$$\int \mathrm{d}^3\boldsymbol{r}\int \mathrm{d}^3\boldsymbol{v}\, F(t,\boldsymbol{r},\boldsymbol{v}) = N, \tag{5.1}$$

the total number of particles (rather than 1). Clearly, the *0-th velocity moment* of F is the (position- and time-dependent) particle number density:

$$\int \mathrm{d}^3\boldsymbol{v}\, F(t,\boldsymbol{r},\boldsymbol{v}) = n(t,\boldsymbol{r}), \tag{5.2}$$

which integrates to the total particle number:

$$\int \mathrm{d}^3\boldsymbol{r}\, n(t,\boldsymbol{r}) = N \tag{5.3}$$

(the $\boldsymbol{r}$ integrals are always over the system's volume V). Note that in a homogeneous system,

$$n(\boldsymbol{r}) = n = \text{const} \quad \text{and} \quad F(\boldsymbol{r},\boldsymbol{v}) = F(\boldsymbol{v}) = nf(\boldsymbol{v}), \tag{5.4}$$

which gets us back to our old familiar homogeneous velocity pdf $f(\boldsymbol{v})$ (which integrates to 1 over the velocity space).

If we know $F(t,\boldsymbol{r},\boldsymbol{v})$, we can calculate other bulk properties of the gas, besides its density (5.2), by taking *moments* of F, i.e., velocity integrals of F multiplied by various powers of $\boldsymbol{v}$.

Thus, the *first moment*,

$$\int \mathrm{d}^3\boldsymbol{v}\, m\boldsymbol{v}F(t,\boldsymbol{r},\boldsymbol{v}) = mn(t,\boldsymbol{r})\boldsymbol{u}(t,\boldsymbol{r}), \tag{5.5}$$

is the *mean momentum density*, where $\boldsymbol{u}(t,\boldsymbol{r})$ is the mean velocity of the gas flow (without the factor of m, this expression, $n\boldsymbol{u}$, is the *mean particle flux*).

A *second moment* gives the *mean energy density*:

$$\begin{aligned}
\int \mathrm{d}^3\boldsymbol{v}\, \frac{mv^2}{2}\, F(t,\boldsymbol{r},\boldsymbol{v}) &= \int \mathrm{d}^3\boldsymbol{w}\, \frac{m|\boldsymbol{u}+\boldsymbol{w}|^2}{2}\, F \\
&= \frac{mu^2}{2} \underbrace{\int \mathrm{d}^3\boldsymbol{w}\, F}_{=n(t,\boldsymbol{r})} + m\boldsymbol{u} \cdot \underbrace{\int \mathrm{d}^3\boldsymbol{w}\, \boldsymbol{w}F}_{\substack{=0 \text{ by} \\ \text{definition} \\ \text{of } \boldsymbol{w}}} + \int \mathrm{d}^3\boldsymbol{w}\, \frac{mw^2}{2}\, F \\
&= \underbrace{\frac{mnu^2}{2}}_{\substack{\text{energy density of} \\ \text{mean motions;} \\ \boldsymbol{u}(t,\boldsymbol{r}) \text{ given by} \\ \text{Eq. (5.5)}}} + \underbrace{\left\langle \frac{mw^2}{2} \right\rangle n}_{\substack{\equiv \varepsilon(t,\boldsymbol{r}), \\ \text{internal-energy} \\ \text{density (motions} \\ \text{around the mean)}}},
\end{aligned} \tag{5.6}$$

where we have utilised the decomposition of particle velocities into the mean and peculiar, $\boldsymbol{v} = \boldsymbol{u}(t,\boldsymbol{r}) + \boldsymbol{w}$ (cf. §1.2), where $\boldsymbol{u}$ is defined by Eq. (5.5). The total "ordered" energy and the total internal ("disordered") energy are (cf. Eq. (1.9))

$$K = \int \mathrm{d}^3\boldsymbol{r}\, \frac{mnu^2}{2} \quad \text{and} \quad U = \int \mathrm{d}^3\boldsymbol{r}\, \varepsilon(t,\boldsymbol{r}), \tag{5.7}$$

respectively.

So how do we calculate $F(t,\boldsymbol{r},\boldsymbol{v})$?

5.2. *Local Maxwellian Equilibrium*

Recall that we attributed the dependence of F on $\boldsymbol{r}$ and t to certain *macroscopic* inhomogeneities of the system (open windows etc.). It is reasonable, for a wide class of systems, to assume that the spatial (l) and temporal (t) scales of these inhomogeneities

are much greater than $\lambda_{\rm mfp}$ and $\tau_{\rm c}$ in our gas:†

$$l \gg \lambda_{\rm mfp}, \quad t \gg \tau_{\rm c}. \tag{5.8}$$

Then we can break up our gas into "fluid elements" of size Δl and consider them for a time Δt such that

$$l \gg \Delta l \gg \lambda_{\rm mfp}, \quad t \gg \Delta t \gg \tau_{\rm c}. \tag{5.9}$$

Clearly, on these "intermediate" scales, the fluid elements will behave as little homogeneous systems, with locally constant n, moving at some locally constant $\boldsymbol{u}$. We can then go to the frame moving with this local velocity $\boldsymbol{u}$ (i.e., following the fluid element) and expect that all our old results derived for a homogeneous static volume of gas will apply—in particular, we should expect the gas comprising each fluid element to attain, on the collisional time scale $\tau_{\rm c}$, the *local Maxwellian equilibrium:*

$$\boxed{F_{\rm M}(t,\boldsymbol{r},\boldsymbol{v}) = n(t,\boldsymbol{r})\left[\frac{m}{2\pi k_{\rm B}T(t,\boldsymbol{r})}\right]^{3/2} \exp\left[-\frac{m|\boldsymbol{v}-\boldsymbol{u}(t,\boldsymbol{r})|^2}{2k_{\rm B}T(t,\boldsymbol{r})}\right] = \frac{n}{(\sqrt{\pi}v_{\rm th})^3}\, e^{-w^2/v_{\rm th}^2}} \tag{5.10}$$

(n and $v_{\rm th} = \sqrt{2k_{\rm B}T/m}$ here are both functions of t and $\boldsymbol{r}$).

Everything is as before, but now locally: e.g., the pressure is [cf. Eq. (1.28)]

$$P(t,\boldsymbol{r}) = n(t,\boldsymbol{r})k_{\rm B}T(t,\boldsymbol{r}) = \frac{2}{3}\,\varepsilon(t,\boldsymbol{r}) \tag{5.11}$$

and, therefore, the *local temperature* is, by definition, 2/3 of the mean internal energy per particle:

$$k_{\rm B}T(t,\boldsymbol{r}) = \frac{2}{3}\frac{\varepsilon(t,\boldsymbol{r})}{n(t,\boldsymbol{r})} = \frac{2}{3}\left\langle\frac{mw^2}{2}\right\rangle = \langle mw_x^2\rangle \tag{5.12}$$

[cf. Eqs. (2.22) and (2.23)].

It is great progress to learn that *only three functions on a 3D space ($\boldsymbol{r}$), viz., n, $\boldsymbol{u}$ and T, completely describe the particle distribution in the 6D phase space ($\boldsymbol{v}$, $\boldsymbol{r}$).*‡ How then do we determine these three functions?

Thermodynamics gives us a hint as to how they will evolve in time. We know that if we put in contact two systems with different T, their temperatures will tend to equalise—so *temperature gradients* between fluid elements must tend to relax—and this should be a collisional process because that is how contact between particles with different energies is made. Same is true about *velocity gradients* (we will prove this thermodynamically in §10.4). But Thermodynamics just tells us that everything must tend from local to *global equilibrium* (no gradients)—not *how fast* that happens or what the intermediate stages in this evolution look like. Kinetic Theory will allow us to describe this *route to equilibrium* quantitatively.¶

But before bringing the full power of Kinetic Theory to bear on this problem (in §6), we will first consider what can be said *a priori* about the evolution of n, $\boldsymbol{u}$ and T.

† So we are now treating the limit opposite to what we were considering when discussing effusion (§3).

‡ NB: in §6.2, we will learn that in fact, fluxes of momentum and energy—and, therefore, transport phenomena—come from small deviation of F from the local Maxwellian.

¶ We will also see what happens when systems are constantly *driven* out of equilibrium (§§5.6.4–5.6.6).

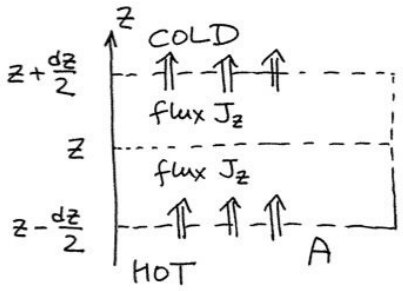


FIGURE 11. Heat flux, see Eq. (5.18).

5.3. *Conservation Laws*

Clearly, the evolution of n, $\boldsymbol{u}$ and T must be constrained by *conservation laws*. Indeed, if our system is closed and insulated, whatever happens in it must respect the conservation of the total number of particles:

$$\int \mathrm{d}^3\boldsymbol{r}\, n = N = \text{const}, \tag{5.13}$$

of the total momentum:

$$\int \mathrm{d}^3\boldsymbol{r}\, mn\boldsymbol{u} = 0 = \text{const} \tag{5.14}$$

(0 because we can work in the frame moving with the centre of mass of the system), and of the total energy:

$$\int \mathrm{d}^3\boldsymbol{r}\left(\frac{mnu^2}{2} + \underbrace{\varepsilon}_{=\frac{3}{2}nk_\mathrm{B}T}\right) = K + U = \text{const}. \tag{5.15}$$

Without knowing any Kinetic Theory, can we establish from these constraints the general form of the evolution equations for n, $\boldsymbol{u}$ and T? Yes, we can!

5.3.1. *Temperature*

For simplicity, let us first consider a situation in which nothing moves on average ($\boldsymbol{u} = 0$) and $n = \text{const}$ globally.† Then all energy in the system is internal energy,

$$\varepsilon = \frac{3}{2}\, nk_\mathrm{B}T(t, \boldsymbol{r}) \tag{5.16}$$

and only temperature is inhomogeneous.

To simplify even further, consider a 1D problem, where $T = T(t, z)$ varies in one direction only. Internal energy (heat) will flow from hot to cold regions (as we know from Thermodynamics), so there will be a *heat flux*:

$J_z(z) =$ internal energy flowing along z per unit time through unit area perpendicular to the z axis.

Then the rate of change of internal energy in a small volume $A \times [z - \mathrm{d}z/2, z + \mathrm{d}z/2]$ (A

† The latter means we are considering an "incompressible" medium—generally, a pretty good approximation for gas unless it is moving (super)sonically and a really good approximation for liquids and solids. In fact, everything can be derived without these approximations, I am making them only to minimise the amount of algebra.

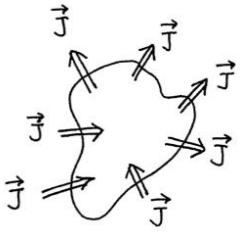


FIGURE 12. Heat flows into or out of an arbitrary volume.

is area; see Fig. 11) is‡

$$\underbrace{\frac{\partial}{\partial t}\frac{3}{2}nk_{\mathrm{B}}T\cdot A\mathrm{d}z}_{\substack{\text{energy in the}\\ \text{volume } A\mathrm{d}z}} = \underbrace{J_z\left(z-\frac{\mathrm{d}z}{2}\right)\cdot A}_{\substack{\text{energy}\\ \text{flowing in}}} - \underbrace{J_z\left(z+\frac{\mathrm{d}z}{2}\right)\cdot A}_{\substack{\text{energy}\\ \text{flowing out}}}. \tag{5.17}$$

This instantly gives

$$\frac{\partial}{\partial t}\frac{3}{2}nk_{\mathrm{B}}T = -\frac{J_z\left(z+\frac{\mathrm{d}z}{2}\right)-J_z\left(z-\frac{\mathrm{d}z}{2}\right)}{\mathrm{d}z} = -\frac{\partial J_z}{\partial z}, \quad \text{as } \mathrm{d}z\to 0. \tag{5.18}$$

It is very easy to generalise this to a 3D situation. The rate of change of internal energy in an arbitrary volume V is

$$\underbrace{\frac{\partial}{\partial t}\int_V \mathrm{d}^3\boldsymbol{r}\,\frac{3}{2}nk_{\mathrm{B}}T}_{\substack{\text{energy in}\\ \text{volume } V}} = -\underbrace{\int_{\partial V}\mathrm{d}\boldsymbol{A}\cdot\boldsymbol{J}}_{\substack{\text{flux through}\\ \text{the boundary}\\ \text{of the volume}}} = -\int_V \mathrm{d}^3\boldsymbol{r}\,\boldsymbol{\nabla}\cdot\boldsymbol{J}, \tag{5.19}$$

where we have used Gauss's theorem. The heat flux is now a vector, $\boldsymbol{J}$, pointing in the direction in which the heat flows (Fig. 12). Since V can be chosen completely arbitrarily, the integral relation (5.19) becomes a differential one:

$$\boxed{\frac{\partial}{\partial t}\frac{3}{2}nk_{\mathrm{B}}T = -\boldsymbol{\nabla}\cdot\boldsymbol{J}}\,. \tag{5.20}$$

This is of course just a local statement of energy conservation.

Thus, *if we can calculate the heat flux, $\boldsymbol{J}$, we can determine the evolution of T.*

Exercise 5.1. Electromagnetism: Charge and Energy Conservation. a) Prove (from Maxwell's equations) that the Coulomb charge density ρ and the current density $\boldsymbol{j}$ (which is the flux of charge) are related by

$$\frac{\partial\rho}{\partial t} = -\boldsymbol{\nabla}\cdot\boldsymbol{j}. \tag{5.21}$$

Show that this also follows from the conservation of charge.

b) The energy density of the electromagnetic field is (in Gauss units)

$$\varepsilon = \frac{E^2+B^2}{8\pi}. \tag{5.22}$$

‡ Note that incompressibility (n = const) is useful here as it allows us not to worry about the net flux of matter into (or out of) our volume. In the more general, compressible, case, this contribution to the rate of change of internal energy turns up in the form of the $\boldsymbol{\nabla}\cdot\boldsymbol{u}$ term in Eq. (5.37).

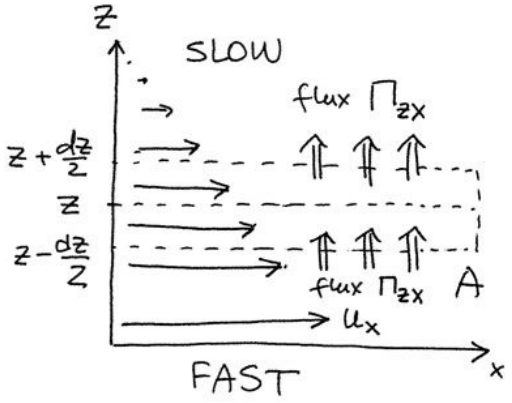


FIGURE 13. Momentum flux, see Eq. (5.26).

What is the flux of this energy (in terms of $\boldsymbol{E}$ and $\boldsymbol{B}$)? Is the electromagnetic energy conserved? (if not, where does it go?)

This is an opportunity to check whether you understand E&M.

Exercise 5.2. Continuity Equation. Now consider a gas with some mean flow velocity $\boldsymbol{u}(t,\boldsymbol{r})$ and density $n(t,\boldsymbol{r})$, both varying in (3D) space and time. What is the flux of particles through a surface within such a system? Use the requirement of particle conservation to derive *the continuity equation*

$$\frac{\partial n}{\partial t} = -\boldsymbol{\nabla}\cdot(n\boldsymbol{u}). \tag{5.23}$$

Note that this equation can be rewritten as

$$\left(\frac{\partial}{\partial t} + \boldsymbol{u}\cdot\boldsymbol{\nabla}\right) n = -n\boldsymbol{\nabla}\cdot\boldsymbol{u}. \tag{5.24}$$

The left-hand side is the so-called *convective time derivative* of n—the rate of change of density in a fluid element moving with velocity $\boldsymbol{u}$ (think about why that is; we will use similar logic in §6.3). The above equation then means that a negative divergence of the gas flow, $\boldsymbol{\nabla}\cdot\boldsymbol{u} < 0$, implies local compression, whereas positive divergence, $\boldsymbol{\nabla}\cdot\boldsymbol{u} > 0$, implies local rarefaction.

In fact, you know all this from your first-year maths.

5.3.2. *Velocity*

We can handle momentum conservation in a similar fashion. Let us again assume $n = \text{const}$, but allow a z-dependent flow velocity in the x direction (this is called a *shear flow*):

$$\boldsymbol{u} = u_x(t,z)\hat{\boldsymbol{x}} \tag{5.25}$$

In this system, momentum will flow from fast- to slow-moving layers of the gas (because, as we will learn below, they experience friction against each other, due to particle collisions). We define *momentum flux*

$\Pi_{zx}(z)$ = momentum in the x direction flowing along z per unit time through unit area perpendicular to the z axis.

Then, analogously to Eq. (5.17) (see Fig. 13),

$$\frac{\partial}{\partial t}\underbrace{mnu_x\cdot A\mathrm{d}z}_{\substack{\text{momentum in}\\ \text{the volume}\\ A\mathrm{d}z}} = \underbrace{\Pi_{zx}\left(z-\frac{\mathrm{d}z}{2}\right)\cdot A}_{\substack{\text{momentum}\\ \text{flowing in}}} - \underbrace{\Pi_{zx}\left(z+\frac{\mathrm{d}z}{2}\right)\cdot A}_{\substack{\text{momentum}\\ \text{flowing out}}}, \tag{5.26}$$

whence

$$\boxed{\frac{\partial}{\partial t}mnu_x = -\frac{\partial \Pi_{zx}}{\partial z}}\,. \tag{5.27}$$

Thus, *in order to determine the evolution of velocity, we must calculate the momentum flux.*

Let us generalise this calculation. Let $n(t,\boldsymbol{r})$ and $\boldsymbol{u}(t,\boldsymbol{r})$ both be functions of space and time. Considering an arbitrary volume V of the gas, we can write the rate of change of momentum in it as

$$\frac{\partial}{\partial t}\int_V \mathrm{d}^3\boldsymbol{r}\, mn\boldsymbol{u} = -\int_{\partial V}\mathrm{d}\boldsymbol{S}\cdot\boldsymbol{\Pi}, \tag{5.28}$$

or, in tensor notation,

$$\frac{\partial}{\partial t}\int_V \mathrm{d}^3\boldsymbol{r}\, mnu_j = -\int_{\partial V}\mathrm{d}S_i\Pi_{ij}. \tag{5.29}$$

The momentum flux is now a tensor (also knows as the *stress tensor*): Π_{ij} is the flux of the j-th component of momentum in the i direction (in the case of the shear flow, this tensor only had one non-zero component, Π_{zx}). Application of Gauss's Theorem gives us

$$\frac{\partial}{\partial t}mnu_j = -\partial_i\Pi_{ij}. \tag{5.30}$$

The momentum flux consists of three parts:

—one (*"convective"*) due to the fact that the boundary of a fluid element containing the same particles itself moves with velocity $\boldsymbol{u}$: the flux of the j-th component of the momentum, mnu_j, due to this effect is $mnu_j\boldsymbol{u}$ and so

$$\Pi_{ij}^{(\text{convective})} = mnu_iu_j, \tag{5.31}$$

i.e., momentum "carries itself" (just like it carries particle density: recall the flux of particles being $n\boldsymbol{u}$ in Eq. (5.23));

—one due to the fact that there is pressure in the system and pressure is also momentum flux, viz., the flux of each component of the momentum in the direction of that component (recall §1.4: particles with velocity component v_z transfer momentum in the z direction to the wall perpendicular to z—in our current calculation, this pressure acts on the boundary of our chosen volume V); thus, the pressure part of the momentum flux is diagonal:

$$\Pi_{ij}^{(\text{pressure})} = P\delta_{ij}; \tag{5.32}$$

—and, finally, one due to friction between layers of gas moving at different velocities; as we have seen in §5.3.2, this part of the momentum-flux tensor, $\Pi_{ij}^{(\text{viscous})}$, will contain off-diagonal elements, but we have not yet worked out how to calculate them.

Substituting these three contributions, viz.,

$$\Pi_{ij} = mnu_iu_j + P\delta_{ij} + \Pi_{ij}^{(\text{viscous})}, \tag{5.33}$$

into Eq. (5.30), we get

$$\frac{\partial}{\partial t}mn\boldsymbol{u} = -\boldsymbol{\nabla}\cdot(mn\boldsymbol{u}\boldsymbol{u}) - \boldsymbol{\nabla}P - \boldsymbol{\nabla}\cdot\boldsymbol{\Pi}^{(\text{viscous})}, \tag{5.34}$$

or, after using Eq. (5.23) to express $\partial n/\partial t$,

$$mn\left(\frac{\partial \boldsymbol{u}}{\partial t}+\boldsymbol{u}\cdot\boldsymbol{\nabla}\boldsymbol{u}\right)=-\boldsymbol{\nabla}P-\boldsymbol{\nabla}\cdot\boldsymbol{\Pi}^{(\mathrm{viscous})}. \tag{5.35}$$

This is the desired generalisation of Eq. (5.27)—the evolution equation for the mean flow velocity $\boldsymbol{u}(t,\boldsymbol{r})$. This equation says that fluid elements move around at their own velocity (the convective time derivative in the left-hand side) and are subject to forces arising from pressure gradients and friction (the right-hand side); if there are any other forces in the system, e.g., gravity, those have to be put into the right-hand side of Eq. (5.35). Obviously, we still need to calculate $\boldsymbol{\Pi}^{(\mathrm{viscous})}$ in order for this equation to be useful in actual calculations.

Exercise 5.3. Energy Flows. Generalise Eq. (5.20) to the case of non-zero flow velocity $\boldsymbol{u}(t,\boldsymbol{r})\neq 0$ and non-constant $n(t,\boldsymbol{r})$. Consider the total energy density of the fluid,

$$\frac{mnu^2}{2}+\frac{3}{2}\,nk_{\mathrm{B}}T, \tag{5.36}$$

and calculate the rate of change of the total energy inside a volume V due to two contributions: energy being carried by the flow $\boldsymbol{u}$ through the boundary ∂V + work done by pressure on that boundary. If you use Eqs. (5.23) and (5.35) to work out the time derivatives of n and $\boldsymbol{u}$, in the end you should be left with the following evolution equation for T:

$$\frac{3}{2}\,nk_{\mathrm{B}}\left(\frac{\partial T}{\partial t}+\boldsymbol{u}\cdot\boldsymbol{\nabla}T\right)=-\boldsymbol{\nabla}\cdot\boldsymbol{J}-nk_{\mathrm{B}}T\boldsymbol{\nabla}\cdot\boldsymbol{u}-\Pi_{ij}^{(\mathrm{viscous})}\partial_i u_j. \tag{5.37}$$

Interpret all the terms and identify the conditions under which the gas behaves *adiabatically*, i.e., satisfies

$$\left(\frac{\partial}{\partial t}+\boldsymbol{u}\cdot\boldsymbol{\nabla}\right)\frac{P}{n^{5/3}}=0. \tag{5.38}$$

5.4. *Thermal Conductivity and Viscosity*

So, we have evolution equations for temperature and velocity, Eqs. (5.18) and (5.27) (sticking with the 1D case), which are local expressions of energy and momentum conservation laws containing the as yet unknown fluxes J_z and Π_{zx}. In (relatively) short order, we will learn how to calculate these fluxes from kinetic theory (i.e., from particle distributions)—we can be optimistic about being able to do this in view of our experience calculating fluxes in the effusion problem (§3; effusion was transport on scales $\ll\lambda_{\mathrm{mfp}}$, what we need now is transport on scales $\gg\lambda_{\mathrm{mfp}}$). However, first let us ask *a priori* what the answer should look like.

From thermodynamics (heat flows from hot to cold), we expect that

- $J_z\neq 0$ only if $\partial T/\partial z\neq 0$,
- J_z has the opposite sign to $\partial T/\partial z$.

Similarly, $\Pi_{zx}\neq 0$ only if $\partial u_x/\partial z\neq 0$ and also has the opposite sign. It is then a plausible conjecture that *fluxes will just be proportional to (minus) gradients*—indeed this is more or less inevitable if the gradients are in some (to be determined; see §5.6.3) sense not very large, because we can simply Taylor-expand the fluxes, which are clearly functions

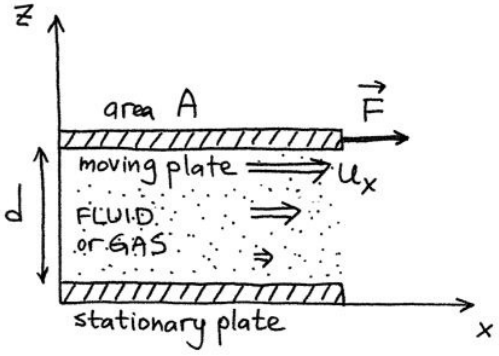


FIGURE 14. (Gedanken) experiment to define and determine viscosity; see Eq. (5.43).

of the gradients, around zero values of these gradients:

$$J_z\left(\frac{\partial T}{\partial z}\right) = \underbrace{J_z(0)}_{=0} + J_z'(0)\frac{\partial T}{\partial z} + \cdots \approx -\varkappa\,\frac{\partial T}{\partial z}, \tag{5.39}$$

$$\Pi_{zx}\left(\frac{\partial u_x}{\partial z}\right) = \underbrace{\Pi_{zx}(0)}_{=0} + \Pi_{zx}'(0)\frac{\partial u_x}{\partial z} + \cdots \approx -\eta\,\frac{\partial u_x}{\partial z}, \tag{5.40}$$

where we have introduced two (we expect, positive) *transport coefficients:*

- *the thermal conductivity* $\varkappa$,
- *the dynamical viscosity* η.

In 3D, the heat flux is (obviously)

$$\boldsymbol{J} = -\varkappa\boldsymbol{\nabla}T, \tag{5.41}$$

whereas the viscous part of the stress tensor [appearing in Eq. (5.35)] is

$$\boldsymbol{\Pi}^{(\text{viscous})} = -\eta\left[\boldsymbol{\nabla}\boldsymbol{u} + (\boldsymbol{\nabla}\boldsymbol{u})^T - \frac{2}{3}\,\mathsf{I}\,\boldsymbol{\nabla}\cdot\boldsymbol{u}\right], \tag{5.42}$$

where I is a unit matrix. The latter expression is not immediately obvious, but you might want to investigate on your own why it holds.

The proportionalities between fluxes and gradients expressed by Eqs. (5.39) and (5.40) do indeed turn out to hold, experimentally, in a good range of physical parameters (n, P, T) and for very many substances (gases or fluids). The coefficients $\varkappa$ and η can be experimentally measured and tabulated even if we know nothing of kinetics or microphysics. It is thus that physics—and certainly engineering!—very often progress to workable models without necessarily achieving complete understanding right away.

For example, viscosity can be introduced and measured as follows. Set up an experiment with two horizontal plates of area A at a vertical distance d from each other and a fluid (or gas) between them, the lower plate stationary, the upper one being moved at a horizontal velocity u_x (Fig. 14). If one measures the force F that one needs to apply to the upper plate in order to maintain a constant u_x, one discovers that, for small enough d,

$$\boxed{\frac{F}{A} = \eta\,\frac{u_x}{d} \approx \eta\,\frac{\partial u_x}{\partial z}}\,, \tag{5.43}$$

where η is a dimensional coefficient independent of u_x, d, or A and approximately constant in a reasonable range of physical conditions for any particular type of inter-plate substance used. By definition, η is the *dynamical viscosity* of that substance. The left-hand side of Eq. (5.43) is

force (=momentum per time) per area, which is the momentum flux downward from the upper plate to the lower, $F/A = -\Pi_{zx}$, and so Eq. (5.40) is recovered.

The physics of momentum transport here is straightforward: the upper plate moves, the molecules of gas immediately adjacent to that plate collide with it, receive some momentum, eventually make their way some distance downward, collide with molecules in a lower layer of gas, pass some momentum to them, those in turn collide with molecules further down, etc. (in the case of a fluid, we would talk about the layer immediately adjacent to the moving plate sticking to it and passing momentum to the next layer lower down via friction, etc.).

Note that the relationships (5.39) and (5.40) are valid much more broadly than will be the upcoming expressions for $\varkappa$ and η that we will derive for ideal gas. Thus, we can talk about the viscosity of water or thermal conductivity of a metal, although neither obviously can be viewed as a collection of non-interacting billiard-ball particles on any level of simplification.

5.5. *Transport Equations*

If we now substitute Eqs. (5.39) and (5.40) into Eqs. (5.18) and (5.27), we obtain closed equations for T and u_x:

$$\frac{\partial}{\partial t}\frac{3}{2}nk_{\rm B}T = \varkappa\frac{\partial^2 T}{\partial z^2}, \tag{5.44}$$

$$\frac{\partial}{\partial t}mnu_x = \eta\frac{\partial^2 u_x}{\partial z^2}. \tag{5.45}$$

These are the *transport equations* that we were after.

Note that in pulling $\varkappa$ and η out of the z derivative, we assumed them to be independent of z: this is fine even though they depend on T (which depends on z) as long as the temperature gradients and, therefore, the temperature differences are not large on the scales that we are considering and so $\varkappa$ and η can be approximated by constant values taken at some reference temperature.

Let us make this quantitative. Let $\varkappa = \varkappa(T)$ and assume that $T = T_0 + \delta T$, where $T_0 = \text{const}$ and all the temperature variation is contained in the small perturbation $\delta T(t,z) \ll T_0$. This is indeed a commonplace situation: temperature variations in our everyday environment rarely exceed $\sim 10\%$ of the absolute temperature $T \sim 300$ K. Then

$$\varkappa(T) \approx \varkappa(T_0) + \varkappa'(T_0)\delta T \tag{5.46}$$

and so, from Eqs. (5.18) and (5.39),

$$\frac{\partial}{\partial t}\frac{3}{2}nk_{\rm B}T = \frac{\partial}{\partial z}\varkappa(T)\frac{\partial T}{\partial z} \approx \varkappa(T_0)\frac{\partial^2\delta T}{\partial z^2} + \varkappa'(T_0)\frac{\partial}{\partial z}\delta T\frac{\partial\delta T}{\partial z}, \tag{5.47}$$

but the second term is quadratic in the small quantity δT and so can be neglected, giving us back Eq. (5.44) (after δT is replaced by T, which is legitimate because the constant part T_0 vanishes under gradients).

5.6. *Relaxation to Global Equilibrium*

Let us now form some idea of the nature of the solutions to the transport equations (5.44) and (5.45): what do they tell us about the time evolution of temperature and velocity? Recall that the motivation of this entire line of inquiry was our expectation that the gas would get to *local* Maxwellian equilibrium (§5.2) over a few collision times and then slowly evolve towards *global* Maxwellian equilibrium, in which all spatial gradients in n, $\boldsymbol{u}$ or T would be erased. We are about to see that this is exactly the behaviour that Eqs. (5.44) and (5.45) describe.

It is apposite to notice here that *these equations have the same mathematical structure*:

they are both *diffusion equations* (why that is, physically, will be discussed in §5.7). Let us write them explicitly in this form:

$$\frac{\partial T}{\partial t} = D_T \frac{\partial^2 T}{\partial z^2}, \tag{5.48}$$

$$\frac{\partial u_x}{\partial t} = \nu \frac{\partial^2 u_x}{\partial z^2}. \tag{5.49}$$

The temperature (equivalently, energy) diffusion coefficient is the *thermal diffusivity*, related to the thermal conductivity in a simple fashion:

$$D_T = \frac{\varkappa}{n c_1}, \tag{5.50}$$

where $c_1 = 3k_\mathrm{B}/2$ is the heat capacity per particle (we have written the expression for D_T in this way because everything we have done so far is in fact valid also for gases or other substances with different values of c_1). Similarly, the velocity (or momentum) diffusion coefficient is the *kinematic viscosity*, related to the dynamical viscosity as

$$\nu = \frac{\eta}{mn}. \tag{5.51}$$

Since the Laplacian (∇^2, or $\partial^2/\partial z^2$ in 1D) is a negative definite operator, Eqs. (5.48) and (5.49) describe gradual relaxation with time of temperature and velocity gradients, provided $D_T > 0$ and $\nu > 0$—relaxation to a global, homogeneous Maxwellian equilibrium.

5.6.1. *Initial-Value Problem: Fourier Decomposition*

The simplest way to see this and to estimate the time scales on which this relaxation will occur is to consider an initial-value problem for, e.g., Eq. (5.48) with some boundary conditions that allow decomposition of the initial condition and the solution into a Fourier series (or, more generally, a Fourier integral). If the initial temperature distribution is

$$T(t=0,z) = \sum_k \hat{T}_0(k) e^{ikz}, \tag{5.52}$$

then the solution of Eq. (5.48),

$$T(t,z) = \sum_k \hat{T}(t,k) e^{ikz}, \tag{5.53}$$

satisfies†

$$\frac{\partial \hat{T}}{\partial t} = -D_T k^2 \hat{T} \quad \Rightarrow \quad \hat{T}(t,k) = \hat{T}_0(k) e^{-D_T k^2 t}. \tag{5.54}$$

Thus, *spatial variations ($k \neq 0$) of temperature relax exponentially fast in time on the diffusion time scale:*

$$\hat{T}(t,k) \propto e^{-t/\tau_\mathrm{diff}}, \quad \tau_\mathrm{diff} = \frac{1}{D_T k^2} \sim \frac{l^2}{D_T}, \tag{5.55}$$

where $l \sim k^{-1}$ is the typical spatial scale of the variation and τ_diff is, therefore, its typical time scale.

Recall that in arguing for a local Maxwellian, we required the assumption that *these scales were much greater than the spatial and time scales of particle collisions*, λ_mfp and τ_c [see Eq. (5.8)]. Are they? Yes, but to show this (and to be able to solve practical

† For full reconstruction of the solution $T(t,z)$, see §5.7.3.

problems), we still have to derive explicit expressions for D_T and ν (the velocity diffusion governed by Eq. (5.49) is entirely analogous to the temperature diffusion we have just discussed).

In what follows, we will do this not once but four times, in four different ways (which highlight different aspects of the problem):

- a dimensional guess, a scoundrel's last (or, in our case, first) resort (§5.6.2),
- an estimate based on modelling collisions as particle diffusion, a physically important insight (§5.7),
- a "pseudo-kinetic" derivation, dodgy but nice and simple (§6.1),
- a "real" kinetic derivation, more involved, but also more systematic, mathematically appealing and showing how more complicated problems are solved (the rest of §6).

5.6.2. *Dimensional Estimate of Transport Coefficients*

As often happens, the quickest way to get the answer (or an answer) is a dimensional guess. The dimensionality of diffusion coefficients is

$$[D_T] = [\nu] = \frac{\text{length}^2}{\text{time}}. \tag{5.56}$$

Clearly, transport of energy and momentum from one part of the system to another is due to particles colliding. Therefore, both the energy- and momentum-diffusion coefficients must depend on some quantities characterising particle collisions. We need a length and a time: to wit, the mean free path $\lambda_{\rm mfp}$ and the collision time $\tau_{\rm c}$. Then [using Eq. (4.5)]

$$\boxed{D_T \sim \nu \sim \frac{\lambda_{\rm mfp}^2}{\tau_{\rm c}} \sim v_{\rm th}^2 \tau_{\rm c} \sim v_{\rm th}\lambda_{\rm mfp}}\,. \tag{5.57}$$

This is indeed true (properly proved in §6), although of course we cannot calculate numerical prefactors.

5.6.3. *Separation of Scales*

Armed with the estimate Eq. (5.57), we can now ascertain that there indeed is a separation of scales between collisional relaxation to local equilibrium and diffusive relaxation to the global one: the diffusion time, Eq. (5.55), becomes

$$\tau_{\rm diff} \sim \left(\frac{l}{\lambda_{\rm mfp}}\right)^2 \tau_{\rm c} \gg \tau_{\rm c} \quad \text{if} \quad l \gg \lambda_{\rm mfp}. \tag{5.58}$$

To be precise, this says that spatial-scale separation implies time-scale separation, i.e., if we set up some macroscopic ($l \gg \lambda_{\rm mfp}$) temperature gradients or shear flows in the system, they will relax slowly compared to the collision time, with the system evolving through a sequence of local Maxwellian equilibria [Eq. (5.10)] as T or $\boldsymbol{u}$ gradually become uniform.

5.6.4. *Sources, Sinks and Boundaries*

We have so far only discussed the situation where some *initial* non-equilibrium state relaxes freely towards global equilibrium. In the real world, there often are external circumstances that mathematically amount to *sources* or *sinks* in the transport equations and keep the system out of equilibrium even as it ever strives towards it. In such systems, the transport equations can have *steady-state* (time-independent; §5.6.5) or *stationary time-dependent* (periodic; §5.6.6) solutions.

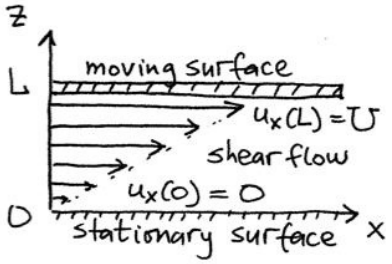


FIGURE 15. Linear shear flow, Eq. (5.62).

Thus, in the heat diffusion equation (5.44), there can be *heating* and *cooling* terms:

$$\frac{\partial}{\partial t}\frac{3}{2}nk_{\rm B}T = \varkappa\frac{\partial^2 T}{\partial z^2} + H - C. \tag{5.59}$$

Here the *heating rate* H represents some form of distributed heating, e.g., viscous [the last term in Eq. (5.37); see also §6.4.3] or Ohmic (if current can flow through the medium); the *cooling rate* C can, e.g., be due to radiative cooling (usually in very hot gases/plasmas).

Similarly, the momentum equation (5.45) can include external *forces:*

$$\frac{\partial}{\partial t}mnu_x = \eta\frac{\partial^2 u_x}{\partial z^2} + f_x, \tag{5.60}$$

where f_x is the "body force," or force density, in the x direction. Common examples of body forces (for all of which, however, one requires the more general, 3D version of the momentum equation; see Eq. (5.35)) are the pressure gradient, gravity, Coriolis and centrifugal forces in rotating systems, Lorentz force in conducting media, buoyancy force in stratified media, etc.

Sources or sinks of heat and momentum can also take the form of *boundary conditions*, e.g.,

—a surface kept at some fixed temperature,

—a given heat flux constantly pumped through a surface (perhaps via the latter being in contact with a heat source generating heat at a given rate),

—a rate of cooling at a surface specified in terms of its temperature (e.g., Newton's law of cooling: cooling rate proportional to the temperature difference between the surface of a body and the environment),

—a surface moving at a given velocity, etc.

5.6.5. *Steady-State Solutions*

Steady-state solutions arise when sources, sinks and/or boundary conditions are constant in time and so cause time-independent temperature or velocity profiles to emerge.

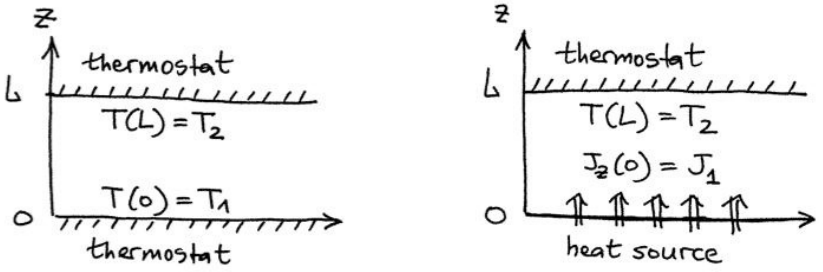
For example, the *force balance*

$$\eta\frac{\partial^2 u_x}{\partial z^2} + f_x = 0 \tag{5.61}$$

will imply some profile $u_x(z)$, given the spatial dependence of the force $f_x(z)$ and some boundary conditions on u_x (since the diffusion equation is second-order in z, two of those are needed). The simplest case is $f_x = 0$, $u_x(0) = 0$, $u_x(L) = U$, which instantly implies, for $z \in [0, L]$

$$u_x(z) = U\frac{z}{L}, \tag{5.62}$$

a solution known as *linear shear flow* (Fig. 15).



(a) (b)

FIGURE 16. Boundary conditions for the heat diffusion equation: (a) two thermostatted surfaces, (b) a thermostatted surface and a heat source.

Similarly, looking for steady-state solutions of Eq. (5.44) subject to both ends of the domain being kept at fixed temperatures, $T(0) = T_1$ and $T(L) = T_2$ (Fig. 16a), we find

$$\frac{\partial^2 T}{\partial z^2} = 0 \quad \Rightarrow \quad T(z) = T_1 + (T_2 - T_1)\frac{z}{L}. \tag{5.63}$$

Note that the simple linear profiles (5.62) and (5.63) are entirely independent of the transport coefficients $\varkappa$ and η.

A slightly more sophisticated example is a set up where, say, the bottom surface of the system is heated at some known fixed rate, i.e., the heat flux through the $z = 0$ boundary is specified, $J_z(0) = J_1$, while the top surface is in contact with a fixed-temperature thermostat, $T(L) = T_2$ (Fig. 16b). Then Eq. (5.44) or, indeed, already Eq. (5.18) gives, in steady state,

$$\frac{\partial J_z}{\partial z} = 0 \quad \Rightarrow \quad J_z = \text{const} = J_1. \tag{5.64}$$

Since $J_z = -\varkappa \partial T/\partial z$ [Eq. (5.39)],

$$T(z) = \frac{J_1}{\varkappa}(L - z) + T_2. \tag{5.65}$$

From this we learn what the temperature at the bottom boundary is, viz., $T(0) = J_1 L/\varkappa + T_2$, and, therefore, the overall temperature contrast that can be maintained by injection of a given power J_1, viz., $\Delta T = J_1 L/\varkappa$.

Exercise 5.4. Work out the steady-state temperature profile $T(r)$ that will be maintained at the radii $r \in [r_1, r_2]$ in an axisymmetric system where $T(r_1) = T_1$ and $T(r_2) = T_2$.

Note that steady-state profiles of the kind described above, even though they are solutions of the transport equations, are not necessarily *stable* solutions. Time-dependent motions can develop as a result of small perturbations of the steady state (e.g., for convection, given large enough temperature contrasts, the so-called Rayleigh-Bénard problem; see, e.g., Chandrasekhar 2003). Indeed, it is very common for Nature to find such ways of relaxing gradients via instabilities and resulting motions (*turbulence*) when the gradients (deviations from global equilibrium) are strong and collisional/diffusive transport is relatively slow (Nature tends to be impatient with non-equilibrium set-ups).

5.6.6. *Time-Periodic Solutions*

If we now consider a situation in which the boundary condition is time-dependent in some periodic way, e.g., the surface of the system is subject to some seasonal temperature changes, then the solution that will emerge will be time-periodic. Physically, this describes

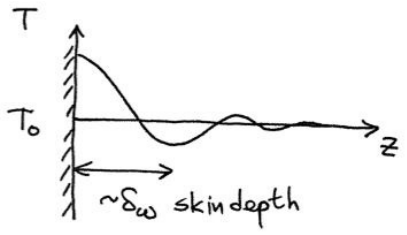


FIGURE 17. Temperature perturbation with given frequency ω penetrates $\sim$ a skin depth δ_ω into heat-conducting medium, Eq. (5.70).

a system whose state is a result of the external conditions constantly driving it out of equilibrium and the heat diffusion constantly pushing it back to equilibrium.

The treatment of such cases is analogous to what we did with the relaxation of an initial inhomogeneity in §5.6.1, but now the Fourier transform is in time rather than in space. So, consider a semi-infinite domain, $z \in [0, \infty)$, with the boundary condition

$$T(t, z = 0) = \sum_\omega \hat{T}_0(\omega) e^{-i\omega t} \tag{5.66}$$

(say a building with the outer wall at $z = 0$ exposed to the elements, with ω's being the frequencies of daily, annual, millennial etc. temperature variations). Then the solution of Eq. (5.48) can be sought in the form

$$T(t, z) = \sum_\omega \hat{T}(\omega, z) e^{-i\omega t}, \tag{5.67}$$

where $\hat{T}(\omega, z)$ must satisfy

$$-i\omega\hat{T} = D_T \frac{\partial^2 \hat{T}}{\partial z^2}, \quad \hat{T}(\omega, z = 0) = \hat{T}_0(\omega), \quad \hat{T}(\omega, z = \infty) = 0. \tag{5.68}$$

The last condition assumes that temperature variations decay at infinity, far away from the source of the disturbance. The solution is

$$\hat{T}(\omega, z) = C_1 e^{ikz} + C_2 e^{-ikz}, \quad k = (1 + i)\sqrt{\frac{|\omega|}{2D_T}}, \tag{5.69}$$

where $C_1 = \hat{T}_0(\omega)$ and $C_2 = 0$ to satisfy the boundary conditions at $z = 0$ and $z = \infty$. Finally, from Eq. (5.67),

$$\boxed{T(t, z) = \sum_\omega \hat{T}_0(\omega) \exp\left[-i\left(\omega t - \frac{z}{\delta_\omega}\right) - \frac{z}{\delta_\omega}\right], \quad \delta_\omega = \sqrt{\frac{2D_T}{|\omega|}}} \,, \tag{5.70}$$

where δ_ω is the typical scale on which temperature perturbations with frequency ω decay, known as the *skin depth*—the further away from the boundary (and the higher the frequency), the more feeble is the temperature variation that manages to penetrate there (Fig. 17). Note that it also arrives to $z > 0$ with a time delay $\Delta t = z/\delta_\omega|\omega|$.

This was an example of "relaxation to equilibrium" effectively occurring in space rather than in time.

You see that once we have a diffusion equation for heat or momentum, solving it—and, therefore, working out how systems return (or strive) to global equilibrium—becomes a

problem in applied mathematics rather than in physics (although interpreting the answer still requires some physical insight). Returning to physics, the key piece of unfinished business that remains is to calculate the diffusion coefficients D_T and ν (or $\varkappa$ and η) based on some theory of particle motion and collisions in an ideal gas (and we will restrict these calculations to ideal gas only).

5.7. *Diffusion*

Before we make good on our promise of a proper kinetic calculation, it is useful to discuss what fundamental property of moving particles in a collisional gas that the diffusion equations encode.

Let us forget about transport equations for a moment, consider an ideal-gas system and imagine that there is a sub-population of particles in this gas, with number density n^*, that carry some identifiable property: e.g., they might be labelled in some way (e.g., be particles of a different species than the rest). Non-rigorously, we will argue this property they carry might also be mean energy or momentum and so the evolution equation for n^* that we are about to derive should have the same form as the evolution equations for mean momentum or energy density (temperature) of the gas.

5.7.1. *Derivation of the Diffusion Equation*

Suppose that at time t, the mean number density of the labelled particles at the location z is $n^*(t,z)$ (we will work in 1D, assuming that only in the z direction is there a macroscopic variation of n^*). What will it be at the same location after a (short) time Δt? During that time, some particles will move from z to other places and other particles will arrive to z from elsewhere. Therefore,

$$n^*(t+\Delta t,z) = \langle n^*(t,z-\Delta z)\rangle\,, \tag{5.71}$$

where $z-\Delta z$ are positions where the particles that arrive to z at $t+\Delta t$ were at time t and the average is over all these individual particle displacements (Δz is a random variable). *This is the essence of "transport": the particle density, or, more generally, some quantity carried by particles, is brought ("transported") to a given location at a given time by particles arriving at that location at that time from elsewhere, so the mean density at z and $t+\Delta t$ is determined by what the density was at t in all those earlier particle locations.*

Let us take Δt to be small enough so the corresponding particle displacements are much smaller than the scale of spatial variation of $n^*(z)$, viz.,

$$\Delta z \ll \left(\frac{1}{n^*}\frac{\partial n^*}{\partial z}\right)^{-1}. \tag{5.72}$$

Then we can Taylor-expand Eq. (5.71) in small Δz:

$$n^*(t+\Delta t,z) = \left\langle n^*(t,z) - \Delta z\,\frac{\partial n^*}{\partial z} + \frac{\Delta z^2}{2}\frac{\partial^2 n^*}{\partial z^2} + \ldots\right\rangle \approx n^*(t,z) + \frac{\langle\Delta z^2\rangle}{2}\frac{\partial^2 n^*}{\partial z^2}, \tag{5.73}$$

where we have assumed $\langle\Delta z\rangle = 0$ (no mean motion in the z direction; **Exercise**: work out what happens when $\langle\Delta z\rangle = u_z \neq 0$). Rearranging this equation, we get

$$\frac{n^*(t+\Delta t,z) - n^*(t,z)}{\Delta t} = \frac{\langle\Delta z^2\rangle}{2\Delta t}\frac{\partial^2 n^*}{\partial z^2}. \tag{5.74}$$

In the limit $\Delta t \to 0$, we find that n^* satisfies a diffusion equation, similar to Eqs. (5.48) and (5.49):

$$\boxed{\frac{\partial n^*}{\partial t} = D\frac{\partial^2 n^*}{\partial z^2}, \quad D = \lim_{\Delta t\to 0}\frac{\langle\Delta z^2\rangle}{2\Delta t}}\,, \tag{5.75}$$

assuming that the limit exists and so D is finite.

5.7.2. *Random-Walk Model*

So, is D finite? It is important to understand that "$\Delta t \to 0$" here means that Δt is small compared to the time scales on which the diffusive evolution of n^* occurs ($\Delta t \ll \tau_{\rm diff}$ in the

notation of §5.6.3), but it can still be a long time compared to the collision time, $\Delta t \gg \tau_c$. Let us model particle motion as a succession of free flights, each lasting for a time τ_c and followed by a random kick—a collision with another particle—as a result of which the direction of the particle's motion in the z direction may be reversed. Mathematically, we may write this model for particle displacements as follows

$$\Delta z = \sum_{i=1}^{N} \delta z_i, \tag{5.76}$$

where δz_i are independent random displacements with mean $\langle \delta z_i \rangle = 0$ and variance $\langle \delta z_i^2 \rangle = \lambda_{\rm mfp}^2$, and $N = \Delta t/\tau_c$ is the number of collisions over time Δt. By the Central Limit Theorem (see, e.g., Sinai 1992), in the limit $N \to \infty$, the quantity

$$X = \sqrt{N} \left(\frac{1}{N} \sum_{i=1}^{N} \delta z_i - \langle \delta z_i \rangle \right) \tag{5.77}$$

will have a normal (Gaussian) distribution with zero mean and variance $\langle \delta z_i^2 \rangle - \langle \delta z_i \rangle^2 = \lambda_{\rm mfp}^2$:

$$f(X) = \frac{1}{\lambda_{\rm mfp}\sqrt{2\pi}}\, e^{-X^2/2\lambda_{\rm mfp}^2}. \tag{5.78}$$

Since $\Delta z = X\sqrt{N}$, we conclude that, for $N = \Delta t/\tau_c \gg 1$,

$$D = \frac{\langle \Delta z^2 \rangle}{2\Delta t} = \frac{\langle X^2 \rangle}{2\tau_c} = \frac{\lambda_{\rm mfp}^2}{2\tau_c}, \tag{5.79}$$

so we recover the dimensional guess (5.57), up to a numerical factor, of course.

The model of the particle motion that we have used to obtain this result—a sequence of independent random increments—is known as *Brownian motion*, or *random walk*, and describes random meandering of a particle being bombarded by other particles of the gas and thus undergoing a sequence of random kicks. The density of such particles—or of any quantity they carry, such as energy or momentum—always satisfies a diffusion equation, as follows from the above derivation.

In the context of a diffusive spreading of an admixture of particles of a distinct species in an ambient gas, Eq. (5.75) is called *Fick's law*. In the expression for the diffusion coefficient, Eq. (5.79), $\lambda_{\rm mfp}$ and τ_c are the mean free path and the collision time of the labelled species (which, if this species has different mass than the ambient one, are not the same as the ambient mean free path and collision time; see Exercise 4.3).

If we were to use the model above to understand transport of energy or momentum, while this is fine qualitatively, we ought to be cognizant of an important nuance. Implicitly, if we treat n^* as energy density ($3nk_BT/2$) or momentum density (mnu_x) and carry out exactly the same calculation, we are assuming that particles that have random-walked through many collisions from $z - \Delta z$ to z have not, through all these collisions, changed their energy or momentum. This is, of course, incorrect—in fact, in each collision, energy and momentum are exchanged and so the velocity of each particle receives a random kick uncorrelated with the particle's previous history. Thus, the particle random-walks not just in position space z but also in velocity space $\boldsymbol{v}$. The reason the above calculation is still fine is that we can think of the particles it describes not literally as particles but as units of energy or momentum random-walking from place to place—and also from particle to particle!—and thus effectively diffusing from regions with higher average u_x or T to regions with lower such averages.

5.7.3. *Diffusive Spreading*

Thus, we have argued that the diffusion equation—Eq. (5.75), (5.48) or (5.49)—is a macroscopic manifestation of particles (or energy, or momentum) random walking and, roughly speaking, covering distances that scale as $\sqrt{t}$ with the time it takes to cover them. This would suggest that an initial spot of concentration of a particle admixture, energy or momentum would spread in space as $\sqrt{t}$. This is indeed the case, as can be inferred from the solution to the initial-value problem we worked out in §5.6.1.

Using Eqs. (5.53) and (5.54), we can reconstruct the full solution of Eq. (5.48), given the

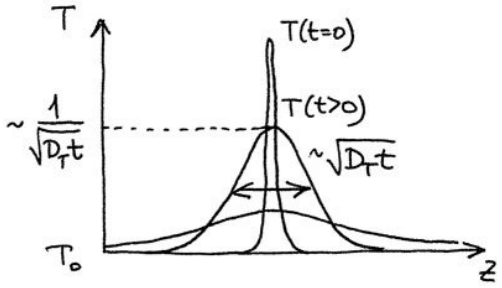


FIGURE 18. Diffusive spreading of an initial hot spike, Eq. (5.83).

initial perturbation $T(t=0,z)$:

$$T(t,z) = \sum_k \hat{T}_0(k)\, e^{ikz - D_T k^2 t}. \tag{5.80}$$

Here $\hat{T}_0(k)$ can be expressed as the inverse Fourier transform of $T(t=0,z)$, from Eq. (5.52):

$$\hat{T}_0(k) = \frac{1}{L}\int \mathrm{d}z'\, T(t=0,z')\, e^{-ikz'}, \tag{5.81}$$

where L is the length of the domain in z. Substituting this into Eq. (5.80) and replacing the sum over k with an integral (which we can do if we notice that, in a periodic domain of size L, the "mesh" size in k is $2\pi/L$),

$$\sum_k = \frac{L}{2\pi}\int \mathrm{d}k, \tag{5.82}$$

we get

$$\begin{aligned} T(t,z) &= \frac{1}{2\pi}\int \mathrm{d}z'\, T(t=0,z') \int \mathrm{d}k\, e^{ik(z-z') - D_T k^2 t} \\ &= \int \mathrm{d}z'\, T(t=0,z')\, \frac{1}{\sqrt{4\pi D_T t}} \exp\left[-\frac{(z-z')^2}{4D_T t}\right], \end{aligned} \tag{5.83}$$

where we have done the k integral by completing the square in the exponential. This formula (which is an example of a *Green's-function solution* of a partial differential equation) describes precisely what we anticipated: a diffusive (random-walk-like) spreading of the initial perturbation with $z - z' \sim \sqrt{D_T t}$. The easiest way to see this is to imagine that the initial perturbation is a sharp spike at the origin, $T(t=0,z) = \delta(z)$. After time t, this spike turns into a Gaussian-shaped profile with rms width $= \sqrt{2D_T t}$ (Fig. 18).

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6. Kinetic Calculation of Transport Coefficients

6.1. *A Nice but Dodgy Derivation*

[Literature: Blundell & Blundell (2009), §9]

6.1.1. *Viscosity*

Given a shear flow profile, $u_x(z)$, we wish to calculate the momentum flux Π_{zx} through the plane defined by a fixed value of the coordinate z (Fig. 19). The number of particles that cross that plane per unit time per unit area is given by Eq. (3.2):

$$\mathrm{d}\Phi(\boldsymbol{v}) = n v_z f(\boldsymbol{v}) \mathrm{d}^3\boldsymbol{v} = n v^3 f(v) \mathrm{d}v \cos\theta \sin\theta \mathrm{d}\theta \mathrm{d}\phi. \tag{6.1}$$

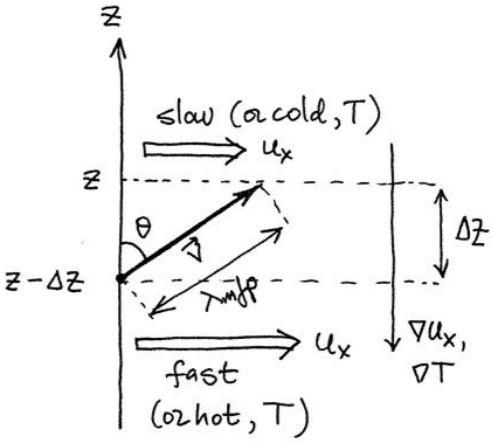


FIGURE 19. Physics of transport: particles wander from faster-moving (or hotter) regions to slower (or colder) ones, bring with then extra momentum (or energy). This gives rise to net momentum (or heat) flux and so to the viscosity (thermal conductivity) of the gas.

These particles have travelled the distance $\lambda_{\rm mfp}$ since their last collision—i.e., since they last "communicated" with the gas as a collective. This was at the position $z-\Delta z$, where $\Delta z = \lambda_{\rm mfp}\cos\theta$ (because they are flying at angle θ to the z axis). But, since u_x is a function of z, the mean momentum of the particles at $z-\Delta z$ is different than it is at z and so a particle that last collided at $z-\Delta z$ brings with it to z some extra momentum:

$$\Delta p = mu_x(z-\Delta z) - mu_x(z) \approx -m\,\frac{\partial u_x}{\partial z}\Delta z = -m\,\frac{\partial u_x}{\partial z}\lambda_{\rm mfp}\cos\theta, \tag{6.2}$$

assuming that $\Delta z \ll l$ (l is the scale of variation of u_x). The flux of momentum through z is then simply

$$\Pi_{zx} = \int \mathrm{d}\Phi(\boldsymbol{v})\Delta p = -mn\,\frac{\partial u_x}{\partial z}\,\lambda_{\rm mfp}\underbrace{\int_0^\infty \mathrm{d}v\, v^3 f(v)}_{=\langle v\rangle/4\pi}\underbrace{\int_0^\pi \mathrm{d}\theta\cos^2\theta\sin\theta}_{=2/3}\underbrace{\int_0^{2\pi}\mathrm{d}\phi}_{=2\pi}$$

$$= -\frac{1}{3}\,mn\lambda_{\rm mfp}\langle v\rangle\frac{\partial u_x}{\partial z}. \tag{6.3}$$

Note that, unlike in our effusion (§3) or pressure (§1.4) calculations, the integral is over all θ because particles come from $z-\Delta z$, where $\Delta z = \lambda_{\rm mfp}\cos\theta$ can be positive or negative.

Comparing Eq. (6.3) with Eq. (5.40), we read off the expression for dynamical viscosity:

$$\boxed{\eta = \frac{1}{3}\,mn\lambda_{\rm mfp}\langle v\rangle = \frac{2}{3\sqrt{\pi}}\,mn\lambda_{\rm mfp}v_{\rm th} = \frac{2}{3\sigma}\sqrt{\frac{2mk_{\rm B}T}{\pi}}}\,. \tag{6.4}$$

We have recovered the dimensional guess (5.57), with a particular numerical coefficient (which is, however, wrong, as we are about to explain). Note that the assumption $\Delta z \sim \lambda_{\rm mfp} \ll l$ is justified *a posteriori:* once we have Eq. (6.4), we can confirm scale separation as in §5.6.3.

The last expression in Eq. (6.4), to obtain which we used Eqs. (4.5) and (2.21), emphasises the fact that the dynamical viscosity depends on the temperature but not the number density of the gas.

Exercise 6.1. What is going on physically? Why does it make sense that the rate of momentum transport should be independent of the density of particles that transport it? Robert Boyle discovered this in 1660 when he put a pendulum inside a vessel from which he proceeded to pump out the air. The rate at which the pendulum motion was damped did not change.

If Boyle had had a really good vacuum pump and continued pumping the air out, at what pressure would have he started detecting a change in the pendulum's damping rate? Below that pressure, estimate the momentum flux from the pendulum, given the pendulum's typical velocity u and any other parameters that you might reasonably expect to know.

6.1.2. *Thermal Conductivity*

In order to obtain the heat flux J_z, given some temperature profile $T(z)$, we go through a completely analogous calculation: particles that arrive at z after having last experienced a collision at $z-\Delta z$ bring to z some extra energy:

$$\Delta E = c_1 T(z-\Delta z) - c_1 T(z) \approx -c_1 \frac{\partial T}{\partial z}\,\lambda_{\rm mfp}\cos\theta, \tag{6.5}$$

where $c_1 = 3k_{\rm B}/2$ is the heat capacity per particle. Therefore, the flux of energy (heat flux) is

$$J_z = \int \mathrm{d}\Phi(\boldsymbol{v})\Delta E = -\frac{1}{3}\,nc_1\lambda_{\rm mfp}\langle v\rangle\frac{\partial T}{\partial z}, \tag{6.6}$$

whence, upon comparison with Eq. (5.39), we infer the heat capacity:

$$\boxed{\varkappa = \frac{1}{3}\,nc_1\lambda_{\rm mfp}\langle v\rangle = \frac{2}{3\sqrt{\pi}}\,nc_1\lambda_{\rm mfp}v_{\rm th} = \frac{2c_1}{3\sigma}\sqrt{\frac{2k_{\rm B}T}{\pi m}}}\,. \tag{6.7}$$

This again is consistent with the dimensional expression (5.57).

Exercise 6.2. Fick's Law of Diffusion. Given the number density $n^*(z)$ and the mean free path $\lambda_{\rm mfp}$ of an admixture of labelled particles, as well as the temperature of the ambient gas, calculate the flux of the labelled species, Φ_z^*, and derive Fick's Law of Diffusion, Eq. (5.75).

6.1.3. *Why This Derivation is Dodgy*

Our new expressions (6.4) and (6.7) for the transport coefficients and their derivation *look* more quantitative and systematic than what we had before, but in fact they are not. It is useful to understand why that is, in order to appreciate the need for and the structure of the better derivation that is to follow.

Much of the appearance of rigour in the derivation of Eqs. (6.4) and (6.7) came from taking into account the fact that particles ending up at location z at time t might have travelled at an angle to the z axis. Integrating over the resulting combination of sines and cosines produced numerical factors that had a veneer of quantitative precision. However, while precisely integrating over the particles' angle distribution, we blithely assumed that they all had travelled exactly the same distance $\lambda_{\rm mfp}$ between collisions and carried exactly the same excess momentum and energy—but surely all of these things must in fact depend on the particles' velocities, which are random variables and so have to be properly averaged over? To illustrate the imprecise nature of these assumptions, imagine that instead of what we did, we assumed, equally plausibly (as far as we can tell at this stage), that all particles travelled the same time $\tau_{\rm c}$ between collisions. Then we would have to replace $\lambda_{\rm mfp} \to v\tau_{\rm c}$ in our calculations, leading to

$$\frac{\eta}{mn} = \frac{\varkappa}{nc_1} = \frac{1}{3}\langle v^2\rangle\tau_{\rm c} = \frac{1}{2}\,v_{\rm th}^2\tau_{\rm c} = \frac{1}{2}\,\lambda_{\rm mfp}v_{\rm th}. \tag{6.8}$$

This has the same dependence on $\lambda_{\rm mfp}$ and $v_{\rm th}$ as our previous attempts, but a different numerical coefficient (which is as wrong—or as qualitatively irrelevant—as all the other ones we have calculated so far).

Thus, the derivation given in this section is in fact no more rigorous then the random-walk model of §5.7.2 or even the dimensional estimate of §5.6.2—although it does highlight the essential fact that we need some sort of kinetic (meaning based on the velocity distribution) calculation of the fluxes.

Another, somewhat more formalistic, objection to our last derivation is that the homogeneous Maxwellian $f(v)$ was used, despite the fact that we had previously made quite a lot of fuss about only having a local Maxwellian $F(t,\boldsymbol{r},\boldsymbol{v})$ [Eq. (5.10)] depending on z via $T(z)$ and $u_x(z)$. In fact, this might have been OK because the scale of inhomogeneities was long ($l \gg \lambda_{\rm mfp}$) and the flow velocity small ($u_x \ll v_{\rm th}$), but we certainly did not set up systematically whatever expansion around a homogeneous distribution that might have justified this approach.

You will find some further critique of the derivation above, as well as the quantitatively correct formulae for the transport coefficients, in Blundell & Blundell (2009), §9.4. The derivation of these formulae can be found, e.g., in Chapman & Cowling (1991).

Clearly, if we are to claim that we really can do better than our unapologetically qualitative arguments in §5, we must provide a more systematic algorithm for calculating transport coefficients. We shall do this now and in the process learn how to solve (kinetic) problems involving scale separation—a useful piece from the theoretical toolbox of physics.

6.2. *Kinetic Expressions for Fluxes*

Let us go back to basics. Suppose we know the particle distribution $F(z,\boldsymbol{v})$ (we continue to stick to the 1D case). What are the fluxes of momentum and energy? They are

$$\Pi_{zx}(z) = \int \mathrm{d}^3\boldsymbol{v}\, m v_x \cdot v_z \cdot F(z,\boldsymbol{v}), \tag{6.9}$$

$$J_z(z) = \int \mathrm{d}^3\boldsymbol{v}\, \frac{mv^2}{2} \cdot v_z \cdot F(z,\boldsymbol{v}) \tag{6.10}$$

(in the latter expression, we assumed $\boldsymbol{u}=0$ for simplicity, an assumption that will be lifted in §6.4.3). But if F is a local Maxwellian,

$$F_{\rm M}(z,\boldsymbol{v}) = \frac{n}{\left[\sqrt{\pi}v_{\rm th}(z)\right]^3}\,\exp\left\{-\frac{[v_x-u_x(z)]^2+v_y^2+v_z^2}{v_{\rm th}^2(z)}\right\}, \quad v_{\rm th}(z)=\sqrt{\frac{2k_{\rm B}T(z)}{m}}, \tag{6.11}$$

then $\Pi_{zx}=0$ and $J_z=0$ because they both have a single power of v_z under the integral and $F_{\rm M}$ is even in v_z! This means that non-zero fluxes come from the distribution function in fact *not* being *exactly* a local Maxwellian:

$$F(z,\boldsymbol{v}) = F_{\rm M}(z,\boldsymbol{v}) + \delta F(z,\boldsymbol{v}), \tag{6.12}$$

and we must now find δF.

In order to do this, we need an *evolution equation* for F, the argument for a local Maxwellian (§5.2) is no longer enough.

6.3. *Kinetic Equation*

The simplest derivation of the kinetic equation goes as follows. The particles found at location $\boldsymbol{r}$ with velocity $\boldsymbol{v}$ at time $t+\Delta t$ are the particles moving at velocity $\boldsymbol{v}$ that arrived

to this location from $\boldsymbol{r}-\boldsymbol{v}\Delta t$, where they were at time t, plus those that got scattered into this $\boldsymbol{v}$ by collisions that they experienced during the time interval $[t,t+\Delta t]$, at the location $\boldsymbol{r}$:

$$F(t+\Delta t,\boldsymbol{r},\boldsymbol{v})=F(t,\boldsymbol{r}-\boldsymbol{v}\Delta t,\boldsymbol{v})+\Delta F_{\rm c}\approx F(t,\boldsymbol{r},\boldsymbol{v})-\boldsymbol{v}\cdot\boldsymbol{\nabla}F(t,\boldsymbol{r},\boldsymbol{v})\Delta t+\Delta F_{\rm c}, \quad (6.13)$$

where we have expanded in $\boldsymbol{v}\Delta t$ assuming small enough Δt, viz., $v\Delta t\ll|\boldsymbol{\nabla}\ln F|^{-1}$. Dividing through by Δt and taking the limit $\Delta t\to 0$, we get the *kinetic equation:*

$$\boxed{\frac{\partial F}{\partial t}+\boldsymbol{v}\cdot\boldsymbol{\nabla}F=C[F]}\,, \quad (6.14)$$

where the right-hand side, $C[F]=\lim_{\Delta t\to 0}\Delta F_{\rm c}/\Delta t$, is called the *collision operator*, whereas the left-hand side expresses conservation of particle density in phase space: indeed, our equation can be written as $\partial F/\partial t=-\boldsymbol{\nabla}\cdot(\boldsymbol{v}F)+C[F]$, where $\boldsymbol{v}F$ is the flux of particles with velocity $\boldsymbol{v}$.

Exercise 6.3. Kinetic Equation for a Plasma. We have assumed that no forces act on particles, apart from collisions. Work out the form of the kinetic equation if some external force $m\boldsymbol{a}$ acts on each particle, e.g., gravity $\boldsymbol{a}=\boldsymbol{g}$, or Lorentz force $\boldsymbol{a}=(q/m)(\boldsymbol{E}+\boldsymbol{v}\times\boldsymbol{B}/c)$ (q is the particle charge). The kinetic equation for the latter case is the Vlasov–Landau equation describing an ionised particle species in a plasma (see. e.g., Schekochihin 2015, and references therein).

Eq. (6.14) might appear rather less than satisfactory as we have not specified what $C[F]$ is. Thinking about what it might be is depressing as it is clearly quite a complicated object:
—collisions leading to a change in the local number of particles with velocity $\boldsymbol{v}$ must have involved particles that had other velocities $\boldsymbol{v}'$ before they collided, so $C[F]$ is likely to be a integral operator depending on $F(t,\boldsymbol{r},\boldsymbol{v}')$ integrated over a range of $\boldsymbol{v}'$;
—assuming predominantly binary collisions, $C[F]$ is also likely to be a quadratic operator in F because the probability of getting a particle with velocity $\boldsymbol{v}$ after a collision must depend on the joint probability of two particles with some suitable velocities meeting.

In §6.5, we will happily avoid these complications by introducing a very simple model of $C[F]$,† but first let us show what can be done without knowing the explicit form of $C[F]$ (in the process, we will also learn some important properties of the collision operator).

6.4. *Local Conservation Equations Rederived*

The kinetic equation (6.14) in principle contains full information about the evolution of the system, so we ought to be able to recover from it the conservation equations Eqs. (5.18) and (5.27), which we originally derived on general grounds.

Let us return to our 1D case where the local equilibrium quantities only depend on one coordinate z, the density is constant in space and the mean velocity is along the x direction:

$$F=F(t,z,\boldsymbol{v}), \quad (6.15)$$

$$n=\int \mathrm{d}^3\boldsymbol{v}\,F=\text{const}, \quad (6.16)$$

$$\boldsymbol{u}=\frac{1}{n}\int \mathrm{d}^3\boldsymbol{v}\,\boldsymbol{v}F=u_x(t,z)\hat{\boldsymbol{x}}. \quad (6.17)$$

† See, e.g., Dellar (2015) for the derivation of the full Boltzmann collision operator.

The kinetic equation (6.14) is, in 1D,

$$\frac{\partial F}{\partial t} + v_z \frac{\partial F}{\partial z} = C[F]. \tag{6.18}$$

We will now find the evolution equations for n, u_x and T by taking moments of Eq. (6.18).

6.4.1. *Number Density*

The zeroth moment:

$$\frac{\partial n}{\partial t} = \int \mathrm{d}^3\boldsymbol{v}\, \frac{\partial F}{\partial t} = \int \mathrm{d}^3\boldsymbol{v} \left(-v_z \frac{\partial F}{\partial z} + C[F]\right)$$
$$= -\frac{\partial}{\partial z} \underbrace{\int \mathrm{d}^3\boldsymbol{v}\, v_z F}_{\substack{=0 \\ \text{because} \\ u_z = 0}} + \underbrace{\int \mathrm{d}^3\boldsymbol{v}\, C[F]}_{=0} = 0. \tag{6.19}$$

The second term vanishes because, whatever the explicit form of the collision operator is, it cannot lead to any change in the number of particles—*elastic collisions conserve particle number:*†

$$\int \mathrm{d}^3\boldsymbol{v}\, C[F] = 0. \tag{6.20}$$

Eq. (6.19) implies that density does not change and if we start with uniform $n = \text{const}$, it will stay that way—it is good to know that our assumption to that effect was consistent with our equations.

6.4.2. *Momentum Density*

The first moment:

$$\frac{\partial}{\partial t} mnu_x = \int \mathrm{d}^3\boldsymbol{v}\, mv_x \frac{\partial F}{\partial t} = \int \mathrm{d}^3\boldsymbol{v}\, mv_x \left(-v_z \frac{\partial F}{\partial z} + C[F]\right)$$
$$= -\frac{\partial}{\partial z} \underbrace{\int \mathrm{d}^3\boldsymbol{v}\, mv_z v_x F}_{\substack{=\Pi_{zx}, \\ \text{as defined} \\ \text{in Eq. (6.9)}}} + \underbrace{\int \mathrm{d}^3\boldsymbol{v}\, mv_x C[F]}_{=0} = -\frac{\partial \Pi_{zx}}{\partial z}. \tag{6.21}$$

Similarly to Eq. (6.19), the second term vanishes because, again, whatever the explicit form of the collision operator might be, it cannot lead to any change in the mean momentum of particles—*elastic collisions conserve momentum:*

$$\int \mathrm{d}^3\boldsymbol{v}\, m\boldsymbol{v}\, C[F] = 0. \tag{6.22}$$

Thus, we have recovered the momentum-conservation equation (5.27). Note that since $\Pi_{zx} = 0$ for $F = F_\mathrm{M}$ (§6.2),

$$\boxed{\Pi_{zx} = m \int \mathrm{d}^3\boldsymbol{v}\, v_z v_x \delta F = m \int \mathrm{d}^3\boldsymbol{w}\, w_z w_x \delta F}\,. \tag{6.23}$$

† In Eq. (7.2), ΔF_c represents collisions between particles at the point $\boldsymbol{r}$ in space. The only effect of these collisions is a redistribution of particle velocities—any movements of particles between different points in space are accounted for in the $\boldsymbol{v} \cdot \boldsymbol{\nabla} F$ term. Therefore, ΔF_c cannot change the total number of particles at $\boldsymbol{r}$ and so $\int \mathrm{d}^3\boldsymbol{v}\, \Delta F_\mathrm{c} = 0$. Similar considerations apply to the conservation of momentum, Eq. (6.22), and energy, Eq. (6.26).

The second expression, in terms of peculiar velocities $w_x = v_x - u_x$ and $w_z = v_z$, will be useful shortly. It holds because $\int \mathrm{d}^3\boldsymbol{w}\, w_z \delta F = 0$ by definition and because, according to Eq. (6.32), the deviation from the local Maxwellian cannot contain any mean flow.

6.4.3. *Energy Density*

The second moment, corresponding to energy, contains both the bulk and internal motion because we are keeping $\boldsymbol{u} = u_x\hat{\boldsymbol{x}}$ in this calculation: as in Eq. (5.6),

$$\langle E\rangle = \int \mathrm{d}^3\boldsymbol{v}\,\frac{mv^2}{2}\,F = \frac{mnu_x^2}{2} + \int \mathrm{d}^3\boldsymbol{w}\,\frac{mw^2}{2}\,F = \frac{mnu_x^2}{2} + \frac{3}{2}\,nk_\mathrm{B}T. \tag{6.24}$$

The first term is the kinetic energy of the mean motion and the second is the internal energy, related to temperature in the usual way, given by Eq. (5.11) [recall that according to Eq. (6.32), the deviation of F from the local Maxwellian F_M is not allowed to contribute to energy, so we can use the mean-energy formula derived for a local Maxwellian].

Therefore,

$$\begin{aligned}\frac{\partial}{\partial t}\frac{3}{2}\,nk_\mathrm{B}T &= \int \mathrm{d}^3\boldsymbol{v}\,\frac{mv^2}{2}\frac{\partial F}{\partial t} - \frac{\partial}{\partial t}\frac{mnu_x^2}{2}\\ &= \int \mathrm{d}^3\boldsymbol{v}\,\frac{mv^2}{2}\left(-v_z\,\frac{\partial F}{\partial z} + C[F]\right) - u_x\,\underbrace{\frac{\partial}{\partial t}\,mnu_x}_{\substack{\text{use}\\ \text{Eq. (6.21)}}}\\ &= -\frac{\partial}{\partial z}\int \mathrm{d}^3\boldsymbol{v}\,\frac{mv^2}{2}\,v_z F + \underbrace{\int \mathrm{d}^3\boldsymbol{v}\,\frac{mv^2}{2}\,C[F]}_{=0} + u_x\frac{\partial \Pi_{zx}}{\partial z}.\end{aligned} \tag{6.25}$$

Similarly to Eqs. (6.19) and (6.21), the second term vanishes because the collision operator cannot lead to any change in the mean energy of particles—*elastic collisions conserve energy:*

$$\int \mathrm{d}^3\boldsymbol{v}\,\frac{mv^2}{2}\,C[F] = 0. \tag{6.26}$$

The first term in Eq. (6.25) looks very much like the divergence of the heat flux, Eq. (6.10), but we must be careful as heat is only the random part of the motions, while $\boldsymbol{v}$ now also contains the mean flow $\boldsymbol{u} = u_x\hat{\boldsymbol{x}}$. Breaking up $\boldsymbol{v} = u_x\hat{\boldsymbol{x}} + \boldsymbol{w}$ as before, where $\int \mathrm{d}^3\boldsymbol{v}\,\boldsymbol{w}F = 0$, we get (noting that $v_z = w_z$)

$$\int \mathrm{d}^3\boldsymbol{v}\,\frac{mv^2}{2}\,v_z F = \int \mathrm{d}^3\boldsymbol{w}\left(\frac{mu_x^2}{2} + mu_x w_x + \frac{mw^2}{2}\right) w_z F = u_x\Pi_{zx} + J_z. \tag{6.27}$$

In the second expression above, the first term vanished because $\int \mathrm{d}^3\boldsymbol{w}\, w_z F = 0$, the second term gave $u_x\Pi_{zx}$ by virtue of Eq. (6.23), and the last term is the heat flux:

$$\boxed{J_z = \int \mathrm{d}^3\boldsymbol{w}\,\frac{mw^2}{2}\,w_z\delta F}\,, \tag{6.28}$$

where only δF is left because $J_z = 0$ for $F = F_\mathrm{M}$ (see §6.2). Assembling Eqs. (6.27) and (6.25), we get

$$\frac{\partial}{\partial t}\frac{3}{2}\,nk_\mathrm{B}T = -\frac{\partial J_z}{\partial z} - \Pi_{zx}\frac{\partial u_x}{\partial z}. \tag{6.29}$$

Our old energy conservation equation (5.18) is recovered if we set $u_x = 0$ (which is the

assumption under which we derived it in §5.3.1). But if we do retain the flow, we see that a new term has appeared, which looks like a heating term—it will indeed turn out to represent viscous heating when we calculate Π_{zx} (§6.7.1).

So, we have recovered the local conservation laws, Eqs. (6.21) and (6.29), which contain the fluxes (6.23) and (6.28), in order to calculate which, we need to know δF. For that, we must solve the kinetic equation (6.18) and in order to solve it, we must have an explicit expression for $C[F]$.

6.5. *Collision Operator*

For ideal gas, the explicit expression for collision operator (integral, quadratic in F, as per the discussion at the end of §6.3) was derived by Boltzmann (see, e.g., Boltzmann 1995; Chapman & Cowling 1991; Dellar 2015). I will not present this derivation here, but instead use the basic criteria that must be satisfied by the collision operator to come up with a very simple model for it (not quantitatively correct, but good enough for our purposes).

• Firstly, the effect of collisions must be *to drive the particle distribution towards local thermodynamic equilibrium*, i.e., the local Maxwellian (5.10). Once this distribution is achieved, since the fast-time-scale effect of collisions is local to any given fluid element, the collisions should not change the local Maxwellian:

$$C[F_{\mathrm{M}}] = 0. \tag{6.30}$$

If one derives the collision operator based on an explicit microphysical model of particle collisions, one can then *prove* that $C[F] = 0$ *implies* $F = F_{\mathrm{M}}$ and also that collisions always drive the distribution towards F_{M}. This property is associated with the so-called *Boltzmann's H-Theorem*, which is the law of entropy increase for kinetic systems. This belongs to a more advanced course of kinetic theory (e.g., Dellar 2015).

• Secondly, *the relaxation to the local Maxwellian must occur on the collisional time scale* $\tau_{\mathrm{c}} = (\sigma n v_{\mathrm{th}})^{-1}$ [see Eq. (4.4); this depends on n and T, so in general τ_{c} is a function of $\boldsymbol{r}$; in a more quantitative theory, it transpires that it can also be a function of $\boldsymbol{v}$].

• Thirdly, as we have already explained in §6.4, *elastic collisions must not change the total number, momentum or energy density of the particles* and so the collision operator satisfies the *conservation properties* (6.20), (6.22) and (6.26).

Arguably the simplest possible form of the collision operator that satisfies these criteria is the so-called *Krook operator* (also known as the *BGK operator*, after Bhatnagar–Gross–Krook):

$$\boxed{C[F] = -\frac{F - F_{\mathrm{M}}}{\tau_{\mathrm{c}}} = -\frac{1}{\tau_{\mathrm{c}}}\,\delta F}\,, \tag{6.31}$$

where, to satisfy the conservation laws (6.20), (6.22) and (6.26), it is demanded that

$$\int \mathrm{d}^3\boldsymbol{v}\,\delta F = 0, \quad \int \mathrm{d}^3\boldsymbol{v}\,m\boldsymbol{v}\delta F = 0, \quad \int \mathrm{d}^3\boldsymbol{v}\,\frac{mv^2}{2}\,\delta F = 0. \tag{6.32}$$

This means that we are committed to considering only such deviations from the local Maxwellian that contain no perturbation of n, $\boldsymbol{u}$ or energy (and, therefore, of T). This does not cause any loss of generality because any changes in n, $\boldsymbol{u}$ and T that do occur can always be absorbed into the local Maxwellian F_{M}.

Namely, consider any arbitrary pdf F. Let $F_{\rm M}$ be a local Maxwellian [Eq. (5.10)] such that its density n, mean velocity $\boldsymbol{u}$ and mean energy $\varepsilon = 3nk_{\rm B}T/2$ are the same as the density, mean velocity and mean energy of F (as defined in §5.1). Then we can always write

$$F = F_{\rm M} + \underbrace{F - F_{\rm M}}_{\equiv\, \delta F}, \tag{6.33}$$

where δF obviously satisfies Eqs. (6.32).

The Krook operator is, of course, grossly simplified and inadequate for many kinetic calculations—and it certainly will not give us quantitatively precise values of transport coefficients. However, where it loses in precision it compensates in analytical simplicity and it is amply sufficient to show the basic idea of the calculation of these coefficients. The process of enlightened guesswork (also known as *modelling*) that we followed in devising it is also quite instructive as an illustration of how one comes up with a simple physically sensible model where the exact nature of the underlying process (in this case, collisions) might be unknown or too difficult to incorporate precisely, but it is clear what criteria must be respected by any sensible model.

6.6. *Solution of the Kinetic Equation*

The kinetic equation (6.18) with the Krook collision operator (6.31) is

$$\frac{\partial F}{\partial t} + v_z \frac{\partial F}{\partial z} = -\frac{F - F_{\rm M}}{\tau_{\rm c}} = -\frac{1}{\tau_{\rm c}}\,\delta F. \tag{6.34}$$

Suppose that $\delta F \ll F_{\rm M}$ and also that the spatiotemporal variations of δF occur on the same (large) scales as those of $F_{\rm M}$ (we will confirm in a moment that these are self-consistent assumptions). Then, in the left-hand side of Eq. (6.34), we can approximate $F \approx F_{\rm M}$. This instantly gives us an expression for the deviation from the Maxwellian:

$$\delta F \approx -\tau_{\rm c}\left(\frac{\partial F_{\rm M}}{\partial t} + v_z \frac{\partial F_{\rm M}}{\partial z}\right). \tag{6.35}$$

We can now estimate the size of δF:

$$\tau_{\rm c} v_z \frac{\partial F_{\rm M}}{\partial z} \sim \frac{\tau_{\rm c} v_{\rm th}}{l}\, F_{\rm M} \sim \frac{\lambda_{\rm mfp}}{l}\, F_{\rm M} \ll F_{\rm M}, \tag{6.36}$$

just as we assumed (provided $l \gg \lambda_{\rm mfp}$, i.e., all spatial variations of u_x and T are macroscopic). Turning to the other term in Eq. (6.35), if we anticipate that, as in §5.6.3,

$$\tau_{\rm c} \frac{\partial F_{\rm M}}{\partial t} \sim \frac{\tau_{\rm c}}{\tau_{\rm diff}}\, F_{\rm M} \sim \left(\frac{\lambda_{\rm mfp}}{l}\right)^2 F_{\rm M}, \tag{6.37}$$

we conclude that this term is not only small compared to $F_{\rm M}$, as per our *a priori* assumption, but also compared to the term (6.36)—and so must be neglected entirely if we are expanding in the small parameter $\lambda_{\rm mfp}/l$. You might object that, technically speaking, we do not yet know that, for macroscopic quantities, $\partial/\partial t \sim v_{\rm th}\lambda_{\rm mfp}\partial^2/\partial z^2 \sim 1/\tau_{\rm diff}$, but the idea here is to "order" the time derivative in this way and then confirm that the resulting approximate solution will satisfy this ordering. This may well be your first experience of this kind of skulduggery, but this is how serious things are done and it is worth learning how to do them!

Thus, our solution (6.35) is now quite compact:

$$\begin{aligned}\delta F &\approx -\tau_{\rm c} v_z \frac{\partial F_{\rm M}}{\partial z} \\ &= -\tau_{\rm c} v_z \left[\frac{m|\boldsymbol{v}-u_x\hat{\boldsymbol{x}}|^2}{2k_{\rm B}T^2}\frac{\partial T}{\partial z} - \frac{3}{2}\frac{1}{T}\frac{\partial T}{\partial z} + \frac{m}{2k_{\rm B}T}\,2(\boldsymbol{v}-u_x\hat{\boldsymbol{x}})\cdot\hat{\boldsymbol{x}}\,\frac{\partial u_x}{\partial z}\right] F_{\rm M} \\ &= -\tau_{\rm c} w_z \left[\left(\frac{w^2}{v_{\rm th}^2}-\frac{3}{2}\right)\frac{1}{T}\frac{\partial T}{\partial z} + \frac{2w_x}{v_{\rm th}^2}\frac{\partial u_x}{\partial z}\right] F_{\rm M}, \end{aligned} \tag{6.38}$$

where we have taken $\partial/\partial z$ of the expression for the local Maxwellian, Eq. (6.11). Thus, we have solved the kinetic equation and found the small deviation of the particle distribution function from the local Maxwellian caused by mean velocity and temperature gradients. The very first line of Eq. (6.38) is perhaps the most transparent as to the mechanism of this deviation: δF is simply the result of taking a local Maxwellian and letting it evolve ballistically for a time $\tau_{\rm c}$, with all particles flying in straight lines at their initial velocities. Because $\tau_{\rm c}$ is small, they only have an opportunity to do this for a short time, before collisions restore local equilibrium, and so the local Maxwellian gets only slightly perturbed.

Note that δF is *neither Maxwellian nor isotropic*—as indeed ought to be the case as it arises from the global equilibrium being broken by the presence of flows (which have a direction, in our case x) and gradients (which also have a direction, in our case z). The deviation from the Maxwellian is small because the departures from the equilibrium—the gradients—are macroscopic (i.e., the corresponding time and spatial scales are long compared to collisional scales $\tau_{\rm c}$ and $\lambda_{\rm mfp}$).

If our collision operator had been a more realistic and, therefore, much more complicated, integral operator than the Krook model one, solving the kinetic equation would have involved quite a lot of hard work inverting this operator—while with the Krook operator, that inversion was simply multiplication by $\tau_{\rm c}$, which took us painlessly from Eq. (6.34) to Eq. (6.35). You will find the strategies for dealing with the true Boltzmann collision operator in Chapman & Cowling (1991) or Lifshitz & Pitaevskii (1981).

Exercise 6.4. Check that the solution (6.38) satisfies the particle, momentum and energy conservation conditions (6.32).

6.7. *Calculation of Fluxes*

Finally, we use the solution (6.38) in Eqs. (6.23) and (6.28) to calculate the fluxes.

6.7.1. *Momentum Flux*

Eq. (6.23) gives

$$\begin{aligned}\Pi_{zx} &= \int {\rm d}^3\boldsymbol{w}\, m w_z w_x \delta F \\ &= -m\tau_{\rm c}\int {\rm d}^3\boldsymbol{w}\, w_z^2 w_x \left[\left(\frac{w^2}{v_{\rm th}^2}-\frac{3}{2}\right)\frac{1}{T}\frac{\partial T}{\partial z} + \frac{2w_x}{v_{\rm th}^2}\frac{\partial u_x}{\partial z}\right] F_{\rm M}(w) \\ &= -\left[\frac{2m\tau_{\rm c}}{v_{\rm th}^2}\int {\rm d}^3\boldsymbol{w}\, w_z^2 w_x^2 F_{\rm M}(w)\right]\frac{\partial u_x}{\partial z} \equiv -\eta\,\frac{\partial u_x}{\partial z}, \end{aligned} \tag{6.39}$$

where the term involving $\partial T/\partial z$ vanished because its integrand was odd in w_x. Satisfyingly, we have found that the momentum flux is proportional to the mean-velocity gradient, as we have previously argued it must be [see Eq. (5.40)]. The coefficient of proportionality between them is, by definition, the *dynamical viscosity*, the expression

for which is, therefore,

$$\eta = \frac{2m\tau_\mathrm{c}}{v_\mathrm{th}^2}\int \mathrm{d}^3\boldsymbol{w}\, w_z^2 w_x^2 F_\mathrm{M}(w)$$
$$= \frac{2m\tau_\mathrm{c}}{v_\mathrm{th}^2}\underbrace{\int_0^\infty \mathrm{d}w\, w^6 \frac{n}{(\sqrt{\pi}v_\mathrm{th})^3}\, e^{-w^2/v_\mathrm{th}^2}}_{=15nv_\mathrm{th}^4/16\pi}\underbrace{\int_0^\pi \mathrm{d}\theta\, \sin^3\theta\cos^2\theta}_{=4/15}\underbrace{\int_0^{2\pi}\mathrm{d}\phi\, \cos^2\phi}_{=\pi}$$
$$= \frac{1}{2}\, mnv_\mathrm{th}^2\tau_\mathrm{c} = \frac{1}{2}\, mn\lambda_\mathrm{mfp}v_\mathrm{th}. \tag{6.40}$$

No surprises here: the same dependence on λ_mfp and temperature (via v_th) as in Eq. (6.4), but a different numerical coefficient.† This coefficient depends on the form of the collision operator and so, since the collision operator we used is only a crude model, the coefficient is order-unity wrong. It is progress, however, that we now know what to do to calculate viscosity precisely for any given model of collisions. You will find many such precise calculations in, e.g., Chapman & Cowling (1991).

6.7.2. *Heat Flux*

A similar calculation of the velocity integral in Eq. (6.28) gives us heat flux:

$$J_z = \int \mathrm{d}^3\boldsymbol{w}\, \frac{mw^2}{2}\, w_z \delta F$$
$$= -\frac{m\tau_\mathrm{c}}{2}\int \mathrm{d}^3\boldsymbol{w}\, w_z^2 w^2 \left[\left(\frac{w^2}{v_\mathrm{th}^2}-\frac{3}{2}\right)\frac{1}{T}\frac{\partial T}{\partial z} + \frac{2w_x}{v_\mathrm{th}^2}\frac{\partial u_x}{\partial z}\right] F_\mathrm{M}(w)$$
$$= -\left[\frac{m\tau_\mathrm{c}}{2T}\int \mathrm{d}^3\boldsymbol{w}\, w_z^2 w^2\left(\frac{w^2}{v_\mathrm{th}^2}-\frac{3}{2}\right)F_\mathrm{M}(w)\right]\frac{\partial T}{\partial z} \equiv -\varkappa\frac{\partial T}{\partial z}, \tag{6.41}$$

where the term involving $\partial u_x/\partial z$ vanished because its integrand was odd in w_x. The heat flux turns out to be proportional to the temperature gradient, as expected [see Eq. (5.39)]. The expression for the *thermal conductivity* is, therefore,

$$\varkappa = \frac{m\tau_\mathrm{c}}{2T}\int \mathrm{d}^3\boldsymbol{w}\, w_z^2 w^2\left(\frac{w^2}{v_\mathrm{th}^2}-\frac{3}{2}\right)F_\mathrm{M}(w)$$
$$= \frac{k_\mathrm{B}\tau_\mathrm{c}}{v_\mathrm{th}^2}\underbrace{\int_0^\infty \mathrm{d}w\, w^6\left(\frac{w^2}{v_\mathrm{th}^2}-\frac{3}{2}\right)\frac{n}{(\sqrt{\pi}v_\mathrm{th})^3}\, e^{-w^2/v_\mathrm{th}^2}}_{=15nv_\mathrm{th}^4/8\pi}\underbrace{\int_0^\pi \mathrm{d}\theta\, \sin\theta\cos^2\theta}_{=2/3}\underbrace{\int_0^{2\pi}\mathrm{d}\phi}_{=2\pi}$$
$$= \frac{5}{2}\, nk_\mathrm{B}v_\mathrm{th}^2\tau_\mathrm{c} = \frac{5}{3}\, nc_1\lambda_\mathrm{mfp}v_\mathrm{th}, \quad c_1 = \frac{3}{2}\, k_\mathrm{B}. \tag{6.42}$$

Again, we have the same kind of expression as in Eq. (6.7), but with a different prefactor. You now have enough experience to spot that these prefactors come from the averaging of various angle and speed dependences over the underlying Maxwellian distribution—and the prefators are nontrivial basically because of intrinsic correlations between, e.g., in this case, particle energy, the speed and angle at which it moves (transport), and the form of the non-Maxwellian correction to the local equilibrium which is caused by the temperature gradient and enables heat to flow on average.

† Note that the angle dependence of the integrand in Eq. (6.3) that we so proudly worked out in §6.1 was in fact wrong. However, the derivation in §6.1, while "dodgy," was not useless: it highlighted much better than the present, more systematic, one that momentum and energy are transported because of particles wondering between regions of gas with different u_x and T.

Since we now have the heat equation including also viscous heating, Eq. (6.29), it is worth writing out its final form: using Eqs. (6.42) and (6.39), we have

$$\boxed{\frac{\partial}{\partial t}\frac{3}{2}\,nk_{\mathrm{B}}T = \varkappa\,\frac{\partial^2 T}{\partial z^2} + \eta\left(\frac{\partial u_x}{\partial z}\right)^2}\,. \tag{6.43}$$

The viscous term is manifestly positive, so does indeed represent heating. Eq. (6.43) and the momentum equation (6.21) combined with Eq. (6.39),

$$\boxed{\frac{\partial}{\partial t}mnu_x = \eta\,\frac{\partial^2 u_x}{\partial z^2}}\,, \tag{6.44}$$

form a closed system, completely describing the evolution of our gas.

Exercise 6.5. Show that the total energy of the gas

$$\langle E\rangle = \int \mathrm{d}^3\boldsymbol{r}\left(\frac{mnu_x^2}{2} + \frac{3}{2}\,nk_{\mathrm{B}}T\right) \tag{6.45}$$

is conserved by Eqs. (6.43) and (6.44).

Exercise 6.6. Fick's Law of Diffusion. a) Starting from the kinetic equation for the distribution function $F^*(t,\boldsymbol{r},\boldsymbol{v})$ of some labelled particle admixture in a gas, derive the diffusion equation

$$\frac{\partial n^*}{\partial t} = D\,\frac{\partial^2 n^*}{\partial z^2}$$

for the number density $n^*(t,z) = \int \mathrm{d}^3\boldsymbol{v}\,F^*(t,z,\boldsymbol{v})$ of the labelled particles (assuming n^* changes only in the z direction). Derive also the expression for the diffusion coefficient D, given
—the molecular mass m^* of the labelled particles,
—the temperature T of the ambient gas (assume T is uniform),
—collision frequency ν_{c}^* of the labelled particles with the ambient ones.
Assume that the ambient gas is static (no mean flows), that the density of the labelled particles is so low that they only collide with the unlabelled particles (and not each other) and that the frequency of these collisions is much larger than the rate of change of any mean quantities. Use the Krook collision operator, assuming that collisions relax the distribution of the labelled particles to a Maxwellian F_{M}^* with density n^* and the same velocity (zero) and temperature (T) as the ambient unlabelled gas.

Hint. Is the momentum of the labelled particles conserved by collisions? You should discover that self-diffusion is related to the mean velocity u_z^* of the labelled particles (you can assume $u_z^* \ll v_{\mathrm{th}}$). You can calculate this velocity either directly from $\delta F^* = F^* - F_{\mathrm{M}}^*$ or from the momentum equation for the labelled particles.

b) Derive the momentum equation for the mean flow u_z^* of the labelled particles and obtain the result you have known since school: that the friction force (the collisional drag exerted on labelled particles by the ambient population) is proportional to the mean velocity of the labelled particles. What is the proportionality coefficient (the "drag coefficient")? This, by the way, is the "Aristotelian equation of motion"—Aristotle thought force was generally proportional to velocity. It took a while for another brilliant man to figure out the more general formula.

Show from the momentum equation you have derived that the flux of the labelled particles is proportional to their pressure gradient:

$$\Phi_z^* = n^*u_z^* = -\frac{1}{m^*\nu_{\mathrm{c}}^*}\frac{\partial P^*}{\partial z}. \tag{6.46}$$

PART III

Foundations of Statistical Mechanics

7. From Microphysics to Macrophysics

7.1. *What Are We Trying to Do?*

Thermodynamics was all about flows of energy, which we formalised in two ways:

$$\mathrm{d}U = \underbrace{\delta Q}_{\text{heat}} - \underbrace{\delta W}_{\text{work}} = T\mathrm{d}S - P\mathrm{d}V. \tag{7.1}$$

Note that T and S were *introduced* via their relationship with heat in reversible processes. All this was completely general. But to calculate anything specific, we needed two further pieces of information:

1) equation of state $P = P(T, V)$, e.g., $PV = Nk_{\mathrm{B}}T$ for ideal gas,

2) energy $U = U(T, V)$, e.g., $U = \frac{3}{2}Nk_{\mathrm{B}}T$ for monatomic ideal gas.

It was also useful to be able to calculate $S(T, V)$ and various other functions of state, but all that could be obtained from thermodynamics once the two ingredients above were in place.

Working these out always required some microphysical model of the substance that the system was made of (e.g., classical ideal gas). Similarly, for non-PV systems, we always had some model of the stuff they were made of in order to determine, e.g., tension $f(T, L)$ as a function of length L and temperature T, magnetisation $\boldsymbol{M}(T, \boldsymbol{B})$ as a function of magnetic field $\boldsymbol{B}$ and temperature T, etc. (there was always T because there was always energy—a special quantity in Statistical Physics).

So, the goal is, *given a system with certain known microphysical properties (exactly what needs to be known, we will see shortly), to learn how to construct its equation of state and the relationship between its energy and temperature (as well as other parameters, e.g., V).*

To work out a specific algorithm for the construction of the thermodynamics of any given system, recall that the *free energy* satisfies

$$\mathrm{d}F = -S\mathrm{d}T - P\mathrm{d}V \tag{7.2}$$

and so, if we just know $F(T, V)$, we can calculate everything particularly straightforwardly:

$$P = -\left(\frac{\partial F}{\partial V}\right)_T, \quad \text{equation of state,} \tag{7.3}$$

$$S = -\left(\frac{\partial F}{\partial T}\right)_V, \quad \text{entropy,} \tag{7.4}$$

$$U = F + TS, \quad \text{energy,} \tag{7.5}$$

$$C_V = T\left(\frac{\partial S}{\partial T}\right)_V, \quad \text{heat capacity, etc.} \tag{7.6}$$

Thus, *our formal programme is to learn how to calculate $F(T, V)$.*

NB: We are talking exclusively about systems *in equilibrium*. If we also want to know

how they get there, we need a lot more than just $F(V,T)$! Kinetic Theory dealt with such questions, Statistical Mechanics will *not*.

7.2. *The System and Its States*

What does it mean to have a microphysical model (description) of a physical system? And what exactly *is* a system? Well, *any physical system is a quantum-mechanical system and a quantum-mechanical system is something that can be in a number of quantum states*—we will call them *microstates*, they are defined by a set of quantum numbers (eigenvalues of a complete set of commuting variables).

NB: For systems with many degrees of freedom, e.g., many particles, we are talking about *collective* states of the system, *not* simply or necessarily of a direct superposition of the states of individual particles. E.g., to anticipate §11.8, a state of a box of ideal gas will be characterised by a set of numbers telling us how many particles occupy each single-particle state (characterised by discrete values of spin and momentum allowed in the box)—not by a list of which single-particle state each particle sits in.

Let us enumerate the system's microstates:

$$\alpha = 1,\ 2,\ 3,\ \ldots,\ \Omega \ggg 1 \tag{7.7}$$

(the total number of possible microstates is huge for a large system). For each such state, there is a certain probability of the system being in it:

$$p_1,\ p_2,\ p_3,\ \ldots,\ p_\alpha,\ \ldots,\ p_\Omega, \quad \sum_{\alpha=1}^{\Omega} p_\alpha = 1. \tag{7.8}$$

Each state has a certain energy:

$$E_1,\ E_2,\ E_3,\ \ldots,\ E_\alpha,\ \ldots,\ E_\Omega. \tag{7.9}$$

They might also have momenta, angular momenta, spins, etc.

If we knew all these things, we could then calculate various macrophysical quantities as averages over the distribution $\{p_\alpha\}$, e.g., the mean energy

$$\boxed{U = \langle E_\alpha \rangle = \sum_\alpha p_\alpha E_\alpha}\ . \tag{7.10}$$

Thus, it is easy to define macroscopic counterparts to quantities that exist on the microscopic level, but it is not as yet clear what such thermodynamical quantities not obviously related to microphysics as P, S, T are. In fact, the question of what pressure is can be cleared up without delay.

7.3. *Pressure*

The concept of pressure arises in connection with changing the volume of the system. In most of what follows, we will treat volume as an *exact external parameter* (as opposed to some mean property to be measured). Let us consider deformations that occur very very slowly. We know from Quantum Mechanics (e.g., Binney & Skinner 2013, §12.1) that if an external parameter (here volume) is changed slowly in an otherwise isolated system, the system will stay in the same eigenstate (say, α) with its energy, $E_\alpha(V)$, changing slowly. This process is called *adiabatic* (we will learn soon that this meaning of "adiabatic" is equivalent to the thermodynamical one).

Since the system's microstates $\{\alpha\}$ do not change in an adiabatic process, neither do

their probabilities $\{p_\alpha\}$. The corresponding change in the mean energy is then

$$\mathrm{d}U_{\mathrm{ad}} = \left(\frac{\partial U}{\partial V}\right)_{p_1,\ldots,p_\Omega} \mathrm{d}V = \sum_\alpha p_\alpha \frac{\partial E_\alpha}{\partial V}\,\mathrm{d}V. \tag{7.11}$$

But the slow change of energy in a system due exclusively to change in its volume can be related to the work done *on* the system by whatever force is applied to effect the change. This work is, of course, equal to minus the work done *by* the system against that force:

$$\mathrm{d}U_{\mathrm{ad}} = \mathrm{d}W_{\mathrm{ad}} = -P\mathrm{d}V, \tag{7.12}$$

and so we may *define* pressure as

$$\boxed{P = -\sum_\alpha p_\alpha \frac{\partial E_\alpha}{\partial V} = -\left\langle \frac{\partial E_\alpha}{\partial V} \right\rangle}. \tag{7.13}$$

Similarly, in non-PV systems,

$$f = \sum_\alpha p_\alpha \frac{\partial E_\alpha}{\partial L}, \qquad \text{tension,} \tag{7.14}$$

$$\boldsymbol{M} = -\sum_\alpha p_\alpha \frac{\partial E_\alpha}{\partial \boldsymbol{B}}, \quad \text{magnetisation, etc.} \tag{7.15}$$

Thus, if we know $\{p_\alpha\}$ and $\{E_\alpha\}$ (the latter as functions of V or other external parameters), then we can calculate pressure and/or its non-PV analogs.

It is clear that we cannot make any progress calculating $\{E_\alpha\}$ without specifying what our system is made of and how it is constituted. So the determination of the energies is a job for the microphysical (in general, quantum) theory. Normally, exact solution will only be possible for simple models (like the ideal gas). The amazing thing, however, is that *in equilibrium*, we will be able to determine $\{p_\alpha\}$ as functions of $\{E_\alpha\}$ in a completely general way—without having to solve a Ω-dimensional Schrödinger equation for our system (which would clearly be a hopeless quest).

NB: When I say "determine $\{p_\alpha\}$," what I really mean is *find a set of probabilities $\{p_\alpha\}$ such that upon their insertion into averages such (7.10) or (7.13), correct (experimentally verifiable) macroscopic quantities will be obtained.* This does *not* mean that these probabilities will literally be solutions of the Schrödinger equation for our system (many different sets of probabilities give the same averages, so, e.g., getting the correct mean energy does not imply—or, indeed, require—that the true probabilities be used).

To learn how to determine these $\{p_\alpha\}$, we will make a philosophical leap and learn to calculate things not on the basis of what we *know*, but on the basis of what we *don't know!*

8. Principle of Maximum Entropy

8.1. *Quantifying Ignorance*

In order to make progress, we will adopt the following, perhaps surprising (in view of your experience so far of school and undergraduate physics) attitude to the probabilities $\{p_\alpha\}$: we will think of them as measuring our *ignorance* about the true microstate the system is really in. Just how ignorant we are depends on what information we do possess

(or reliably expect to be able to obtain). The probabilities that we assign to various possible microstates of the system will then be the *likelihoods* for the system to be in those microstates, given the information that we have.†

8.1.1. *Complete Ignorance*

Suppose first that we know nothing at all about the system. Then the only fair way of assigning probabilities to microstates is to make them all equally likely:

$$p_\alpha = \frac{1}{\Omega}. \tag{8.1}$$

This principle of fairness (in acknowledging that we have no basis for discriminating between microstates) can be given the status of a postulate, known as *the fundamental postulate of Statistical Mechanics*, a.k.a. *equal a priori probabilities postulate*, due to Boltzmann. Usually it is phrased as follows:

For an isolated system in equilibrium, all microstates are equally likely.

Here "isolated" means that the system is not in contact with anything—which is consistent with us knowing nothing about it (to know something, we must measure something and to measure something, we would need to interfere with the system, which then would not be completely isolated anymore). "In equilibrium" means that $\{p_\alpha\}$ are not changing, the system is assumed to have settled in some statistically steady state.

In Boltzmann's Statistical Mechanics, this postulate serves as a starting point for the whole construction (see §12.2.2), but here we quickly move on to a more interesting situation.

8.1.2. *Some Knowledge*

The reason the no-knowledge case is *not* interesting is that ultimately, we are building this theory so we can *predict results of measurements*. This means we do in fact expect to know something about our system—namely, the quantities that we will measure. Those will typically be macroscopic quantities, e.g., the mean energy:

$$U = \sum_\alpha p_\alpha E_\alpha. \tag{8.2}$$

Clearly, any particular measured value of U will be consistent with lots of different microstates, so knowing U, while not generally consistent with equal probabilities (8.1), will not constrain the values of p_α's very strongly: indeed, there are $\Omega \ggg 1$ p_α's and only one equation (8.2) that they are required to satisfy (plus the normalisation $\sum_\alpha p_\alpha = 1$). We may be able to measure other quantities and so have more information in the form of equations like Eq. (8.2), but it is clear that the amount of information we are ever likely to have (or want) falls hugely short of uniquely fixing every p_α.

8.1.3. *Assignment of Likelihoods*

[Literature: Jaynes (2003), §11.4]

We would now like to find a way of *assigning values to* $\{p_\alpha\}$ *systematically, taking into account strictly the information we have and nothing more.* We will adopt the following algorithm (Jaynes 2003, §11.4).

† Adopting the view of probabilities as likelihoods—as opposed to the *frequencies* with which the system is supposed to visit those microstates—is a controversial move, which will be further discussed in §12.3.

We have Ω microstates and need to assign probabilities $p_1, \ldots, p_\Omega$ to them, subject to $\sum_\alpha p_\alpha = 1$ and whatever constraints are imposed by our information.

• Choose some integer $\mathcal{N} \gg \Omega$ and embark on assigning $\mathcal{N}$ "quanta" of probability, each of magnitude $1/\mathcal{N}$, to the Ω microstates (imagine tossing $\mathcal{N}$ pennies into Ω boxes in an equiprobable way). After we have used up all $\mathcal{N}$ quanta, suppose we find

$$\begin{aligned} &\mathcal{N}_1 \text{ quanta in microstate } 1,\\ &\mathcal{N}_2 \text{ quanta in microstate } 2,\\ &\ldots\\ &\mathcal{N}_\Omega \text{ quanta in microstate } \Omega, \end{aligned}$$

which corresponds to the assignment of probabilities

$$p_\alpha = \frac{\mathcal{N}_\alpha}{\mathcal{N}}, \quad \alpha = 1, \ldots, \Omega. \tag{8.3}$$

• Check whether this set $\{p_\alpha\}$ satisfies the constraint(s) imposed by the available information, e.g., Eq. (8.2). If it does not, reject this assignment of probabilities and repeat the experiment. Keep going until a satisfactory set $\{p_\alpha\}$ is found.

What is the most likely outcome of this game? The number of ways W in which an assignment (8.3) can be obtained is the number of ways of choosing $\mathcal{N}_1, \ldots, \mathcal{N}_\Omega$ quanta out of a set of $\mathcal{N}$, viz.,

$$W = \frac{\mathcal{N}!}{\mathcal{N}_1! \cdots \mathcal{N}_\Omega!}. \tag{8.4}$$

All outcomes are equiprobable, so the most likely assignment $\{\mathcal{N}_\alpha\}$ is the one that *maximises W subject to the constraints imposed by the available information.*†

Note that we were at liberty to choose $\mathcal{N}$ as large as we liked and so we may assume that all $\mathcal{N}_\alpha \gg 1$ and use *Stirling's formula* to evaluate factorials:

$$\ln \mathcal{N}! = \mathcal{N} \ln \mathcal{N} - \mathcal{N} + O(\ln \mathcal{N}). \tag{8.5}$$

Then, using also $\sum_\alpha \mathcal{N}_\alpha = \mathcal{N}$,

$$\begin{aligned} \ln W &= \underbrace{\mathcal{N} \ln \mathcal{N} - \mathcal{N}}_{\sum_\alpha (\mathcal{N}_\alpha \ln \mathcal{N} + \cancel{\mathcal{N}_\alpha})} + O(\ln \mathcal{N}) - \sum_\alpha [\mathcal{N}_\alpha \ln \mathcal{N}_\alpha - \cancel{\mathcal{N}_\alpha} + O(\ln \mathcal{N}_\alpha)] \\ &= -\sum_\alpha \mathcal{N}_\alpha \ln \frac{\mathcal{N}_\alpha}{\mathcal{N}} + O(\ln \mathcal{N}) \\ &= -\mathcal{N} \left[\sum_\alpha p_\alpha \ln p_\alpha + O\left(\frac{\ln \mathcal{N}}{\mathcal{N}} \right) \right]. \end{aligned} \tag{8.6}$$

More precisely, if $\mathcal{N}, \mathcal{N}_\alpha \to \infty$ as $\mathcal{N}_\alpha/\mathcal{N} \to p_\alpha = \text{const}$, then there is a finite limit

$$\frac{1}{\mathcal{N}} \ln W \to -\sum_\alpha p_\alpha \ln p_\alpha \equiv S_{\mathrm{G}}. \tag{8.7}$$

This quantity is called the *Gibbs entropy*, or, in the context of information theory, the *Shannon entropy* ("amount of ignorance" associated with the set of probabilities $\{p_\alpha\}$).

† It is possible to prove that this maximum is very sharp for large $\mathcal{N}$.

Maximising W is the same as maximising $S_{\rm G}$, so the import of this quantity is that *the "fairest" assignment of probabilities* $\{p_\alpha\}$ *subject to some information will correspond to the maximum of* $S_{\rm G}$ *subject to constraints imposed by that information.*

8.1.4. *Some properties of Gibbs–Shannon Entropy*

1) $S_{\rm G}$ depends only on the probabilities $\{p_\alpha\}$, *not* on the quantum numbers (random variables) associated with the microstates that these probabilities describe (e.g., E_α). This means that no change of variables (e.g., $E_\alpha \to f(E_\alpha)$) or rearrangement in the labelling of the microstates $\{\alpha\}$ can change $S_{\rm G}$. In other words, $S_{\rm G}$ is a property of the set of probabilities $\{p_\alpha\}$, not of the states $\{\alpha\}$.

2) Since $0 < p_\alpha \leqslant 1$, $S_{\rm G} \geqslant 0$ always. Note that $p_\alpha > 0$ because $p_\alpha = 0$ would mean that α is not an allowed state of the system; $p_\alpha = 1$ means there is only one state the system can be in, so it must be in it and $S_{\rm G} = 0$—we have perfect knowledge $\Leftrightarrow$ zero ignorance.

3) What is the maximum possible value of $S_{\rm G}$? The number of all possible distributions of $\mathcal{N}$ probability quanta over Ω microstates is $\Omega^{\mathcal{N}}$, which is, therefore, the maximum value that W can take:

$$W_{\max} = \Omega^{\mathcal{N}}. \tag{8.8}$$

Then the maximum possible value of $S_{\rm G}$ is

$$S_{\rm G,max} = \frac{1}{\mathcal{N}} \ln W_{\max} = \ln \Omega. \tag{8.9}$$

This value is attained when our ignorance about the system is total, which means that all microstates are, as far as we are concerned, equiprobable:

$$p_\alpha = \frac{1}{\Omega} \quad \Rightarrow \quad S_{\rm G} = -\sum_\alpha \frac{1}{\Omega} \ln \frac{1}{\Omega} = \ln \Omega = S_{\rm G,max}. \tag{8.10}$$

In this context, the Shannon (1948) definition of the *information content* of a probability distribution is

$$I(p_1, \ldots, p_\Omega) = S_{\rm G,max} - S_{\rm G}(p_1, \ldots, p_\Omega) = \ln \Omega + \sum_\alpha p_\alpha \ln p_\alpha. \tag{8.11}$$

Maximising $S_{\rm G}$ is the same as minimising I. Shannon's paper (Shannon 1948) is an excellent read. I will come back to his results in §12.1.

Exercise 8.1. Tossing a Coin. This example illustrates the scheme for assignment of *a priori* probabilities to microstates discussed in §8.1.3.

Suppose we have a system that only has two states, $\alpha = 1, 2$, and no further information about it is available. We shall assign probabilities to these states in a fair and balanced way: by flipping a coin $\mathcal{N} \gg 1$ times, recording the number of heads $\mathcal{N}_1$ and tails $\mathcal{N}_2$ and declaring that the probabilities of the two states are $p_1 = \mathcal{N}_1/\mathcal{N}$ and $p_2 = \mathcal{N}_2/\mathcal{N}$.

a) Calculate the number of ways, W, in which a given outcome $\{\mathcal{N}_1, \mathcal{N}_2\}$ can happen, find its maximum and prove therefore that the most likely assignment of probabilities will be $p_1 = p_2 = 1/2$. What is the Gibbs entropy of this system?

b) Show that for a large number of coin tosses, this maximum is sharp. Namely, show that the number of ways $W(m)$ in which you can get an outcome with $\mathcal{N}/2 \pm m$ heads (where $\mathcal{N} \gg m \gg 1$) is

$$\frac{W(m)}{W(0)} \approx \exp\left(-2m^2/\mathcal{N}\right), \tag{8.12}$$

where $W(0)$ corresponds to the most likely situation found in (a); hence argue that the relative width of the maximum around $p_{1,2} = 1/2$ is $\delta p \sim 1/\sqrt{\mathcal{N}}$.

8.2. *Method of Lagrange Multipliers*

Mathematically, how does one maximise a function of Ω variables, say, $S_{\rm G}(p_1,\ldots,p_\Omega)$, subject to some constraint that has a general form

$$F(p_1,\ldots,p_\Omega)=0 \tag{8.13}$$

(e.g., Eq. (8.2), which we can write as $F(p_1,\ldots,p_\Omega)\equiv\sum_\alpha p_\alpha E_\alpha - U=0$)?

At the point of maximum (or, to be precise, extremum) of $S_{\rm G}$,

$$\mathrm{d}S_{\rm G}=\frac{\partial S_{\rm G}}{\partial p_1}\,\mathrm{d}p_1+\cdots+\frac{\partial S_{\rm G}}{\partial p_\Omega}\,\mathrm{d}p_\Omega=0, \tag{8.14}$$

but the increments $\{\mathrm{d}p_\alpha\}$ are not independent because $\{p_\alpha\}$ are only allowed to change subject to the constraint (8.13). Thus, F cannot change:

$$\mathrm{d}F=\frac{\partial F}{\partial p_1}\,\mathrm{d}p_1+\cdots+\frac{\partial F}{\partial p_\Omega}\,\mathrm{d}p_\Omega=0. \tag{8.15}$$

From this equation, we can calculate one of $\mathrm{d}p_\alpha$'s in terms of the others: e.g.,

$$\mathrm{d}p_1=-\left(\frac{\partial F/\partial p_2}{\partial F/\partial p_1}\,\mathrm{d}p_2+\cdots+\frac{\partial F/\partial p_\Omega}{\partial F/\partial p_1}\,\mathrm{d}p_\Omega\right). \tag{8.16}$$

Substitute this into Eq. (8.14):

$$\mathrm{d}S_{\rm G}=\left(\frac{\partial S_{\rm G}}{\partial p_2}-\underbrace{\frac{\partial S_{\rm G}/\partial p_1}{\partial F/\partial p_1}}_{\equiv\lambda}\frac{\partial F}{\partial p_2}\right)\mathrm{d}p_2+\cdots+\left(\frac{\partial S_{\rm G}}{\partial p_\Omega}-\underbrace{\frac{\partial S_{\rm G}/\partial p_1}{\partial F/\partial p_1}}_{\equiv\lambda}\frac{\partial F}{\partial p_\Omega}\right)\mathrm{d}p_\Omega. \tag{8.17}$$

In this equation, $\mathrm{d}p_2,\ldots,\mathrm{d}p_\Omega$ are now all independent (we only had one constraint on Ω variables, so $\Omega-1$ of them of them can be independently varied). Therefore, Eq. (8.17) implies that

$$\frac{\partial S_{\rm G}}{\partial p_\alpha}-\lambda\frac{\partial F}{\partial p_\alpha}=0\quad\text{for}\quad\alpha=2,\ldots,\Omega, \tag{8.18}$$

where, by definition of λ,

$$\frac{\partial S_{\rm G}}{\partial p_1}-\lambda\frac{\partial F}{\partial p_1}=0. \tag{8.19}$$

So, we now have $\Omega+1$ variables, $p_1,\ldots,p_\Omega$, λ, and $\Omega+1$ equations for them: Eqs. (8.18), (8.19) and (8.13):

$$\frac{\partial S_{\rm G}}{\partial p_\alpha}-\lambda\frac{\partial F}{\partial p_\alpha}=0\quad\text{for}\quad\alpha=1,\ldots,\Omega, \tag{8.20}$$

$$F(p_1,\ldots,p_\Omega)=0. \tag{8.21}$$

But these are exactly the equations that we would get if we wanted to *maximise* $S_{\rm G}-\lambda F$ *with respect to* $p_1,\ldots,p_\Omega$, λ, *and with no constraints:*

$$\mathrm{d}(S_{\rm G}-\lambda F)=\sum_\alpha\left(\frac{\partial S_{\rm G}}{\partial p_\alpha}-\lambda\frac{\partial F}{\partial p_\alpha}\right)\mathrm{d}p_\alpha-F\mathrm{d}\lambda=0. \tag{8.22}$$

This, then, is the method for conditional maximising (extremising) a function subject to a constraint: add to it the constraint multiplied by $-\lambda$ and maximise the resulting function unconditionally, with respect to the original variables and λ. The additional variable λ is called the *Lagrange multiplier*.

The method is easily generalised to the case of several constraints: suppose, instead of one constraint (8.13), we have m of them:

$$F_i(p_1, \ldots, p_\Omega) = 0, \quad i = 1, \ldots, m. \tag{8.23}$$

To maximise S_G subject to these, introduce m Lagrange multipliers λ_1, …, λ_m and maximise unconditionally

$$\boxed{S_\mathrm{G} - \sum_i \lambda_i F_i \to \max} \tag{8.24}$$

with respect to $\Omega + m$ variables p_1, …, p_Ω, λ_1, … λ_m. Obviously, in order to have a solution, we must have $m < \Omega$—fewer constraints than the system has microstates. But this is not going to be a problem as the number of microstates is usually huge, while the number of things we can possibly hope (or want) to measure very finite indeed.

8.3. *Test of the Method: Isolated System*

Before we do anything nontrivial with our newly acquired technique, let us make sure that we can recover the one case for which we know the solution: equal probabilities for microstates of a system about which we know nothing.

If we know nothing, the only constraint on the probabilities is

$$\sum_\alpha p_\alpha = 1. \tag{8.25}$$

Maximising S_G subject to this constraint is equivalent to unconditionally maximising

$$S_\mathrm{G} - \lambda \left(\sum_\alpha p_\alpha - 1 \right) \to \max. \tag{8.26}$$

This gives

$$\mathrm{d}S_\mathrm{G} - \lambda \sum_\alpha \mathrm{d}p_\alpha - \left(\sum_\alpha p_\alpha - 1 \right) \mathrm{d}\lambda = 0. \tag{8.27}$$

Using the Gibbs formula for S_G, Eq. (8.7), we have

$$\mathrm{d}S_\mathrm{G} = -\sum_\alpha (\ln p_\alpha + 1)\, \mathrm{d}p_\alpha \tag{8.28}$$

and so Eq. (8.27) becomes

$$-\sum_\alpha \underbrace{(\ln p_\alpha + 1 + \lambda)\, \mathrm{d}p_\alpha}_{=0} - \underbrace{\left(\sum_\alpha p_\alpha - 1 \right)}_{=0} \mathrm{d}\lambda = 0. \tag{8.29}$$

In order to satisfy this equation, we must set the coefficient in front of $\mathrm{d}p_\alpha$ to zero, which gives

$$p_\alpha = e^{-(1+\lambda)}. \tag{8.30}$$

Setting also the coefficient in front of $\mathrm{d}\lambda$ to zero (this is just the constraint (8.25)), we find

$$\sum_\alpha e^{-(1+\lambda)} = \Omega\, e^{-(1+\lambda)} = 1 \quad \Rightarrow \quad e^{-(1+\lambda)} = \frac{1}{\Omega}. \tag{8.31}$$

Thus, we recover the equal-probabilities distribution (8.1), with S_G for this distribution taking the maximum possible value [Eq. (8.10)]:

$$p_\alpha = \frac{1}{\Omega}, \quad S_\mathrm{G} = \ln \Omega, \tag{8.32}$$

the state of maximum ignorance. Our method appears to work.

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9. Canonical Ensemble

9.1. *Gibbs Distribution*

We are now going to implement the programme of deriving the probability distribution resulting from maximising entropy subject to a single physical constraint: a fixed value of mean energy,

$$\sum_\alpha p_\alpha E_\alpha = U. \tag{9.1}$$

The set of realisations of a system described by this probability distribution is called the *canonical ensemble*, introduced by J. W. Gibbs (1839–1903), a great American physicist whose name will loom large in everything that follows. Constraints other than (or in addition to) (9.1) will define different ensembles, some of which will be discussed later (see §14 and Exercise 14.5).

As explained in the previous section, in order to find $\{p_\alpha\}$, we must maximise $S_\mathrm{G} = -\sum_\alpha p_\alpha \ln p_\alpha$ subject to the constraint (9.1) and to $\sum_\alpha p_\alpha = 1$ [Eq. (8.25)]. This means that we need two Lagrange multipliers, which we will call λ and β, and an unconditional maximum

$$S_\mathrm{G} - \lambda\left(\sum_\alpha p_\alpha - 1\right) - \beta\left(\sum_\alpha p_\alpha E_\alpha - U\right) \to \max. \tag{9.2}$$

Taking the differential of this,

$$\mathrm{d}S_\mathrm{G} - \lambda\sum_\alpha \mathrm{d}p_\alpha - \left(\sum_\alpha p_\alpha - 1\right)\mathrm{d}\lambda - \beta\sum_\alpha E_\alpha \mathrm{d}p_\alpha - \left(\sum_\alpha p_\alpha E_\alpha - U\right)\mathrm{d}\beta = 0, \tag{9.3}$$

and using Eq. (8.28) for $\mathrm{d}S_\mathrm{G}$, we get

$$-\sum_\alpha \underbrace{(\ln p_\alpha + 1 + \lambda + \beta E_\alpha)}_{=0}\,\mathrm{d}p_\alpha - \underbrace{\left(\sum_\alpha p_\alpha - 1\right)}_{=0}\mathrm{d}\lambda - \underbrace{\left(\sum_\alpha p_\alpha E_\alpha - U\right)}_{=0}\mathrm{d}\beta = 0. \tag{9.4}$$

Setting the coefficients in front of $\mathrm{d}p_\alpha$ individually to zero, we get

$$p_\alpha = e^{-1-\lambda-\beta E_\alpha}. \tag{9.5}$$

The Lagrange multiplier λ, or, equivalently, the normalisation constant $e^{-(1+\lambda)}$, is obtained from

$$\sum_\alpha p_\alpha - 1 = 0 \quad\Rightarrow\quad e^{-(1+\lambda)}\sum_\alpha e^{-\beta E_\alpha} = 1 \quad\Rightarrow\quad e^{-(1+\lambda)} = \frac{1}{Z(\beta)}, \tag{9.6}$$

where

$$Z(\beta) = \sum_\alpha e^{-\beta E_\alpha} \tag{9.7}$$

is called the *partition function* (Z for "Zustandssumme," its German name).† Then the desired probability distribution, Eq. (9.5), is

$$\boxed{p_\alpha = \frac{e^{-\beta E_\alpha}}{Z(\beta)}}, \tag{9.8}$$

† Note the physical interpretation of the partition function as the number of microstates effectively available to the system at a given temperature (see §11.8).

known as the *Gibbs (canonical) distribution*. Finally, the second Lagrange multiplier β is found from the constraint (9.1),

$$\sum_\alpha p_\alpha E_\alpha = \frac{1}{Z(\beta)} \sum_\alpha E_\alpha e^{-\beta E_\alpha} = -\frac{\partial \ln Z}{\partial \beta} = U. \tag{9.9}$$

The latter equality gives us an implicit equation for β in terms of U.

NB: Everything here is also a function of a number of other parameters that we viewed as exactly fixed: e.g., volume V, number of particles N—they enter via the dependence of the energy levels on them, $E_\alpha = E_\alpha(V, N)$. If we instead view them *not* as fixed parameters but as random quantities with some measurable average values, then we will obtain different ensembles: e.g., the *grand canonical ensemble*, usend to describe open systems, where the mean number of particles $\langle N \rangle$ provides a constraint in maximising entropy (§14) or the so-called "pressure ensemble," where it is the average volume of the system, $\langle V \rangle$, that is considered a quantity to be measured (Exercise 14.5).

9.2. *Construction of Thermodynamics*

[Literature: Schrödinger (1990), Ch. II]

I am going to show you that we have solved the problem posed in §7: how to work out all thermodynamically relevant quantities (in particular, free energy) and relationships from just knowing the energy levels $\{E_\alpha\}$ of a given system. To do this, we first need to establish what β means and then how to calculate the thermodynamical entropy S and pressure P.

The Gibbs entropy in the equilibrium given by the Gibbs distribution (9.8) is

$$S_\mathrm{G} = -\sum_\alpha p_\alpha \ln p_\alpha = -\sum_\alpha p_\alpha(-\beta E_\alpha - \ln Z) = \beta U + \ln Z. \tag{9.10}$$

Therefore, in equilibrium,

$$\begin{aligned} \mathrm{d}S_\mathrm{G} &= \beta \mathrm{d}U + U\mathrm{d}\beta + \frac{\mathrm{d}Z}{Z} \\ &= \beta \mathrm{d}U + \cancel{U\mathrm{d}\beta} + \sum_\alpha \underbrace{\frac{e^{-\beta E_\alpha}}{Z}}_{=p_\alpha}(-\beta \mathrm{d}E_\alpha - \cancel{E_\alpha \mathrm{d}\beta}) \\ &= \beta \left(\mathrm{d}U - \sum_\alpha p_\alpha \mathrm{d}E_\alpha \right). \end{aligned} \tag{9.11}$$

Since $E_\alpha = E_\alpha(V)$ (we will hold N to be unchangeable for now), $\mathrm{d}E_\alpha = (\partial E/\partial V)\mathrm{d}V$. Recalling Eq. (7.13), we then identify the second term inside the bracket in Eq. (9.11) as $P\mathrm{d}V$, so

$$\mathrm{d}S_\mathrm{G} = \beta(\mathrm{d}U + P\mathrm{d}V) = \beta \mathrm{d}Q_\mathrm{rev}, \tag{9.12}$$

where $\mathrm{d}Q_\mathrm{rev} = \mathrm{d}U - \mathrm{d}W_\mathrm{ad}$ is the *definition* of reversible heat, the difference between the change in internal energy and the adiabatic work $\mathrm{d}W_\mathrm{ad} = -P\mathrm{d}V$ done on the system. The left-hand side of Eq. (9.12) is a full differential of S_G, which is clearly a function of state. So we have found that *β is an integrating factor of heat in thermal equilibrium—Kelvin's definition of (inverse) thermodynamical temperature!*

Thus, it must be the case that

$$\boxed{\beta = \frac{1}{k_\mathrm{B} T}}, \tag{9.13}$$

i.e., $1/\beta$ differs from the thermodynamical temperature at most by a constant factor, which we choose to be the Boltzmann constant simply to convert from energy units (β multiplies E_α in the exponentials, so its units are inverse energy) to degrees Kelvin, a historical (in)convenience. Then Eq. (9.12) immediately implies the relationship between the thermodynamical entropy S and the Gibbs–Shannon entropy $S_{\rm G}$:

$$\boxed{S = k_{\rm B} S_{\rm G}}\,. \tag{9.14}$$

(see §§9.3 and 9.4 for a more formal proof of these results).

With Eqs. (9.13) and (9.14), Eq. (9.12) turns into the familiar *fundamental equation of thermodynamics:*

$$\boxed{T\mathrm{d}S = \mathrm{d}U + P\mathrm{d}V}\,. \tag{9.15}$$

We are done: introducing as usual the *free energy*

$$F = U - TS, \tag{9.16}$$

we can calculate everything (see §7.1): equation of state, entropy, energy, etc.:

$$P = -\left(\frac{\partial F}{\partial V}\right)_T, \quad S = -\left(\frac{\partial F}{\partial T}\right)_V, \quad U = F + TS, \ldots \tag{9.17}$$

The progress we have made is that we now know the explicit expression for F in terms of energy levels of the systems: namely, combining Eqs. (9.10), (9.13) and (9.14), we get

$$\frac{S}{k_{\rm B}} = \frac{U}{k_{\rm B}T} + \ln Z, \tag{9.18}$$

whence, via Eq. (9.16),

$$\boxed{F = -k_{\rm B}T \ln Z, \quad \text{where} \quad Z = \sum_\alpha e^{-E_\alpha/k_B T}}\,. \tag{9.19}$$

This means, by the way, that *if we know the partition function, we know about the system everything that is needed to describe its equilibrium thermodynamics.*

Note that from Eq. (9.19) follows a nice way to write the Gibbs distribution (9.8):

$$Z = e^{-\beta F} \quad \Rightarrow \quad p_\alpha = \frac{e^{-\beta E_\alpha}}{Z} = e^{\beta(F-E_\alpha)}. \tag{9.20}$$

9.3. *Some Mathematical Niceties*

[Literature: Schrödinger (1990), Ch. II]

If you thought the derivation of Eqs. (9.13) and (9.14) in §9.2 was a little cavalier, mathematically, here is a more formal proof.

We had derived, using only the principle of maximum entropy, Eq. (8.7) (Gibbs–Shannon entropy, which at that point had nothing to do with the thermodynamic entropy, heat engines or any of that), and the definition of pressure, Eq. (7.13), that [Eq. (9.12)]

$$\mathrm{d}S_{\rm G} = \beta \mathrm{d}Q_{\rm rev}. \tag{9.21}$$

From Thermodynamics, we knew the *thermodynamic* entropy S, *thermodynamic* temperature T and the reversible heat to be related by

$$\mathrm{d}S = \frac{1}{T}\,\mathrm{d}Q_{\rm rev}. \tag{9.22}$$

Therefore,

$$\mathrm{d}S = \frac{1}{\beta T}\,\mathrm{d}S_{\mathrm{G}}. \tag{9.23}$$

Since the left-hand side of this equation is a full differential, so is the right-hand side. Therefore, $1/\beta T$ is a function of S_{G} only:

$$\frac{1}{\beta T} = f(S_{\mathrm{G}}) \quad \Rightarrow \quad \mathrm{d}S = f(S_{\mathrm{G}})\mathrm{d}S_{\mathrm{G}} \quad \Rightarrow \quad S = \varphi(S_{\mathrm{G}}), \tag{9.24}$$

i.e., thermodynamic entropy is some function (obtained by integration of f) of Gibbs entropy and only of it.

But S is an additive function (we know this from Thermodynamics) and so is S_{G} (see proof in §10.1). Therefore, if we consider two systems, 1 and 2, and the combined system 12, we must have

$$S_{\mathrm{G},1} + S_{\mathrm{G},2} = S_{\mathrm{G},12}, \quad S_1 + S_2 = S_{12}, \tag{9.25}$$

whence

$$\varphi_1(S_{\mathrm{G},1}) + \varphi_2(S_{\mathrm{G},2}) = \varphi_{12}(S_{\mathrm{G},1} + S_{\mathrm{G},2}), \tag{9.26}$$

whence

$$\varphi_1'(S_{\mathrm{G},1}) = \varphi_{12}'(S_{\mathrm{G},1} + S_{\mathrm{G},2}), \tag{9.27}$$

$$\varphi_2'(S_{\mathrm{G},2}) = \varphi_{12}'(S_{\mathrm{G},1} + S_{\mathrm{G},2}). \tag{9.28}$$

Therefore,

$$\varphi_1'(S_{\mathrm{G},1}) = \varphi_2'(S_{\mathrm{G},2}) = \mathrm{const} \equiv k_{\mathrm{B}} \tag{9.29}$$

("separation constant"), giving

$$\varphi'(S_{\mathrm{G}}) = f(S_{\mathrm{G}}) = k_{\mathrm{B}} \quad \Rightarrow \quad \frac{1}{k_{\mathrm{B}}T} = f(S_{\mathrm{G}}) = k_{\mathrm{B}} \quad \Rightarrow \quad \beta = \frac{1}{k_{\mathrm{B}}T}, \tag{9.30}$$

the desired Eq. (9.13), q.e.d. This implies, finally [see Eq. (9.24)],

$$\mathrm{d}S = k_{\mathrm{B}}\mathrm{d}S_{\mathrm{G}} \quad \Rightarrow \quad S = k_{\mathrm{B}}S_{\mathrm{G}} + \mathrm{const}. \tag{9.31}$$

Setting const = 0 gives Eq. (9.14), q.e.d. It remains to discuss this choice of the integration consant, which has a physical meaning.

9.4. *Third Law*

[Literature: Schrödinger (1990), Ch. III]

From Eq. (9.10), the Gibbs entropy in thermal equilibrium is

$$S_{\mathrm{G}} = \ln \sum_{\alpha} e^{-\beta E_\alpha} + \beta\, \frac{\sum_\alpha E_\alpha e^{-\beta E_\alpha}}{\sum_\alpha e^{-\beta E_\alpha}}. \tag{9.32}$$

Consider what happens to this quantity in the limit $T \to 0$, or $\beta \to \infty$. Suppose the lowest energy level is E_1 and the lowest m microstates have this energy, viz.,

$$E_\alpha = E_1 \text{ for } \alpha = 1, \ldots, m \quad \text{and} \quad E_\alpha > E_1 \text{ for } \alpha > m. \tag{9.33}$$

Then

$$
\begin{aligned}
S_{\rm G} &= \ln\left(me^{-\beta E_1}+\sum_{\alpha>m}e^{-\beta E_\alpha}\right)+\beta\,\frac{mE_1e^{-\beta E_1}+\sum_{\alpha>m}E_\alpha e^{-\beta E_\alpha}}{me^{-\beta E_1}+\sum_{\alpha>m}e^{-\beta E_\alpha}},\\
&= \ln\left[me^{-\beta E_1}\left(1+\frac{1}{m}\sum_{\alpha>m}e^{-\beta(E_\alpha-E_1)}\right)\right]+\beta E_1\,\frac{1+\frac{1}{m}\sum_{\alpha>m}\frac{E_\alpha}{E_1}e^{-\beta(E_\alpha-E_1)}}{1+\frac{1}{m}\sum_{\alpha>m}e^{-\beta(E_\alpha-E_1)}}\\
&\approx \ln m-\cancel{\beta E_1}+\frac{1}{m}\sum_{\alpha>m}\cancel{e^{-\beta(E_\alpha-E_1)}}+\beta E_1\left[\cancel{1}+\frac{1}{m}\sum_{\alpha>m}\left(\frac{E_\alpha}{E_1}-\cancel{1}\right)e^{-\beta(E_\alpha-E_1)}\right]\\
&= \ln m+\frac{\beta}{m}\sum_{\alpha>m}E_\alpha e^{-\beta(E_\alpha-E_1)}. \qquad (9.34)
\end{aligned}
$$

The second term is exponentially small as $\beta\to\infty$, so

$$
\boxed{S_{\rm G}\to\ln m\quad\text{as}\quad T\to 0}\,, \qquad (9.35)
$$

where m is the degeneracy of the lowest energy level. Physically, this makes sense: at zero temperature, the system will be in one of its m available lowest-energy states, all of which have equal probability.

Setting const $=0$ in Eq. (9.31) means that also the thermodynamic entropy

$$
S\to k_{\rm B}\ln m\quad\text{as}\quad T\to 0. \qquad (9.36)
$$

Recall that *the 3-rd Law of Thermodynamics* said that $S\to 0$ as $T\to 0$. This is not a contradiction because $k_{\rm B}\ln m$ is very small compared to typical values that S can have: indeed, since S is additive, it will generally be proportional to the number of particles in the system, $S\propto k_{\rm B}N$ (see §11.9), whereas obviously $\ln m\ll N$ except for very strange systems. Thus, *the choice* const $=0$ *in Eq. (9.31) is basically the statement of the 3-rd Law.* You will find further discussion of this topic in Chapter III of Schrödinger (1990).

NB: In any event, these details do not matter very much because what is important is that the constant in Eq. (9.31) is a constant, independent of the parameters of the system, so all entropy *differences* are independent of it—and related via $k_{\rm B}$ when expressed in terms of S and $S_{\rm G}$.

9.5. *Part I Obviated, Road Ahead Clear*

Thus, I have proved that the statistical-mechanical T and S are the same as the thermodynamical T and S. This was a nice exercise, but, strictly speaking, unnecessary. Instead, I could have *defined*

$$
S\equiv k_{\rm B}S_{\rm G}\quad\text{and}\quad T\equiv\frac{1}{k_{\rm B}\beta} \qquad (9.37)
$$

(with a historical factor of $k_{\rm B}$ to show respect for tradition) and then *constructed all of Thermodynamics as a consequence of Statistical Mechanics*, without ever having to go through all those heat engines, Carnot cycles, etc. Indeed, with the definitions (9.37), we get the entire thermodynamic calculus, based on Eq. (9.15), the specific expression for F (or Z), Eq. (9.19), and the expressions (9.17) for everything else in terms of F.

So, the way it all has been presented to you is *chronological*, rather than *logical:* Thermodynamics was worked out in the 19-th century before Statistical Mechanics finally emerged in its modern form in the early 20-th. Logically, we no longer need a separate construction of Thermodynamics, except as an intellectual exercise and a beautiful example of how to set up an empirical theory of physical phenomena whose microscopic nature ones does not understand.

In principle, we are ready now to apply the scheme for calculating thermodynamic equilibria worked out in §9.2 to various specific cases: the classical monatomic ideal gas



FIGURE 20. A model of elastic chain (Exercise 9.1).

(§11), diatomic gases, magnetic systems, etc. (Part IV). But before we can in good faith embark on these practical calculations, we must deal with some conceptual issues:
— conditions for thermodynamic equilibrium (which, in our new language, means the state of maximum entropy subject to measurable constraints),
— its stability (if $\mathrm{d}S_{\mathrm{G}} = 0$, how do we know it is a maximum, rather than a minimum?),
— 2-nd Law,
— the meaning of probabilities, information, its loss etc.
The first two of these are more mundane and will be dealt with in §10; the last two are rather tricky and are postponed to the "postscript" sections §§12 and 13.

Exercise 9.1. Elastic Chain. A very simplistic model of an elastic chain is illustrated in Fig. 20. This is a 1D chain consisting of N segments, each of which can be in one of two (non-degenerate) states: horizontal (along the chain) or vertical. Let the length of the segment be a when it is horizontal and 0 when it is vertical. Let the chain be under fixed tension γ and so let the energy of each segment be 0 when it is horizontal and γa when it is vertical. The temperature of the chain is T.

a) What are the microstates of the chain? Using the canonical ensemble, work out the single-segment partition function and hence the partition function of the entire chain.

b) Work out the relationship between mean energy U and mean length L of the chain and hence calculate the mean length as a function of γ and T. Under what approximation do we obtain Hooke's law:

$$L = L_0 + \frac{A\gamma}{T}, \tag{9.38}$$

where L_0 and A are constants?

c) Calculate the heat capacity for this chain and sketch it as a function of temperature (pay attention to what quantity is held constant for the calculation of the heat capacity). Why physically does the heat capacity vanish both at small and large temperatures?

d) *Negative temperature.* If you treat the mean energy U of the chain as given and temperature as the quantity to be found, you will find that temperature can be negative! Sketch T as a function of U and determine under what conditions $T < 0$. Why is this possible in this system and not, say, for the ideal gas? Why does the stability argument from §10.5.2 not apply here?

e) *Superfluous constraints.* This example illustrates that if you have more measurements and so more constraints, you do not necessarily get a different statistical mechanics (so the maximum entropy principle is less subjective than it might seem; see §12.4).

So far we have treated our chain as a canonical ensemble, i.e., we assumed that the only constraint on probabilities would be the mean energy U. Suppose now that we have both a thermometer and a ruler and so wish to maximise entropy subject to two constraints: the mean energy is U and the mean length of the chain is L. Do this and find the probabilities of the microstates α of the chain as functions of their energies E_α and corresponding chain lengths ℓ_α. Show that the maximisation problem only has a solution when U and L are in a specific relationship with each other—so the new constraint is not independent and does not bring in any new physics. Show that in this case one of the Lagrange multipliers is arbitrary (and so can be set to 0—e.g., the one corresponding to the constraint of fixed L; this constraint is superfluous so we are back to the canonical ensemble).

---

10. Thermodynamic Equilibria and Stability

Much of the discussion here will be about systems with different equilibrium characteristics being put in contact with each other and arriving at a new equilibrium.

10.1. *Additivity of Entropy*

Consider two systems:

System 1: microstates α, energy levels $E^{(1)}_\alpha$, probabilities $p^{(1)}_\alpha$,
System 2: microstates α', energy levels $E^{(2)}_{\alpha'}$, probabilities $p^{(2)}_{\alpha'}$,

Now put them together into a composite system, but in such a way that the two constituent systems are in *"loose" thermal contact*, meaning that the microstates of the two systems are independent.† Then the microstates of the composite system are

(α,α') with energy levels $E_{\alpha\alpha'} = E^{(1)}_\alpha + E^{(2)}_{\alpha'}$, probabilities $p_{\alpha\alpha'} = p^{(1)}_\alpha \cdot p^{(2)}_{\alpha'}$.

The Gibbs entropy of this system is

$$\begin{aligned} S_{\rm G} &= -\sum_{\alpha\alpha'} p_{\alpha\alpha'} \ln p_{\alpha\alpha'} = -\sum_{\alpha\alpha'} p^{(1)}_\alpha p^{(2)}_{\alpha'} \ln\left(p^{(1)}_\alpha p^{(2)}_{\alpha'}\right) \\ &= -\sum_\alpha p^{(1)}_\alpha \ln p^{(1)}_\alpha \underbrace{\sum_{\alpha'} p^{(2)}_{\alpha'}}_{=1} - \sum_{\alpha'} p^{(2)}_{\alpha'} \ln p^{(2)}_{\alpha'} \underbrace{\sum_\alpha p^{(1)}_\alpha}_{=1} = S_{{\rm G},1} + S_{{\rm G},2}. \end{aligned} \tag{10.1}$$

Thus, *Gibbs entropy is additive.*‡ So is, of course, mean energy:

$$U = \sum_{\alpha\alpha'} E_{\alpha\alpha'} p_{\alpha\alpha'} = \sum_\alpha E^{(1)}_\alpha p^{(1)}_\alpha \underbrace{\sum_{\alpha'} p^{(2)}_{\alpha'}}_{=1} + \sum_{\alpha'} E^{(2)}_{\alpha'} p^{(2)}_{\alpha'} \underbrace{\sum_\alpha p^{(1)}_\alpha}_{=1} = U_1 + U_2. \tag{10.2}$$

In equilibrium, the Gibbs entropy is the same as the thermodynamical entropy, $k_{\rm B} S_{\rm G} = S$ [Eq. (9.14)], so

$$S = S_1 + S_2. \tag{10.3}$$

Note that in fact, in equilibrium, everything can be derived from the additivity of the energy levels: indeed, $E_{\alpha\alpha'} = E^{(1)}_\alpha + E^{(2)}_{\alpha'}$ implies that *partition functions multiply:* for the composite system at a single temperature (otherwise it would not be in equilibrium; see §10.2),

$$Z(\beta) = \sum_{\alpha\alpha'} e^{-\beta E_{\alpha\alpha'}} = \sum_{\alpha\alpha'} e^{-\beta\left[E^{(1)}_\alpha + E^{(2)}_{\alpha'}\right]} = \left(\sum_\alpha e^{-\beta E^{(1)}_\alpha}\right)\left(\sum_{\alpha'} e^{-\beta E^{(2)}_{\alpha'}}\right) = Z_1(\beta) Z_2(\beta). \tag{10.4}$$

Therefore, the canonical equilibrium probabilities are

$$p_{\alpha\alpha'} = \frac{e^{-\beta E_{\alpha\alpha'}}}{Z} = \frac{e^{-\beta E^{(1)}_\alpha}}{Z_1}\frac{e^{-\beta E^{(2)}_{\alpha'}}}{Z_2} = p^{(1)}_\alpha p^{(2)}_{\alpha'} \tag{10.5}$$

† In the language of Quantum Mechanics, the eigenstates of a composite system are products of the eigenstates of its (two) parts. This works, e.g., for gases or fluids, but not for solids, where states are collective. You will find further discussion of this in Binney & Skinner (2013), §6.1.

‡ Note that if in constructing the expression for entropy we followed the formal route offered by Shannon's Theorem (§12.1), this would be guaranteed automatically (requirement 5 imposed on $S_{\rm G}$ in §12.1).

and also

$$F = -k_{\mathrm{B}}T \ln Z = -k_{\mathrm{B}}T \ln(Z_1 Z_2) = F_1 + F_2, \tag{10.6}$$

$$S = -\left(\frac{\partial F}{\partial T}\right)_V = -\left(\frac{\partial F_1}{\partial T}\right)_V - \left(\frac{\partial F_2}{\partial T}\right)_V = S_1 + S_2, \tag{10.7}$$

$$U = -\frac{\partial \ln Z}{\partial \beta} = -\frac{\partial \ln(Z_1 Z_2)}{\partial \beta} = U_1 + U_2. \tag{10.8}$$

10.2. *Thermal Equilibrium*

We can now derive some consequences of the additivity of entropy coupled with the principle of obtaining the equilibrium state by maximising it.

Consider putting two systems (each in its own equilibrium) into loose thermal contact, but otherwise keeping them isolated (to be precise, we let them exchange energy with each other but not with anything else). Then, to find the new equilibrium, we must keep the total energy constant and maximise entropy:

$$U = U_1 + U_2 = \text{const}, \tag{10.9}$$

$$S = S_1 + S_2 \to \max. \tag{10.10}$$

These conditions are implemented by setting the differentials of both the energy and the entropy to zero under changes in the energies of the two sub-systems:

$$\mathrm{d}U = \mathrm{d}U_1 + \mathrm{d}U_2 = 0 \quad \Rightarrow \quad \mathrm{d}U_2 = -\mathrm{d}U_1, \tag{10.11}$$

$$\mathrm{d}S = \mathrm{d}S_1 + \mathrm{d}S_2 = \frac{\partial S_1}{\partial U_1}\mathrm{d}U_1 + \frac{\partial S_2}{\partial U_2}\mathrm{d}U_2 = \left(\frac{\partial S_1}{\partial U_1} - \frac{\partial S_2}{\partial U_2}\right)\mathrm{d}U_1 = 0. \tag{10.12}$$

From the fundamental equation of thermodynamics [Eq. (9.15)],†

$$\mathrm{d}S = \frac{1}{T}\,\mathrm{d}U + \frac{P}{T}\,\mathrm{d}V, \tag{10.13}$$

we get

$$\frac{1}{T} = \frac{\partial S}{\partial U}, \tag{10.14}$$

so Eq. (10.12) is

$$\mathrm{d}S = \left(\frac{1}{T_1} - \frac{1}{T_2}\right)\mathrm{d}U_1 = 0 \quad \Rightarrow \quad \boxed{T_1 = T_2}\,. \tag{10.15}$$

Thus, *in equilibrium, two systems in loose thermal contact will have equal temperatures.* This is called *thermal equilibrium.*

Note also that out of equilibrium, $\mathrm{d}S > 0$, so $\mathrm{d}U_1 > 0 \Leftrightarrow T_1 < T_2$ and vice versa, i.e., energy flows from hot to cold.

What we have done can be recast formally as a Lagrange multiplier calculation: we are maximising $S_1 + S_2$ subject to $U_1 + U_2 = U$, so, unconditionally,

$$S_1 + S_2 - \lambda(U_1 + U_2 - U) \to \max. \tag{10.16}$$

This gives

$$\left(\frac{\partial S_1}{\partial U_1} - \lambda\right)\mathrm{d}U_1 + \left(\frac{\partial S_2}{\partial U_2} - \lambda\right)\mathrm{d}U_2 + (U_1 + U_2 - U)\mathrm{d}\lambda = 0 \quad \Rightarrow \quad \frac{\partial S_1}{\partial U_1} = \frac{\partial S_2}{\partial U_2} = \lambda = \frac{1}{T}. \tag{10.17}$$

† This equation is only valid for equilibrium states, so its use here means that we are assuming the two subsystems and their composite all to be in equilibrium.

NB: The validity of Eq. (10.14) does *not* depend on the identification of S and T with the entropy and temperature from empirical thermodynamics, the equation holds for the statistical-mechanical entropy (measure of uncertainty in the distribution $\{p_\alpha\}$) and statistical-mechanical temperature (Lagrange multiplier associated with fixed mean energy in the canonical ensemble). The above argument therefore shows that *the statistical-mechanical temperature is a sensible definition of temperature*: it is a scalar function that is the same across a composite system in equilibrium. This property then allows one to introduce a *thermometer* based on this temperature and hence a *temperature scale* (recall that in Thermodynamics, temperature was introduced either via the 0-th Law, as just such a function, which, however, did not have to be unique, or as the universal integrating factor of $\mathrm{d}Q_{\mathrm{rev}}$—Kelvin's definition we used in §9.2 when proving the equivalence between thermodynamical and statistical-mechanical tempeartures). I am stressing this to re-emphasise the point, made in §9.5, that Thermodynamics can be derived entirely from Statistical Mechanics.

10.3. *Physical Interpretation of the Canonical Ensemble*

This is an appropriate moment to discuss what the canonical distribution actually describes physically.

Recall that this distribution followed from stipulating that probabilities of the system's microstates should be maximally unbiased subject only to conspiring to give some fixed (measurable) value of the mean energy U. The resulting Gibbs distribution (9.8) depended on a single parameter β, which we now know is the inverse temperature of the system and which was calculated via the implicit equation (9.9),

$$U = -\frac{\partial \ln Z}{\partial \beta} \quad \Rightarrow \quad \beta = \frac{1}{k_{\mathrm{B}} T} = \beta(U, \dots). \tag{10.18}$$

But both the structure of the theory that has emerged (*implicit* equation for β) and the experience (or anticipation) of the kinds of questions that we are likely to be interested in suggest that in fact it is much preferable to think of the mean energy as a function of temperature, $U = U(T, \dots)$, with T as an "input parameter." This is preferable because the temperature of a system is often known by virtue of the system being in contact with surroundings, a.k.a. *heat reservoir* or *heat bath*, whose temperature is fixed—usually because the system under consideration is small compared to the heat bath and so can draw from or give up to the latter arbitrary amounts of energy without affecting the temperature of the heat bath very much. In equilibrium, $T_{\mathrm{system}} = T_{\mathrm{bath}}$, as we proved in §10.2.

Thus, this is what *the canonical ensemble describes: microstates of a system in thermal contact with a heat bath at fixed temperature.*

One can explicitly construct the Gibbs distribution on this basis if one starts from a (fictional) "closed system" with equal probabilities for all its microstates (the "microcanonical ensemble") and then considers a small part of it. This will be discussed in detail in §12.2.2 (or see, e.g., Blundell & Blundell (2009), §4.6, Landau & Lifshitz (1980), §28).

10.4. *Mechanical and Dynamical Equilibria*

[Literature: Landau & Lifshitz (1980), §§10, 12]

So far, we have focused on energy as the variable quantity exchangeable between systems (or between the system and the heat bath), while treating the volume of the system

as a fixed external parameter and also assuming implicitly that the system was static (neither it not its constituent parts had a velocity). Let us now generalise and consider some number of systems, indexed by i, each having

total energy $\mathcal{E}_i$,
mass m_i,
velocity $\boldsymbol{u}_i$,
centre of mass position $\boldsymbol{r}_i$
and volume V_i.

We now join them all together (in "loose contact," as explained in §10.1, so their microstates remain independent) and allow them to exchange energy, momentum, angular momentum and also push on each other ("exchange volume," but not merge). If we now isolate them and confine them within some volume,† the equilibrium state of the combined system must be the state of maximum entropy subject to the following conservation laws:

$$\sum_i \mathcal{E}_i = \mathcal{E} \qquad \text{total energy,} \tag{10.19}$$

$$\sum_i m_i \boldsymbol{u}_i = \boldsymbol{p} \qquad \text{total momentum,} \tag{10.20}$$

$$\sum_i m_i \boldsymbol{r}_i \times \boldsymbol{u}_i = \boldsymbol{L} \qquad \text{total angular momentum,} \tag{10.21}$$

$$\sum_i V_i = V \qquad \text{total volume.} \tag{10.22}$$

Thus, we must maximise

$$\sum_i S_i - \lambda \left(\sum_i \mathcal{E}_i - \mathcal{E} \right) - \boldsymbol{a} \cdot \left(\sum_i m_i \boldsymbol{u}_i - \boldsymbol{p} \right) - \boldsymbol{b} \cdot \left(\sum_i m_i \boldsymbol{r}_i \times \boldsymbol{u}_i - \boldsymbol{L} \right) - \sigma \left(\sum_i V_i - V \right) \to \max, \tag{10.23}$$

where λ, $\boldsymbol{a}$, $\boldsymbol{b}$ and σ are Lagrange multipliers.

The variables with respect to which we must maximise this expression are $\{\mathcal{E}_i, \boldsymbol{u}_i, V_i\}$. We do not include the masses $\{m_i\}$ in this set because we are assuming that our systems cannot exchange matter—we will see in §14 how to handle the possibility that they might.‡ We also do not include the positions $\{\boldsymbol{r}_i\}$ amongst the variables because the entropy S_i cannot depend on where the system i is—this is because S_i depends only on the probabilities of the system's microstates $\{p_\alpha\}$, which clearly depend only on the internal workings of the system, not on its position in space (unless there is some inhomogeneous external potential in which this entire assemblage resides and which would then affect energy levels—we will not consider this possibility until §14.5).

By the same token, the entropy of each subsystem can depend only on its *internal* energy, not on that of its macroscopic motion, because the probabilities $\{p_\alpha\}$ are, by

† Equivalently, we can simply say that the combined system will have some total energy, momentum, angular momentum and volume, which we expect to be able to measure.

‡ However, if we allowed such an exchange, we would have to disallow something else, for example exchange of volume—otherwise, how would we define where one system ends and another begins?

Galilean invariance, the same in any inertial frame. The internal energy is

$$U_i = \mathcal{E}_i - \frac{m_i u_i^2}{2} \tag{10.24}$$

(because the *total* energy $\mathcal{E}_i$ consists of the internal one, U_i, and the kinetic energy of the system's macroscopic motion, $m_i u_i^2/2$). Therefore,†

$$S_i = S_i(U_i, V_i) = S_i\left(\mathcal{E}_i - \frac{m_i u_i^2}{2}, V_i\right). \tag{10.25}$$

Thus, S_i depends on both $\mathcal{E}_i$ and $\boldsymbol{u}_i$ via its internal-energy dependence.

NB: The reason we treat $\mathcal{E}_i$ and not U_i as variables with respect to which we will be maximising entropy is that only the total energy of the system is constrained by the energy conservation law—it is perfectly fine for energy to be transfered between internal and kinetic as the system seeks equilibrium.

Differentiating the expression (10.23) with respect to $\mathcal{E}_i$, $\boldsymbol{u}_i$ and V_i, and demanding that all these derivatives vanish, we find

$$\frac{\partial S_i}{\partial \mathcal{E}_i} - \lambda = 0 \qquad \text{thermal equilibrium,} \tag{10.26}$$

$$\frac{\partial S_i}{\partial \boldsymbol{u}_i} - m_i\left(\boldsymbol{a} + \boldsymbol{b}\times\boldsymbol{r}_i\right) = 0 \qquad \text{dynamical equilibrium,} \tag{10.27}$$

$$\frac{\partial S_i}{\partial V_i} - \sigma = 0 \qquad \text{mechanical equilibrium.} \tag{10.28}$$

10.4.1. *Thermal Equilibrium*

Using again Eq. (10.14), we find that Eq. (10.26) tells us that in equilibrium, the temperatures of all subsystems must be equal to the same Lagrange multiplier and, therefore, to each other:

$$\frac{\partial S_i}{\partial \mathcal{E}_i} = \frac{\partial S_i}{\partial U_i} = \frac{1}{T_i} \quad \Rightarrow \quad \frac{1}{T_i} = \lambda \equiv \frac{1}{T}. \tag{10.29}$$

This is simply the generalisation to multiple subsystems of the result already obtained in §10.2.

10.4.2. *Mechanical Equilibrium*

Going back to the fundamental equation of thermodynamics, Eq. (10.13), we note that

$$\frac{\partial S_i}{\partial V_i} = \frac{P_i}{T_i}. \tag{10.30}$$

But we already know that all $T_i = T$, so Eq. (10.28) implies that in equilibrium, all pressures are equal as well:

$$\frac{P_i}{T} = \sigma \equiv \frac{P}{T} \tag{10.31}$$

(note that for ideal gas, this Lagrange muliplier is particle density: $\sigma = n k_{\mathrm{B}}$). Physically, this says that in equilibrium, everything is in *pressure balance* (otherwise volumes will expand or shrink to make it so).

† In principle, S_i is also a function of m_i, separately from the energy dependence, viz., $S_i = S_i(U_i, V_i, m_i)$ or $S_i = S_i(U_i, V_i, N_i)$, but we assume all system masses (or numbers of particles N_i) to be fixed.

10.4.3. *Dynamical Equilibrium*

Finally, let us work out what Eq. (10.27) means. In view of Eq. (10.25),

$$\frac{\partial S_i}{\partial \boldsymbol{u}_i} = -m_i \boldsymbol{u}_i \frac{\partial S_i}{\partial \mathcal{E}_i} = -\frac{m_i \boldsymbol{u}_i}{T_i} = -\frac{m_i \boldsymbol{u}_i}{T}. \tag{10.32}$$

Then, from Eq. (10.27),

$$\boldsymbol{u}_i = -T\boldsymbol{a} - T\boldsymbol{b} \times \boldsymbol{r}_i \equiv \boldsymbol{u} + \boldsymbol{\Omega} \times \boldsymbol{r}_i, \tag{10.33}$$

where we have defined $\boldsymbol{u} \equiv -T\boldsymbol{a}$ and $\boldsymbol{\Omega} \equiv -T\boldsymbol{b}$. This means that *the only macroscopic motion in a system in equilibrium can be an overall constant motion of the whole system in some direction plus a rigid-body rotation of the whole system.*

The main implication of these results is that *in a system in equilibrium, there cannot be any temperature or pressure gradients or any internal macroscopic motions (velocity gradients).* Statistical Mechanics does not tell us how this is achieved, but we know from our experience with Kinetic Theory that temperature and velocity gradients will relax to global equilibrium via thermal diffusivity and viscosity, respectively.

A few further observations are in order.

1) In practice, mechanical equilibrium (pressure balance) is often achieved faster than the thermal and dynamical ones are, at least in incompressible systems: pressure imbalances will create uncompensated macroscopic forces, which will give rise to macroscopic motions, which will iron out pressure differences on dynamical time scales.

2) All the arguments above are generalised in an obvious way to non-PV systems.

3) Another type of equilibrium that we might have considered is *particle equilibrium*—by allowing our subsystems to exchange particles, subject to the overall conservation of their total number. This leads to the equalisation of the *chemical potential* across all subsystems—another Lagrange multiplier, which will be introduced in §14, when we study "open systems." Yet another generalisation will be to *phase* and *chemical equilibria*, discussed in §15.

4) In considering quantities other than energy as measurable constraints (momentum, angular, momentum, volume), we went beyond the canonical ensemble—and indeed, other ensembles can be constructed to handle situations where, besides energy, other quantities are considered known: e.g., mean angular momentum ("rotational ensemble"), mean volume ("pressure ensemble"; see Exercise 14.5), mean particle number ("grand canonical ensemble"; see §14), etc. There is no ensemble based on the momentum of translational motion: indeed, if we consider non-rotating systems, Eq. (10.33) says that $\boldsymbol{u}_i = \boldsymbol{u}$ and we can always go to the frame of references in which $\boldsymbol{u} = 0$ and the system is at rest.

10.5. *Stability*

How do we know that when we extremised S, the solution we found was a maximum, not a minimum (or a saddle point)? This is equivalent to asking whether the equilibria we found are *stable*. To check for stability, we need to calculate second derivatives of the entropy.

10.5.1. *Thermal Stability*

From Eqs. (10.26) and (10.29),

$$\frac{\partial^2 S_i}{\partial \mathcal{E}_i^2} = \frac{\partial}{\partial \mathcal{E}_i}\frac{1}{T} = -\frac{1}{T^2}\frac{\partial T}{\partial \mathcal{E}_i} = -\frac{1}{T^2 C_{Vi}} < 0 \tag{10.34}$$

is a necessary condition for stability. Here

$$C_{Vi} = \frac{\partial \mathcal{E}_i}{\partial T} = \frac{\partial U_i}{\partial T} \tag{10.35}$$

is the heat capacity and so, in physics language, this is the requirement that heat capacity should always be positive:

$$\boxed{C_V > 0}\,. \tag{10.36}$$

That this is always so can actually be proven directly by calculating $C_V = \partial U/\partial T$ from $U = -\partial \ln Z/\partial\beta$ and using the explicit Gibbs formula for Z.

Exercise 10.1. Heat Capacity from Canonical Ensemble. Prove the inequality (10.36) by showing that

$$C_V = \frac{\langle \Delta E^2\rangle}{k_{\rm B}T^2} \geqslant 0, \tag{10.37}$$

where $\langle \Delta E^2\rangle$ is the mean square fluctuation of the system's energy around its mean energy U.

10.5.2. *Dynamical Stability*

For simplicity, let us only consider the case with fixed volume and no rotation ($\boldsymbol{\Omega} = 0$). Then, denoting the vector components of the velocity $\boldsymbol{u}_i$ by Greek superscripts and using again the fact that S_i depends on $\boldsymbol{u}_i$ via its internal-energy dependence [Eq. (10.25)], we find another necessary condition for stability:

$$\begin{aligned}\frac{\partial^2 S_i}{\partial u_i^\mu \partial u_i^\nu} &= \frac{\partial}{\partial u_i^\mu}\left(-m_i u_i^\nu \frac{\partial S_i}{\partial \mathcal{E}_i}\right) = -m_i \frac{\partial S_i}{\partial \mathcal{E}_i}\,\delta_{\mu\nu} - m_i u_i^\nu \frac{\partial}{\partial u_i^\mu}\frac{\partial S_i}{\partial \mathcal{E}_i}\\ &= -\frac{m_i}{T}\delta_{\mu\nu} + \cancel{m_i^2 u_i^\mu u_i^\nu \frac{\partial^2 S_i}{\partial \mathcal{E}_i^2}} < 0.\end{aligned} \tag{10.38}$$

The second term can be eliminated because in equilibrium all velocities are the same $\boldsymbol{u}_i = \boldsymbol{u}$ and we can always go the frame where $\boldsymbol{u} = 0$. The condition (10.38) is equivalent to

$$\boxed{T > 0}\,. \tag{10.39}$$

Thus, we have proven that *temperature must be positive*! Systems with negative temperature are unstable.

Another, more qualitative way of arguing this is as follows. The entropy of the composite system is

$$S = \sum_i S_i\left(\mathcal{E}_i - \frac{m_i u_i^2}{2}\right). \tag{10.40}$$

If temperature is negative,

$$\frac{\partial S_i}{\partial \mathcal{E}_i} = \frac{1}{T} < 0, \tag{10.41}$$

then all S_i's would be maximised by decreasing their argument as much as possible, i.e., by increasing all u_i's subject to $\sum_i \boldsymbol{u}_i = 0$. This means that all the parts of the systems would fly in opposite directions (the system would blow up).

NB: This prohibition on negative temperatures can be relaxed if bits of the system are not allowed to move and/or if the system's allowed range of energies is bounded (see Exercise 9.1).†

Note that a similar argument can be made for the *positivity of pressure:* if pressure is negative,

$$P = T\left(\frac{\partial S}{\partial V}\right)_U < 0, \tag{10.42}$$

then entropy in a (closed) system can increase if volume goes down, i.e., the system will shrink to nothing. In contrast, if $P > 0$, then entropy increases as V increases (system expands)—but this is checked by walls or whatever external circumstances maintain the fixed total volume. This argument militates strongly against negative pressures, but it is not, in fact completely prohibitive: negative pressures can exist (although usually in metastable states, to be discussed in Part VII)—this happens for example when cavities form or substances separate from walls, etc.

11. Statistical Mechanics of Classical Monatomic Ideal Gas

We are now going to go through our first example of a statistical mechanical calculation where we start with energy levels for a system, work out Z and, therefore, F, then obtain from it the equation of state, energy and entropy as functions of temperature and volume, hence heat capacities etc.

We shall do this for a familiar system—*classical monatomic ideal gas*, for which we already know all the answers, obtained in Part II from a bespoke theory. Obtaining them again in a new way will help us convince ourselves of the soundness of the much more general formalism that we now have.

Our first objective is to calculate

$$\boxed{Z = \sum_\alpha e^{-\beta E_\alpha}}\,, \tag{11.1}$$

where $\{E_\alpha\}$ are the energy levels of our gas—i.e., of N non-interacting particles in a box of volume V—corresponding to all possible states $\{\alpha\}$ in which these particles can *collectively* find themselves. Thus, in order to compute Z, we must start by working out what are $\{\alpha\}$ and $\{E_\alpha\}$.

11.1. *Single-Particle States*

We do know what the possible states and energies are for a single particle: each of its states is characterised by its momentum $\boldsymbol{p}$ and the corresponding energy is

$$\varepsilon_{\boldsymbol{p}} = \frac{p^2}{2m}, \tag{11.2}$$

where m is the particle's mass. Classically, we might be tempted to say that the states are characterised also by the particle position $\boldsymbol{r}$, but we know from Quantum Mechanics that we cannnot know both $\boldsymbol{p}$ and $\boldsymbol{r}$ exactly. As we are considering our particle to reside in a homogeneous box, the momentum is fixed and the particle can be anywhere—in

† Note, however, a recent objection to the idea of negative temperatures: Dunkel & Hilbert (2014). This paper also has all the relevant references on the subject of negative temperatures; note that what they call "Gibbs entropy" is not the same thing as our Gibbs–Shannon entropy. If you are going to explore this literature, you may want to read §§12 and 13 first.

fact, it is a *monochromatic wave* with wave number $\boldsymbol{k} = \boldsymbol{p}/\hbar$; if the box in which it lives has dimensions $L_x \times L_y \times L_z$, the wave numbers quantise so that an integer number of periods can fit into the box:‡

$$\boldsymbol{k} = \left(\frac{2\pi}{L_x}\, i_x, \frac{2\pi}{L_y}\, i_y, \frac{2\pi}{L_z}\, i_z\right), \tag{11.3}$$

where (i_x, i_y, i_z) are integers. These triplets define a countably infinite number of single-particle states. The corresponding energy levels are

$$\varepsilon_{\boldsymbol{k}} = \frac{\hbar^2 k^2}{2m}. \tag{11.4}$$

11.2. *Down the Garden Path...*

We have N such particles in our box. Since they are non-interacting, you might think, **naïvely**, that this is a case of a composite system containing N subsystems, each with microstates (11.3) and energy levels (11.4). If this were the case, then the collective microstates and energies of the gas in a box would be

$$\alpha = \{\boldsymbol{k}_1, \ldots, \boldsymbol{k}_N\} \quad \Leftrightarrow \quad E_\alpha = \sum_{i=1}^{N} \varepsilon_{\boldsymbol{k}_i}. \tag{11.5}$$

This counting scheme will turn out to be **very wrong**, but let us explore where it leads—we will learn some useful things and later fix it without much extra work.

Under this scheme, the partition function is

$$Z = \sum_{\{\boldsymbol{k}_1,\ldots,\boldsymbol{k}_N\}} e^{-\beta(\varepsilon_{\boldsymbol{k}_1}+\cdots+\varepsilon_{\boldsymbol{k}_N})} = \underbrace{\left[\sum_{\boldsymbol{k}} e^{-\beta\varepsilon_{\boldsymbol{k}}}\right]^N}_{=Z_1} = Z_1^N, \tag{11.6}$$

where Z_1 is the *single-particle partition function*. So, if we can calculate Z_1, we are done.

11.3. *Single-Particle Partition Function*

We do this calculation by approximating $\sum_{\boldsymbol{k}}$ with an integral:

$$Z_1 = \sum_{\boldsymbol{k}} e^{-\beta\varepsilon_{\boldsymbol{k}}} = \sum_{\boldsymbol{k}} \underbrace{\frac{L_x L_y L_z}{(2\pi)^3}}_{\frac{V}{(2\pi)^3}} \underbrace{\frac{2\pi}{L_x}}_{\Delta k_x} \underbrace{\frac{2\pi}{L_y}}_{\Delta k_y} \underbrace{\frac{2\pi}{L_z}}_{\Delta k_z} e^{-\beta\varepsilon_{\boldsymbol{k}}} \approx \frac{V}{(2\pi)^3} \int \mathrm{d}^3\boldsymbol{k}\, e^{-\beta\hbar^2 k^2/2m}, \tag{11.7}$$

where we haved noticed that $\Delta k_{x,y,z} = 2\pi/L_{x,y,z}$ are the spacings between discrete points in the "grid" in $\boldsymbol{k}$ space [see Eq. (11.3)]. The continuous approximation is good as long as the typical scale of variation of k in the integrand is much larger than the '$\boldsymbol{k}$-grid spacing:

$$k \sim \sqrt{\frac{2m}{\beta\hbar^2}} = \frac{\sqrt{2mk_\mathrm{B}T}}{\hbar} \gg \Delta k_{x,y,z} = \frac{2\pi}{L_{x,y,z}} \sim \frac{2\pi}{V^{1/3}}$$
$$\Leftrightarrow \quad T \gg \frac{\hbar^2}{mk_\mathrm{B}V^{1/3}} = \frac{\hbar^2}{mk_\mathrm{B}}\left(\frac{n}{N}\right)^{1/3} = \frac{T_\mathrm{deg}}{N^{1/3}}, \tag{11.8}$$

‡ This works either if we consider the box to have reflecting boundary conditions and so require all our particles to be standing waves (see, e.g., Blundell & Blundell 2009, §21.1) or, even more straightforwardly, if less intuitively, if we use periodic boundary conditions.

where $T_{\rm deg}$ is the degeneration temperature—the lower limit to the temperatures at which the classical approximation can be used, given by Eq. (2.29). The condition (11.8) is easily satisfied, of course (because $T \gg T_{\rm deg}$ and $N \gg 1$).

The triple Gaussian integral in Eq. (11.7) is instantly calculable:

$$Z_1 = \frac{V}{(2\pi)^3}\left(\int \mathrm{d}k_x\, e^{-\beta\hbar^2 k_x^2/2m}\right)^3 = \frac{V}{(2\pi)^3}\left(\frac{2m}{\beta\hbar^2}\,\pi\right)^{3/2} = \frac{V}{\hbar^3}\left(\frac{mk_{\rm B}T}{2\pi}\right)^{3/2} \equiv \frac{V}{\lambda_{\rm th}^3}, \tag{11.9}$$

where we have introduced the *thermal wavelength*

$$\lambda_{\rm th} = \hbar\sqrt{\frac{2\pi}{mk_{\rm B}T}}, \tag{11.10}$$

a quantity that is obviously (dimensionally) convenient here, will continue to prove convenient further on and acquire a modicum of physical meaning in Eq. (11.27).

11.4. *Digression: Density of States*

When we calculate partition functions based on the canonical distribution, only microstates with different energies give different contributions to the sum over states [Eq. (11.1)], whereas microstates whose energies are the same ("degenerate" energy levels) all have the same probabilities and so contribute similarly. Therefore, we can write Z as a weighted integral over energies or over some variable that is in one-to-one correspondence with energy—in the case of energy levels of the ideal gas, $k = |\boldsymbol{k}|$ [Eq. (11.4)]. In this context there arises the quantity called the *density of states*—the number of microstates per k or per ε.

For the classical monatomic ideal gas, we can read off what it is by transforming the integration in Eq. (11.7) to polar coordinates and integrating out the angles in $\boldsymbol{k}$ space:

$$Z_1 = \frac{V}{(2\pi)^3}\int_0^\infty \mathrm{d}k\, 4\pi k^2\, e^{-\beta\hbar^2k^2/2m} \equiv \int_0^\infty \mathrm{d}k\, g(k)\, e^{-\beta\hbar^2k^2/2m}, \quad g(k) = \frac{Vk^2}{2\pi^2}, \tag{11.11}$$

where $g(k)$ is the density of states (per k). The fact that $g(k)$ grows with k says that energy levels are increasingly more degenerate as k goes up (the number of states in a spherical shell of width $\mathrm{d}k$ in $\boldsymbol{k}$ space, $g(k)\mathrm{d}k$, goes up).

Similarly, transforming the integration variable in Eq. (11.11) to $\varepsilon = \hbar^2k^2/2m$, we can write

$$Z_1 = \int_0^\infty \mathrm{d}\varepsilon\, g(\varepsilon)\, e^{-\beta\varepsilon}, \quad g(\varepsilon) = \frac{2}{\sqrt{\pi}}\frac{V}{\lambda_{\rm th}^3}\frac{\sqrt{\varepsilon}}{(k_{\rm B}T)^{3/2}}, \tag{11.12}$$

where $g(\varepsilon)$ is the density of states per ε (*not* the same fuction as $g(k)$, despite, somewhat sloppily, being denoted by the same letter).

Note that the functional form of $g(k)$ or $g(\varepsilon)$ depends on the *dimension of space*.

Exercise 11.1. Density of States in d Dimensions. a) Calculate $g(k)$ and $g(\varepsilon)$ for a classical monatomic ideal gas in d dimensions (also do the $d = 1$ and $d = 2$ cases separately and check that your general formula reduces to the right expressions in 1D, 2D and 3D).

b) Do the same calculation for an ultrarelativistic (i.e., $\varepsilon \gg mc^2$) monatomic ideal gas.

11.5. *Disaster Strikes*

Using Eqs. (11.6) and (11.9), we deduce the N-particle partition function and, therefore, the free energy:

$$Z = Z_1^N = \left(\frac{V}{\lambda_{\rm th}^3}\right)^N \quad \Rightarrow \quad F = -k_{\rm B}T\ln Z = -k_{\rm B}TN\ln\left(\frac{V}{\lambda_{\rm th}^3}\right). \tag{11.13}$$

Hence the entropy:

$$S = -\left(\frac{\partial F}{\partial T}\right)_V = k_{\rm B} N \ln\left(\frac{V}{\lambda_{\rm th}^3}\right) + \frac{3}{2} k_{\rm B} N = k_{\rm B} N \left[\ln N + \frac{3}{2} - \ln(n\lambda_{\rm th}^3)\right], \quad n = \frac{N}{V}. \tag{11.14}$$

The last expression for S is a complete and unmitigated disaster because it is *not additive!!!* Indeed, suppose we double the amount of gas: $N \to 2N$, $V \to 2V$ (density n stays constant), then

$$S_{\rm new} - 2S_{\rm old} = 2k_{\rm B} N \ln 2. \tag{11.15}$$

This is obviously unacceptable as our entire theory was built on additivity of entropy (see §10 and, indeed, §12.1, where Gibbs–Shannon entropy is additive by definition).

So what went wrong?

11.6. *Gibbs Paradox*

To debug our calculation, it is useful to consider the following famous example.

Suppose we have two isolated chambers separated by a partition, one containing gas 1, the other gas 2 (N particles of each). Remove the partition and let the gases mix. Each gas expands into vacuum (Joule expansion), so each picks up $k_{\rm B} N \ln 2$ of entropy and so†

$$\Delta S = 2k_{\rm B} N \ln 2. \tag{11.16}$$

This is certainly true if the two gases are *different*. If, on the other hand, the two gases are *the same*, surely we must have

$$\Delta S = 0 \tag{11.17}$$

because if we reinserted the partition we would be back to *status quo ante!* This inconsistency is called *the Gibbs Paradox*.

As ever, realising there is a paradox helps resolve it.

11.7. *Distinguishability*

It is now clear where the problem came from: when we counted the states of the system (§11.2), we *distinguished* between individual particles: e.g., swapping momenta [$\boldsymbol{k}_i$ and $\boldsymbol{k}_j$, assuming $\boldsymbol{k}_i \neq \boldsymbol{k}_j$, in Eq. (11.5)] between two particles would give a different microstate in our accounting scheme. In the Gibbs set up in §11.6, we got the spurious entropy increase after mixing identical gases by moving "individual" particles from one chamber to another.

In Quantum Mechanics, this problem does not arise because particles are in fact *indistinguishable* (interchanging them amounts to permuting the arguments of some big symmetric wave-function amplitude). One way of explaining this intuitively is to say that distinguishing particles amounts to pointing at them: "this one" or "that one," i.e., identifying their positions. But since their momenta are definite, their positions are in fact completely undeterminable, by the uncertainty principle: they are just waves in a box! In Part IV, you will see that in systems where individual particles are distinguishable, they are often fixed in some spatial positions (e.g., magnetisible spins in a lattice).

Thus, the microstates of a gas in a box should be designated not by lists of momenta

† Another way to derive this result is by arguing that after the partition is removed, there is additional uncertainty for each particle as to whether it ends up in chamber 1 or in chamber 2. These outcomes have equal probabilities 1/2, so the additional entropy per particle is, as per Eq. (8.7), $\Delta S_1 = -k_{\rm B}(\frac{1}{2}\ln\frac{1}{2} + \frac{1}{2}\ln\frac{1}{2}) = k_{\rm B}\ln 2$ and so, for $2N$ particles, we get Eq. (11.16).

of individual particles [Eq. (11.5)], but by

$$\alpha = \{n_{\boldsymbol{k}_1}, n_{\boldsymbol{k}_2}, n_{\boldsymbol{k}_3}, \dots\}, \quad \sum_{\boldsymbol{k}} n_{\boldsymbol{k}} = N, \tag{11.18}$$

where n_i are *occupation numbers* of the single-particle microstates: $n_{\boldsymbol{k}_1}$ particles with wave number $\boldsymbol{k}_1$, $n_{\boldsymbol{k}_2}$ particles with wave number $\boldsymbol{k}_2$, etc., up to the total of N particles. The corresponding *collective* energy levels are

$$E_\alpha = \sum_{\boldsymbol{k}} n_{\boldsymbol{k}} \varepsilon_{\boldsymbol{k}}. \tag{11.19}$$

11.8. *Correct Partition Function*

With this new counting scheme, we conclude that the N-particle partition function really is

$$Z = \sum_{\{n_{\boldsymbol{k}}\}} e^{-\beta \sum_{\boldsymbol{k}} n_{\boldsymbol{k}} \varepsilon_{\boldsymbol{k}}}, \tag{11.20}$$

where the sum is over all possible sequences $\{n_{\boldsymbol{k}}\}$ of occupation numbers, subject to $\sum_{\boldsymbol{k}} n_{\boldsymbol{k}} = N$. Calculating this sum is a somewhat tricky combinatorial problem—we will solve it in §16.2, but for our current purposes, we can use a convenient shortcut.

Suppose we are allowed to neglect all those collective microstates in which more than one particle occupies the same single-particle microstate, i.e.,

$$\text{for any } \boldsymbol{k},\ n_{\boldsymbol{k}} = 0 \text{ or } 1. \tag{11.21}$$

Then the correct, collective microstates (11.18) are the same as our old, wrong ones (11.5) ("particle 1 has wave number $\boldsymbol{k}_1$, particle 2 has wave number $\boldsymbol{k}_2$, etc."; cases where $\boldsymbol{k}_1$, $\boldsymbol{k}_2$, ... are not different are assumed to contribute negligibly to $\sum_\alpha$ in the partition function), except the order in which we list the particles ought not to matter. Thus, we must correct our previous formula for Z, Eq. (11.6), to eliminate the overcounting of the microstates in which the particles were simply permuted—as the particles are indistinguishable, these are in fact *not* different microstates. The necessary correction is, therefore,†

$$Z = \frac{Z_1^N}{N!}. \tag{11.22}$$

Using Eq. (11.9), we have for the classical monatomic ideal gas,

$$Z = \frac{1}{N!}\left(\frac{V}{\lambda_{\text{th}}^3}\right)^N, \quad \lambda_{\text{th}} = \hbar\sqrt{\frac{2\pi}{mk_{\text{B}}T}}. \tag{11.23}$$

Before we use this new formula to calculate everything, let us assess how good the assumption (11.21) is. In order for it to hold, we need that

$$\begin{matrix}\text{number of available}\\ \text{single-particle states}\end{matrix} \quad \gg \quad \begin{matrix}\text{number of}\\ \text{particles } N.\end{matrix} \tag{11.24}$$

† The old formula, $Z = Z_1^N$, is still fine for systems consisting of distinguishable elementary units. It might not be immediately obvious why the validity of the corrected formula (11.22) is restricted to the case (11.21), but breaks down if there are many multiply occupied states. The reason is that our original counting scheme (11.5) distinguished between cases such as "particle 1 has wave number $\boldsymbol{k}_1$, particle 2 has wave number $\boldsymbol{k}_2$, ..." vs. "particle 1 has wave number $\boldsymbol{k}_2$, particle 2 has wave number $\boldsymbol{k}_1$, ..." when $\boldsymbol{k}_1 \neq \boldsymbol{k}_2$—this was wrong and is corrected by the $N!$ factor, which removes all permutations of the particles; however, the scheme (11.5) did *not* distinguish between such cases for $\boldsymbol{k}_1 = \boldsymbol{k}_2$ and so, if they were present in abundance, the factor $N!$ would overcorrect.

The single-particle partition function, Eq. (11.9), gives a decent estimate of the former quantity because the typical energy of the system will be $\varepsilon_{\boldsymbol{k}} \sim k_{\rm B}T$ and the summand in

$$Z_1 = \sum_{\boldsymbol{k}} e^{-\varepsilon_{\boldsymbol{k}}/k_{\rm B}T} \tag{11.25}$$

stays order unity roughly up to this energy, so the sum is simply of order of the number of microstates in the interval $\varepsilon_{\boldsymbol{k}} \lesssim k_{\rm B}T$.† Then the condition (11.26) becomes

$$\frac{V}{\lambda_{\rm th}^3} \gg N \quad \Leftrightarrow \quad \boxed{n\lambda_{\rm th}^3 \ll 1}\,. \tag{11.26}$$

Another popular way of expressing this condition is by stating that the number density of the particles must be much smaller than the "quantum concentration" $n_{\rm Q}$:

$$n \ll n_{\rm Q} \equiv \frac{1}{\lambda_{\rm th}^3}. \tag{11.27}$$

Physically, the quantum concentration is the number of single-particle states per unit volume (this is meaningful because the number of states is an extensive quantity: in larger volumes, there are more wave numbers available, so there are more states).

The condition (11.27) is actually the condition for the *classical limit* to hold, $T \gg T_{\rm deg}$ (see §2.29), guaranteeing the absence of quantum correlations (which have to do with precisely the situation that we wish to neglect: more than one particle trying to be in the same single-particle state; see Part VI). When $n \sim n_{\rm Q}$ or larger, we can no longer use Eq. (11.22) and are in the realm of *quantum gases*. Substituting the numbers, which, e.g., for air at 1 atm and room temperature, gives $n \sim 10^{25}$ m^{-3} vs. $n_{\rm Q} \sim 10^{30}$ m^{-3}, will convince you that we can usefully stay out of that realm for a little longer.

11.9. *Thermodynamics of Classical Ideal Gas*

Finally, let us use Eq. (11.23), to extract the thermodyamics of ideal gas. The free energy is

$$\begin{aligned} F &= -k_{\rm B}T \ln Z = -k_{\rm B}T \left[N \ln\left(\frac{V}{\lambda_{\rm th}^3}\right) - \ln N! \right] \\ &\approx -k_{\rm B}T \left[\cancel{N\ln N} - N\ln(n\lambda_{\rm th}^3) - (\cancel{N\ln N} - N) \right] = -k_{\rm B}TN \left[1 - \ln(n\lambda_{\rm th}^3) \right], \end{aligned} \tag{11.28}$$

where, upon application of Stirling's formula, the non-additive terms have happily cancelled.

The entropy is, therefore,

$$\boxed{S = -\left(\frac{\partial F}{\partial T}\right)_V = k_{\rm B}N \left[\frac{5}{2} - \ln(n\lambda_{\rm th}^3) \right]}, \tag{11.29}$$

the formula known as the *Sackur–Tetrode Equation.* It is nice and additive, no paradoxes.

The mean energy of the gas is

$$U = F + TS = \frac{3}{2}\, k_{\rm B}TN, \tag{11.30}$$

the same as the familiar Eq. (2.23), and hence the heat capacity

$$C_V = \left(\frac{\partial U}{\partial T}\right)_V = \frac{3}{2}\, k_{\rm B}N, \tag{11.31}$$

† In other words, using Eq. (11.12), the number of states that are not exponentially unlikely is $\sim \int_0^{k_{\rm B}T} {\rm d}\varepsilon\, g(\varepsilon) \sim V/\lambda_{\rm th}^3$.

the same as Eq. (2.24).

NB: This formula is for *monatomic* gases. In Part IV, you will learn how to handle diatomic gases, where molecules can have additional energy levels due rotational and vibrational degrees of freedom.

Finally, the equation of state:

$$P = -\left(\frac{\partial F}{\partial V}\right)_T = k_{\mathrm{B}}T\,\frac{N}{V} = nk_{\mathrm{B}}T, \tag{11.32}$$

the same as Eq. (2.19).

NB: The only property of the theory that matters for the equation of state is the fact that $Z \propto V^N$, so neither (in)distinguishability nor the precise form of the single-particle energy levels [Eq. (11.4)] affect the outcome—this will only change when particles start crowding each other out of parts of the volume, as happens for "real" gases (Part VII), or parts of phase space, as happens for quantum ones (Part VI).

Thus, we have recovered from Statistical Mechanics the same thermodynamics for the ideal gas as was constructed empirically in Part I or kinetically in Part II. Note that Eqs. (11.30) and (11.32) constitute the proof that *the kinetic temperature [Eq. (2.20)] and kinetic pressure [Eq. (1.26)] are the same as the statistical mechanical temperature* ($= 1/k_{\mathrm{B}}\beta$) *and statistical mechanical pressure [Eq. (7.13)].*

Exercise 11.2. Adiabatic Law. Using the Sackur–Tetrode equation, show that for a classical monatomic ideal gas undergoing an adiabatic process,

$$PV^{5/3} = \text{const.} \tag{11.33}$$

Exercise 11.3. Relativistic Ideal Gas. a) Show that the equation of state of an ideal gas is still

$$PV = Nk_{\mathrm{B}}T \tag{11.34}$$

even when the gas is heated to such a high temperature that the particles are moving at relativistic speeds. Why is it unchanged?

b) Although the equation of state does not change if particles in a classical ideal gas move at relativistic speeds, show, by explicit calculation of the expression for the entropy, that in the ultrarelativistic limit (i.e., the limit in which the rest energy of the particles is negligible compared to their kinetic energy), the formula for an adiabat is

$$PV^{4/3} = \text{const.} \tag{11.35}$$

c) Show that the pressure of an ultrarelativistic monatomic ideal gas is

$$P = \frac{\varepsilon}{3}, \tag{11.36}$$

where ε is the internal energy density. Is it different than for a nonrelativistic gas? Why?

11.10. *Maxwell's Distribution*

Can we recover from Statistical Mechanics not just the thermodynamical quantities associated with the ideal gas, but also Maxwell's distribution itself? Certainly: the *particle distribution function* as we defined it in Part II is directly related to the *mean occupation number*. Indeed,

$f(\boldsymbol{v})\mathrm{d}^3\boldsymbol{v}$ = mean fraction of particles in microstates with velocities $[\boldsymbol{v}, \boldsymbol{v} + \mathrm{d}^3\boldsymbol{v}]$,

whereas

$\langle n_{\boldsymbol{k}} \rangle$ = mean number of particles in the microstate with wave number $\boldsymbol{k} = m\boldsymbol{v}/\hbar$.

Therefore,

$$f(\boldsymbol{v})\mathrm{d}^3\boldsymbol{v} = \frac{\langle n_{\boldsymbol{k}}\rangle}{N}\frac{V}{(2\pi)^3}\mathrm{d}^3\boldsymbol{k} = \frac{\langle n_{\boldsymbol{k}}\rangle}{n}\left(\frac{m}{2\pi\hbar}\right)^3\mathrm{d}^3\boldsymbol{v} \quad \Rightarrow \quad f(\boldsymbol{v}) = \left(\frac{m}{2\pi\hbar}\right)^3\frac{\langle n_{\boldsymbol{k}}\rangle}{n}. \tag{11.37}$$

We are not yet ready to calculate $\langle n_{\boldsymbol{k}}\rangle$ from Statistical Mechanics—we will do this in §16.3, but in the meanwhile, the anticipation of the Maxwellian $f(\boldsymbol{v})$ tells us what the result ought to be:

$$\langle n_{\boldsymbol{k}}\rangle = n\left(\frac{2\pi\hbar}{m}\right)^3\frac{e^{-mv^2/2k_{\mathrm{B}}T}}{(2\pi k_{\mathrm{B}}T/m)^{3/2}} = (n\lambda_{\mathrm{th}}^3)\,e^{-\beta\varepsilon_{\boldsymbol{k}}}. \tag{11.38}$$

We shall verify this formula in due course (see §16.6).

NB: It is a popular hand-waving shortcut to argue that Maxwell's distribution is the Gibbs distribution for one particle—a system in thermal contact (via collisions) with the rest of the particles, forming the heat bath and thus determining the particle's mean energy.

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12. P.S. More About Entropy

I have tried in the foregoing to take us as quickly as possible from the (perhaps somewhat murky) conceptual underpinnings of the Statistical Mechanics to a state of operational clarity as to how we would compute things. More things will be computed in Part IV.

Fortunately (or sadly), you do not need to *really* understand why maximising the expression $-\sum_\alpha p_\alpha \ln p_\alpha$ works or what it *really* means—you can simply embrace the straightforward Gibbs prescription:

1) compute Z from knowledge of $\{\alpha\}$ and $\{E_\alpha\}$,
2) compute $F = -k_{\mathrm{B}}T\ln Z$,
3) compute P, S, U from that (and C_V, usually),
4) move on to the next problem (in Statistical Mechanics or in life).

From this utilitarian viewpoint, my task of introducing the "Fundamentals of Statistical Mechanics" is complete. Nevertheless, in this section, I wish to discuss the notion of entropy and the meaning of $\{p_\alpha\}$ a little more and also to survey some alternative schemes for setting up Statistical Mechanics (they all eventually lead to the same practical prescriptions). This is for those of you who wish to make sense of the formalism and be convinced that we are on firm ground, intellectually—or are we?

12.1. *Shannon's Theorem*

[Literature: Shannon (1948), §6; Jaynes (2003), §11.3; Binney & Skinner (2013), §6.3.2]

In §8.1.3, I argued that, in order to achieve the "fairest" and most unbiased assignment of probabilities p_α to microstates α, one must maximise the function

$$S_{\mathrm{G}}(p_1,\ldots,p_\Omega) = -\sum_\alpha p_\alpha \ln p_\alpha \tag{12.1}$$

(called Gibbs entropy, Shannon entropy, "information entropy," measure of uncertainty, etc.). I did this by presenting a reasonable and practical scheme for assigning probabilities, which I asked you to agree was the fairest imaginable. In the spirit of formalistic nit-picking, you might be tempted to ask whether the function (12.1) is in any sense unique—could we have invented other "fair games" leading to different definitions of entropy? Here is an argument that addresses this question.

Faced with some set of probabilities $\{p_\alpha\}$ ("a distribution"), let us seek to define a function

$H(p_1, \ldots, p_\Omega)$ that would measure the uncertainty associated with this distribution. In order to be a suitable such measure, H must satisfy certain basic properties:

1) H should be a continuous fuction of p_α's (i.e., changing p_α's a little should not dramatically change the measure of uncertainty associated with them);

2) H should be symmetric with respect to permutations of $\{p_\alpha\}$ (i.e., it should not matter in what order we list the microstates);

3) for any set of probabilities $\{p_\alpha\}$ that are not all equal,

$$H(p_1, \ldots, p_\Omega) < H\left(\frac{1}{\Omega}, \ldots, \frac{1}{\Omega}\right) \equiv H_\Omega \tag{12.2}$$

(the distribution with all equal probabilities corresponds to maximum uncertainty);

4) if $\Omega' > \Omega$, $H_{\Omega'} > H_\Omega$ (more equiprobable microstates $\Rightarrow$ more uncertainty);

5) H should be *additive* and *independent of how we count the microstates*, in the following sense. If the choice of a microstate is broken down into two successive choices—first a subgroup, then the individual state—the total H should be a weighted sum of individual values of H associated with each subgroup. Namely, split the microstates into groups:

$$\alpha = \underbrace{1, \ldots, m_1}_{\substack{\text{group} \\ i=1 \\ \downarrow \\ \text{probability} \\ w_1}} \underbrace{, m_1+1, \ldots, m_1+m_2}_{\substack{\text{group} \\ i=2 \\ \downarrow \\ \text{probability} \\ w_2}} , \ldots, \underbrace{\sum_{i=1}^{M-1} m_i + 1, \ldots, \sum_{i=1}^{M} m_i = \Omega}_{\substack{\text{group} \\ i=M \\ \downarrow \\ \text{probability} \\ w_M}} . \tag{12.3}$$

Clearly, w_i is the sum of p_α's for the states that are in the group i. Within each group, we can assign *conditional* probabilities to all microstates in that group, viz., the probability for the system to be in microstate α within group i if it is given that the system is in one of the microstates in that group, is

$$p_\alpha^{(i)} = \frac{p_\alpha}{w_i}. \tag{12.4}$$

We then want H to satisfy

$$\begin{aligned} \underbrace{H(p_1, \ldots, p_\Omega)}_{\substack{\text{total} \\ \text{uncertainty}}} &= \underbrace{H(w_1, \ldots, w_M)}_{\substack{\text{uncertainty in} \\ \text{the} \\ \text{distribution} \\ \text{of groups}}} + w_1 \underbrace{H(p_1^{(1)}, \ldots, p_{m_1}^{(1)})}_{\substack{\text{uncertainty} \\ \text{within} \\ \text{group 1}}} + w_2 \underbrace{H(p_{m_1+1}^{(2)}, \ldots, p_{m_1+m_2}^{(2)})}_{\substack{\text{uncertainty} \\ \text{within} \\ \text{group 2}}} + \ldots \\ &= H(w_1, \ldots, w_M) + w_1 H\left(\frac{p_1}{w_1}, \ldots, \frac{p_{m_1}}{w_1}\right) + w_2 H\left(\frac{p_{m_1+1}}{w_2}, \ldots, \frac{p_{m_1+m_2}}{w_2}\right) + \ldots \end{aligned} \tag{12.5}$$

Theorem. The only function H with these properties is

$$H(p_1, \ldots, p_\Omega) = -k \sum_\alpha p_\alpha \ln p_\alpha, \tag{12.6}$$

where $k > 0$ is a constant.

Proof. Let us first consider a special case of equal probabilities:

$$\text{all } p_\alpha = \frac{1}{\Omega} \quad \Rightarrow \quad w_i = \frac{m_i}{\Omega}, \quad p_\alpha^{(i)} = \frac{p_\alpha}{w_i} = \frac{1}{m_i}. \tag{12.7}$$

Then the criterion (12.5) becomes

$$H_\Omega \equiv H\left(\frac{1}{\Omega},\ldots,\frac{1}{\Omega}\right) = H(w_1,\ldots,w_M) + \sum_{i=1}^{M} w_i \underbrace{H\left(\frac{1}{m_i},\ldots,\frac{1}{m_i}\right)}_{\equiv H_{m_i}}. \tag{12.8}$$

Therefore,

$$H(w_1,\ldots,w_M) = H_\Omega - \sum_{i=1}^{M} w_i H_{m_i}. \tag{12.9}$$

Now consider the special case of this formula for the situation in which all $m_i = m$ are the same. Then

$$\Omega = mM, \quad w_i = \frac{m}{\Omega}, \tag{12.10}$$

and Eq. (12.9) becomes

$$H_M = H_{mM} - H_m. \tag{12.11}$$

This is a functional equation for $H_m \equiv f(m)$:

$$f(mn) = f(m) + f(n). \tag{12.12}$$

Lemma. The only monotonically increasing† function that satisfies Eq. (12.12) is

$$f(m) = k \ln m, \tag{12.13}$$

where k is a positive constant.

Proof. For any integers $m, n > 1$, we can always find integers r and (an arbitrarily large) s such that

$$\frac{r}{s} < \frac{\ln m}{\ln n} < \frac{r+1}{s} \quad \Rightarrow \quad n^r < m^s < n^{r+1}. \tag{12.14}$$

As f is a monotonically increasing function,

$$f(n^r) < f(m^s) < f(n^{r+1}). \tag{12.15}$$

But Eq. (12.12) implies $f(n^r) = rf(n)$, so the above inequality becomes

$$rf(n) < sf(m) < (r+1)f(n) \quad \Rightarrow \quad \frac{r}{s} < \frac{f(m)}{f(n)} < \frac{r+1}{s}. \tag{12.16}$$

The inequalities (12.14) and (12.16) imply

$$\left|\frac{f(m)}{f(n)} - \frac{\ln m}{\ln n}\right| < \frac{1}{s} \quad \Rightarrow \quad \left|\frac{f(m)}{\ln m} - \frac{f(n)}{\ln n}\right| < \frac{1}{s}\frac{f(n)}{\ln m} \to 0 \tag{12.17}$$

because s can be chosen arbitrarily large. Therefore

$$\frac{f(m)}{\ln m} = \frac{f(n)}{\ln n} = \text{const} = k, \quad \text{q.e.d.} \tag{12.18}$$

The constant is positive, $k > 0$, because $f(m)$ is supposed to be *in*creasing.

Thus, we have proven

$$H_\Omega = k \ln \Omega. \tag{12.19}$$

Substituting this into Eq. (12.9), we get

$$H(w_1,\ldots,w_M) = k\left(\ln\Omega - \sum_{i=1}^{M} w_i \ln m_i\right) = -k\sum_{i=1}^{M} w_i \ln\frac{m_i}{\Omega} = -k\sum_{i=1}^{M} w_i \ln w_i. \tag{12.20}$$

But $\{m_i\}$ and, therefore, $\{w_i\}$, were chosen in a completely general way, subject only to $\sum_i m_i =$

† Which it must be because we need $H_{\Omega'} > H_\Omega$ for $\Omega' > \Omega$; see condition 4 on the H function.

Ω, or $\sum_i w_i = 1$. Therefore, with equal validity,‡

$$H(p_1, \ldots, p_\Omega) = -k \sum_\alpha p_\alpha \ln p_\alpha, \quad \text{q.e.d.} \tag{12.21}$$

Choosing $k = 1$ gives us $H = S_\mathrm{G}$, which we called the Gibbs (or Gibbs–Shannon) entropy (§8.1.3); $k = k_\mathrm{B}$ gives $H = S$, the conventional thermodynamical entropy (in thermal equilibrium); $k = 1/\ln 2$ is the convention for Shannon entropy as used in measuring information content.

12.2. *Boltzmann Entropy and the Ensembles*

12.2.1. *Boltzmann's Formula*

Ludwig Boltzmann's tombstone has a famous formula carved into it:

$$\boxed{S = k \log W}\,, \tag{12.22}$$

the "Boltzmann entropy," where k is a constant (technically speaking, arbitrary $k > 0$, but traditionally k_B, introduced, by the way, by Planck, not Boltzmann) and W is "the number of complexions." What does this mean and how does it relate to what we have discussed so far?

A "complexion" is the same thing as I have so far referred to as a "state" (α). So, Boltzmann's formula is for the entropy of a system in which all states/complexions are equiprobable (this is where the "equal *a priori* probabilities postulate," first mentioned in §8.1.1, comes in), so it is simply the same expression as we found for the case of all $p_\alpha = 1/\Omega$: then Eq. (12.22) has $W = \Omega$ and is the same as the familiar expression

$$S = k_\mathrm{B} \ln \Omega. \tag{12.23}$$

Boltzmann introduced his entropy following somewhat similar logic as we discussed for the Gibbs–Shannon entropy in §12.1: he wanted a function of Ω (the number of equiprobable states) that would be

1) larger for a larger number of states, viz., $S(\Omega') > S(\Omega)$ for $\Omega' > \Omega$,

2) additive for several systems when they are put together (an essential property, as we saw in §10), i.e., since the number of states in a combined system is $\Omega_{12} = \Omega_1 \Omega_2$, we must have

$$S(\Omega_1 \Omega_2) = S(\Omega_1) + S(\Omega_2). \tag{12.24}$$

The proof that *the only such function is given by Eq. (12.23)* is the proof of the Lemma within the proof of Shannon's Theorem in §12.1.† Thus, Boltzmann's entropy simply appears to be a particular case of the Gibbs–Shannon entropy for isolated systems (systems with equiprobable states).

In fact, as we will see in §§12.2.2 and 12.2.3, it is possible to turn the argument around and get the Gibbs entropy (and the Gibbs distribution) from the Boltzmann entropy.

‡ Technically speaking, we have only obtained this formula for p_α's (or w_i's) that are rational numbers. This is OK: if p_α's are irrational, they can be approximated arbitrarily well by rationals and so H still has to be given by Eq. (12.21) because H must be continuous according to criterion 1 imposed on it at the beginning of this section.

† Thus, the uniqueness of the Gibbs–Shannon entropy as an adequate measure of uncertainty for a general probability distribution $\{p_\alpha\}$ is a corollary of the uniqueness of the Boltzmann entropy as an adequate measure of uncertainty for equiprobable outcomes. If you compare the proof of Shannon's theorem in §12.1 with the scheme for getting Gibbs entropy from Boltzmann entropy given at the end of §12.2.2, you will see the connection quite clearly.

12.2.2. *Microcanonical Ensemble*

This is an opportune moment to outline yet another way in which Statistical Mechanics and Thermodynamics can be constructed (as indeed they are, in the majority of textbooks). Effectively, this is an attempt to impart a veneer of "objective reality" to the foundations of the subject, which, in the way I have so far presented them, have perhaps been uncomfortably dependent on such seemingly subjective notions as the observer's information about the system.

Under this new scheme, we start by considering *a completely isolated system* and postulate (as we did in §8.1.1) that *all its microstates are equiprobable.*

Since the system is isolated, its energy is exactly conserved, so those microstates are, in fact, not *all* possible ones, but only those whose energies are equal to the exact energy of the system: $E_\alpha = E$. Their probabilities are then

$$\boxed{p_\alpha = \begin{cases} \dfrac{1}{\Omega(E)} & \text{if } E_\alpha = E, \\ \\ 0 & \text{otherwise,} \end{cases}} \tag{12.25}$$

where $\Omega(E)$ is the total number of microstates with $E_\alpha = E$. This distribution is called *microcanonical* and the underlying ensemble of the realisations of the system, *the microcanonical ensemble.*† The entropy of this distribution is the Boltzmann entropy (§12.2.1):

$$\boxed{S = k_\mathrm{B} \ln \Omega(E)}\,. \tag{12.26}$$

Now, to get the canonical (Gibbs) distribution from Eq. (12.25), pick a small part of the system and ask what is the probability for it to have energy ϵ ($\ll E$)? Using Eq. (12.26) and denoting by

$\Omega_\mathrm{part}(\epsilon)$ the number of microstates of the small part of the system that have energy ϵ,

$\Omega_\mathrm{res}(E-\epsilon)$ the number of microstates of the rest of the system (the reservoir, the heat bath; cf. §10.3) that have energy $E-\epsilon$,

we can express the desired probability as follows:

$$\begin{aligned} p(\epsilon) = \frac{\Omega_\mathrm{part}(\epsilon)\Omega_\mathrm{res}(E-\epsilon)}{\Omega(E)} &= \frac{\Omega_\mathrm{part}(\epsilon)}{\Omega(E)}\,\exp\left\{\frac{S_\mathrm{res}(E-\epsilon)}{k_\mathrm{B}}\right\} \\ &\approx \frac{\Omega_\mathrm{part}(\epsilon)}{\Omega(E)}\,\exp\left\{\frac{1}{k_\mathrm{B}}\left[S_\mathrm{res}(E) - \epsilon\,\underbrace{\frac{\partial S_\mathrm{res}}{\partial E}}_{=1/T} + \dots\right]\right\} \\ &= \underbrace{\frac{e^{S_\mathrm{res}(E)/k_\mathrm{B}}}{\Omega(E)}}_{\substack{\text{norm.}\\ \text{constant}}}\,\Omega_\mathrm{part}(\epsilon)\,e^{-\epsilon/k_\mathrm{B}T}, \end{aligned} \tag{12.27}$$

where T is, *by definition*, the temperature of the reservoir. The prefactor in front of

† This can be generalised in a straightforward fashion to take into account the fact that an isolated system will also conserve its linear and angular momentum. In fact, it is possible to show that *in steady state* (and so, in equilibrium), the distribution can only be a function of globally conserved quantities. As it is usually possible to consider the system in a frame in which it is at rest, E is what matters in Statistical Mechanics (see, e.g., Landau & Lifshitz 1980, §4).

this distribution is independent of ϵ and can be found by normalisation. Thus, we have obtained a variant of the Gibbs distribution (also known as the Boltzmann distribution):

$$p(\epsilon) = \frac{\Omega_{\rm part}(\epsilon)\, e^{-\epsilon/k_{\rm B}}}{Z}, \quad Z = \sum_{\epsilon} \Omega_{\rm part}(\epsilon)\, e^{-\epsilon/k_{\rm B}T}, \tag{12.28}$$

where the normalisation constant has been cast in the familiar form of a partition function, Z. The reason this formula, unlike Eq. (9.8), has the prefactor $\Omega(\epsilon)$ is that this is the probability for the system to have the energy ϵ, not to occupy a particular single state α. Many such states can have the same energy ϵ (to be precise, $\Omega(\epsilon)$ of them will), all with the same probability, so we recover the more familiar formula as follows: for α such that the energy of the subsystem is $E_\alpha = \epsilon$,

$$p_\alpha = \frac{p(\epsilon)}{\Omega_{\rm part}(\epsilon)} = \frac{e^{-\epsilon/k_{\rm B}T}}{Z} = \frac{e^{-\beta E_\alpha}}{Z}, \quad Z = \sum_{\alpha} e^{-\beta E_\alpha}. \tag{12.29}$$

We are done now as we can again calculate everything from this: energy via the usual formula

$$U = -\frac{\partial \ln Z}{\partial \beta} \tag{12.30}$$

and entropy either by showing, as in §9.2, that

$$\mathrm{d}Q_{\rm rev} = \mathrm{d}U + P\mathrm{d}V = T\mathrm{d}\left(\frac{U}{T} + k_{\rm B}\ln Z\right) = T\mathrm{d}S_{\rm part}, \tag{12.31}$$

so T is the thermodynamic temperature and

$$S_{\rm part} = \frac{U}{T} + k_{\rm B}\ln Z \tag{12.32}$$

the thermodynamic entropy of the small subsystem in contact with a reservoir of tempearture T,

or by generalising Boltzmann's entropy in a way reminiscent of the requirement of additivity and indepence of the state-counting protocol (criterion 5 in §12.1). Namely, if we demand

$$\underbrace{S = k_{\rm B}\ln\Omega(E)}_{\substack{\text{total entropy of}\\ \text{isolated system}}} = \underbrace{S_{\rm part}}_{\substack{\text{entropy of}\\ \text{small part}}} + \underbrace{\langle S_{\rm res}(E-\epsilon)\rangle}_{\substack{\text{mean entropy}\\ \text{of reservoir}\\ \text{(over all } \epsilon)}}, \tag{12.33}$$

then

$$\begin{aligned} S_{\rm part} &= k_{\rm B}\ln\Omega(E) - \sum_{\epsilon} p(\epsilon)\, \underbrace{k_{\rm B}\ln\Omega_{\rm res}(E-\epsilon)}_{S_{\rm res}(E-\epsilon)} \\ &= -k_{\rm B}\sum_{\epsilon} \underbrace{p(\epsilon)}_{=\Omega(\epsilon)p_\alpha} \ln \underbrace{\left[\frac{\Omega_{\rm res}(E-\epsilon)}{\Omega(E)}\right]}_{\substack{=p(\epsilon)/\Omega_{\rm part}(\epsilon)\\ =p_\alpha}} = -k_{\rm B}\sum_{\alpha} p_\alpha \ln p_\alpha, \end{aligned} \tag{12.34}$$

and we have thus recovered the Gibbs entropy.

Let me reiterate an important feature of this approach: the microcanonical temperature

was formally defined [see Eq. (12.27)] via the dependence of (Boltzmann) entropy on (exact) energy:

$$\boxed{\frac{1}{T}=\frac{\partial S}{\partial E}}\,. \tag{12.35}$$

This quantity can then be given physical meaning in two (very similar) ways:

either we can repeat the argument of §10.2 replacing mean energies U_1, U_2, U with exact energies E_1, E_2, E and maximising the Boltzmann entropy of two conjoint systems to show that in equilibrium the quantity T defined by Eq. (12.35) must equalise between them—and thus T is a good definition of temperature.

or we note that T defined via Eq. (12.35) is the width of the distribution $p(\epsilon)$ [Eq. (12.28)] and hence enters Eq. (12.31)—thus, $1/T$ is manifestly the integrating factor of reversible heat, so T is the thermodynamic temperature (same argument as in §9.2).

12.2.3. *Alternative Construction of the Canonical Ensemble*

[Literature: Schrödinger (1990)]

Finally, let me outline yet another scheme for constructing the Gibbs canonical ensemble.

Recall that in §8.1.3 we assigned $\mathcal{N}$ "quanta of probability" to Ω microstates in a "fair and balanced" fashion and found that the number of ways in which any particular set of probabilities $p_\alpha = \mathcal{N}_\alpha/\mathcal{N}$ could be obtained was [Eq. (8.4)]

$$W=\frac{\mathcal{N}!}{\mathcal{N}_1!\cdots\mathcal{N}_\Omega!}; \tag{12.36}$$

the entropy then was simply [Eq. (8.7)]

$$S=\frac{k_\mathrm{B}\ln W}{\mathcal{N}}=-k_\mathrm{B}\sum_\alpha p_\alpha\ln p_\alpha \tag{12.37}$$

(with all $\mathcal{N}, \mathcal{N}_\alpha \to \infty$ while keeping $\mathcal{N}_\alpha/\mathcal{N} = p_\alpha = \text{const}$). This was justified as a counting scheme: larger W gave the more probable assignment of p_α's and then it was convenient to take a log to make S additive. At that stage, I presented this scheme simply as a "reasonable" procedure. In §12.1, I removed the need for us to believe in its "reasonableness" by showing that the Gibbs-Shannon expression for S was in some sense the uniquely suitable choice.

This is *not*, in fact, how the Gibbs construction has traditionally been thought about (e.g., by Gibbs—or by Schrödinger 1990, who has a very cogent explanation of the more traditional approach in his lectures). Rather,

- one makes $\mathcal{N}$ mental copies of the system one is interested in and joins them together into one *isolated* über-system (an "ensemble"). The states ("complexions") of this über-system are characterised by

  $\mathcal{N}_1$ copies of the original system being in the state $\alpha = 1$,
  $\mathcal{N}_2$ copies of the original system being in the state $\alpha = 2$,
  ...
  $\mathcal{N}_\Omega$ copies of the original system being in the state $\alpha = \Omega$,

  and so the number of all possible such über-states is W, given by Eq. (12.36).
- Since the über-system is isolated, all these states are equiprobable and the entropy of the über-system is the Boltzmann entropy,

$$S_\mathcal{N}=k_\mathrm{B}\ln W, \tag{12.38}$$

which, if maximised, will give the most probable über-state—this is the equilibrium state of the

über-system. Maximising entropy *per system*,

$$S = \frac{S_{\mathcal{N}}}{\mathcal{N}}, \tag{12.39}$$

which is the same as Gibbs entropy (12.37), is equivalent to maximising $S_{\mathcal{N}}$.

- If $\{N_1, \ldots, N_\Omega\}$ is the most probable über-state, then

$$p_\alpha = \frac{\mathcal{N}_\alpha}{\mathcal{N}} \tag{12.40}$$

are the desired probabilities of the microstates in which one might find a copy of the original system if one picks it randomly from the über-system (the ensemble).

- To complete this construction, one proves that the fluctuations around the most probable über-state vanish as $\mathcal{N} \to \infty$, which is always a good limit because $\mathcal{N}$ is in our head and so can be chosen arbitrarily large (for details, see Schrödinger 1990, Chapter V, VI).

Recall that to get the canonical (Gibbs) distribution (§9.1), we maximised Gibbs entropy [Eq. (12.39), or Eq. (12.37)] subject to fixed *mean energy*

$$\sum_\alpha p_\alpha E_\alpha = U. \tag{12.41}$$

In view of Eq. (12.40), this is the same as

$$\sum_\alpha \mathcal{N}_\alpha E_\alpha = \mathcal{N}U = \mathcal{E}, \tag{12.42}$$

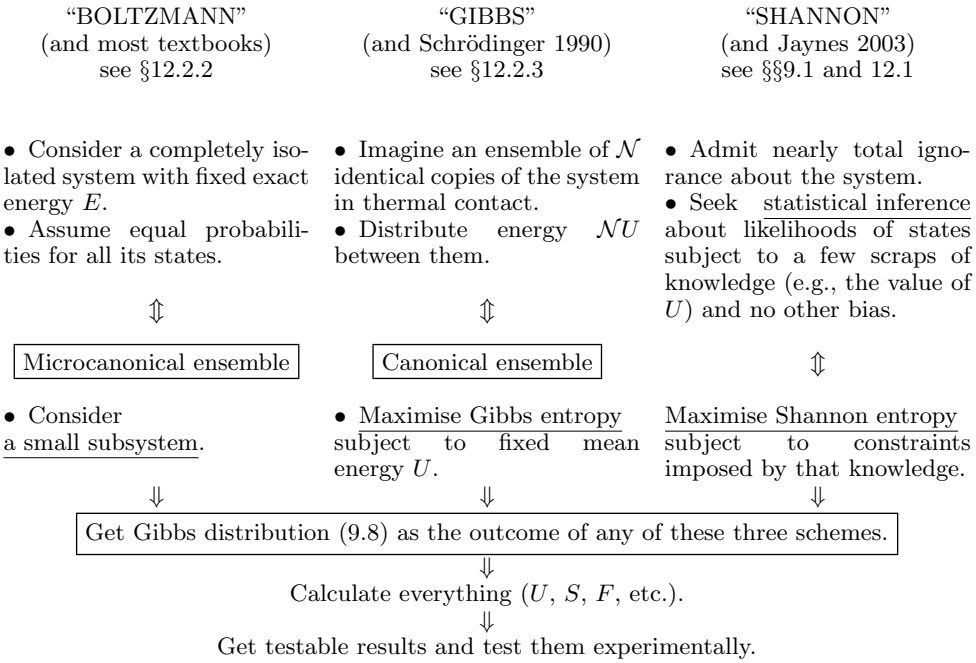
i.e., the (isolated) über-system has the *exact* total energy $\mathcal{E} = \mathcal{N}U$. Thus, *seeking the equilibrium of a system at fixed mean energy U (or, equivalently/consequently, temperature) is the same as seeking the most likely way in which exact energy $\mathcal{N}U$ would distribute itself between very many, $\mathcal{N} \gg 1$, copies of the system, if they were all in thermal contact with each other and isolated from the rest of the world.*

Thus, the canonical ensemble of Gibbs, if interpreted in terms of one "über-system" containing $\mathcal{N}$ copies of the original system with exact total energy $\mathcal{N}U$ is basically a case of microcanonical distribution being applied to this (imaginary) assemblage.

Clearly, a system with mean energy U inside our über-system is a case of *a system in contact with a heat bath* (see §10.3)—in the above construction, the bath is a strange one, as it is made of $\mathcal{N} - 1$ copies of the system itself, but that does not matter because the nature of the heat bath does not matter—what does matter is only the value of the temperature (or, equivalently, mean energy) that it sets for the system.

12.3. *Gibbs vs. Boltzmann and the Meaning of Probabilities*

Let us summarise the three main schemes for the construction of Statistical Mechanics and Thermodynamics that we have learned.



The "Gibbs" and "Shannon" schemes really are versions of one another: whereas the language is different, both the mathematics (cf. §8.1.3sec:ueber) and the philosophy (probabilities as *likelihoods* of finding the system of interest in any given microstates) are the same (one might even argue that the "Shannon" construction is what Gibbs really had in mind). So I will refer to this entire school of thought as "Gibbsian" (perhaps "Gibbsian heresy").

The Boltzmann scheme ("Boltzmannite orthodoxy") is philosophically different: we are invited to think of every step in the construction as describing some form of objective reality, whereas under the Gibbsian approach, we are effectively just trying to come up with the best possible guess, given limited information.

The reality of the Boltzmannite construction is, however, somewhat illusory:

1) *An isolated system with a fixed energy is a fiction:*
—it is impossible to set up practically;
—if set up, it is inaccessible to measurement (because it is isolated!).
So it is in fact just as imaginary as, say, the Gibbsian ensemble of $\mathcal{N}$ identical systems.

2) *What is the basis for assuming equal probabilities?*
The usual view within this school of thought is as follows. As the isolated system in question evolves in time, it samples (repeatedly) its entire phase space—i.e., it visits all possible microstates consistent with its conservation laws ($E = \text{const}$). Thus, *the probability for it to be in any given microstate or set of microstates is simply the fraction of time that it spends in those states.* In other words, *time averages* of any quantities of interest are equal to the *statistical averages*, i.e., the averages over all microstates:

$$\lim_{t\to\infty}\frac{1}{t}\int_0^t \mathrm{d}t'(\text{quantity})(t') = \sum_\alpha p_\alpha(\text{quantity})_\alpha. \tag{12.43}$$

This last statement is known as the *ergodic hypothesis*. To be precise, the assumption is that

$$\begin{pmatrix}\text{time spent in any subset of}\\ \text{microstates}\\ \text{(subvolume of phase space)}\end{pmatrix} = \frac{\text{number of microstates in this subset}}{\text{total number of microstates}}. \tag{12.44}$$

So the idea is that we do all our practical calculations via statistical averages (with $p_\alpha = 1/\Omega$ etc.), but the physical justification for that is that the system is time-averaging itself (we cannot directly calculate time averages because we cannot calculate precise dynamics).

The objection to this view that I find the most compelling is simply that the size of the phase space (the number of microstates) of any macroscopic system is so enormous that it is in fact quite impossible for the system to visit all of it over reasonable times (see Jaynes 2003).

The key divide here is rooted in the old argument about the meaning of probabilities:

—probabilities as frequencies, or "objective" probabilities, measuring how often something actually happens

vs.

—probabilities as (*a priori*) likelihoods, or "subjective" probabilities, measuring our (lack of) knowledge about what happens (this view has quite a respectable intellectual pedigree: Laplace, Bayes, Keynes, Jeffreys, Jaynes... and Binney!).

NB: In choosing to go with the latter view and putting up all these objections to the former, I am not suggesting that one is "right" and the other "wrong." Remember that the falsifiable (and examinable) content of the theory is the same either way, so the issue is which of the logical constructions leading to it makes more sense to me or to you—and I urge you to explore the literature on your own and decide for yourselves whether you are a Gibbsian or a Boltzmannite (either way, you are in good company)—or, indeed, whether you wish to invent a third way!†

12.4. *Whose Uncertainty?*

[Literature: Jaynes (2003), §11.8]

To pre-empt some of the inevitable confusion about the "subjective" nature of maximising uncertainty (*whose* uncertainty?!), let me deal with the common objection that surely, if two observers (call them A and B) have different amounts of information about the same system and so arrive at two different entropy-maximising sets of p_α's, it would be disastrous if those different sets gave different testable predictions about the system! (Heat capacity of a room filled with air cannot depend on who is looking!)

There are three possible scenarios.

- If Mrs B has more constraints (i.e., more knowledge) than Mr A, but her additional constraints are, in fact, *derivable* from Mr A's, then both Mr A and Mrs B will get the same probability distribution $\{p_\alpha\}$ because Mrs B's additional Lagrange multipliers will turn out to be arbitrary and so can be set to 0 (this is easy to see if you work through an example: e.g., Exercise 9.1e).

- If Mrs B's additional constraints are *incompatible* with Mr A's, the method of Lagrange multipliers will produce a set of equations for λ's that will have no real solutions—telling us that the system of constraints is logically contradictory and so no theory exists (this basically means that one of them got their constraints wrong).

- Finally, if Mrs B's additional constraints are neither incompatible with nor derivable from Mr A's, that means that she has discovered new physics: Mrs B's additional constraints will bring in new Lagrange multipliers, which will turn out to have some interesting physical interpretation—usually as some macroscopic thermodynamical quantities (we will see an example of this when we discover chemical potential in §14).

12.5. *Second Law*

[Literature: Jaynes (1965)]

So far in this part of the course, we have not involved time in our considerations: we have always been interested in some eventual equilibrium and the way to calculate it was to maximise $S_{\rm G}$ subject to constraints representing some measurable properties of this equilibrium. This maximisation of $S_{\rm G}$ is *not* the same thing as the 2-nd Law

† This is a bit like the thorny matter of the interpretations of Quantum Mechanics: everyone agrees on the results, but not on why the theory works.

of Thermodynamics, which states, effectively, that *the thermodynamic entropy S of the world (or a closed, isolated system) must not decrease in any process—and so in time.*

This statement is famously replete with hard metaphysical questions (even though it is quite straightforward when it comes to calculating entropy changes in mundane situations)—so it is perhaps useful to see how it emerges within the conceptual framework that I am advocating here. The following proof is what I believe to be an acceptable vulgarisation of an argument due to Jaynes (1965).

Time t:

Consider a closed system (the world) in equilibrium, subject to some set of its properties having just been measured and no other infiormation available. Then our best guess as to its state at this time t is obtained by maximising $S_{\rm G}$ subject to those properties being known at time t. This gives a set of probabilities $\{p_\alpha\}$ that describe this equilibrium. In this equilibrium, the maximum value of $S_{\rm G}$ that we have obtained is equal to the thermodynamical entropy (see proof in §9.2):

$$S(t) = k_{\rm B} S_{\rm G,max}(t). \tag{12.45}$$

Time $t' > t$:

Now consider the evolution of this system from time t to a later time t', starting from the set of states $\{\alpha\}$ and their probabilities $\{p_\alpha\}$ that we inferred at time t and using Hamilton's equations (if the system is classical) or the time-dependent Schrödinger's equation (if it is quantum, as it always really is; see §13.4). During this evolution, *the Gibbs entropy stays constant:*

$$S_{\rm G}(t') = -\sum_\alpha p_\alpha \ln p_\alpha = S_{\rm G}(t). \tag{12.46}$$

Indeed, the Schrödiner equation evolves the states $\{\alpha\}$, but if the system was in some state $\alpha(t)$ at time t with probability p_α, it will be in the descendant $\alpha(t')$ of that state at t' with exactly the same probability; this is like changing labels in the expression for $S_{\rm G}$ while p_α's stay the same—and so does $S_{\rm G}$. Thus,

$$S_{\rm G}(t') = S_{\rm G}(t) = S_{\rm G,max}(t) = \frac{1}{k_{\rm B}}\, S(t). \tag{12.47}$$

Now forget all previous information, make a new set of measurements at time t', work out a new set of probabilities $\{p_\alpha\}$ at t' subject only to these new constraints, by maximising Gibbs entropy, and from it infer the new thermodynamical (equilibrium) entropy:

$$S(t') = \underbrace{k_{\rm B} S_{\rm G,max}(t')}_{\substack{\text{the new } S_{\rm G},\\ \text{maximised at}\\ \text{time } t'}} \geqslant \underbrace{k_{\rm B} S_{\rm G}(t')}_{\substack{\text{the ``true''}\\ S_{\rm G}\text{, evolved}\\ \text{from time } t}} = k_{\rm B} S_{\rm G}(t) = S(t). \tag{12.48}$$

Thus,

$$\boxed{S(t') \geqslant S(t)} \text{ at } t' > t, \text{ q.e.d., Second Law.} \tag{12.49}$$

The meaning of this is that the increase of S reflects our insistence to forget most of the detailed knowledge that we possess as a result of evolving in time any earlier state (even if based on a earlier guess) and to reapply at every later time the rules of statistical

inference based on the very little knowledge we can obtain in later measurements.

If you are sufficiently steeped in quantum ways of thinking by now, you will pounce and ask: who is doing all these measurements?

If it is an external observer or apparatus, then the system isn't really closed and, in particular, the measurement at the later time t' will potentially destroy the identification of all those microstates with their progenitors at time t, so the equality (12.46) no longer holds.†

A further objection is: what if your measurements at t' are much better than at the technologically backward time t? You might imagine an extreme case in which you determine the state of the system at t' precisely and so $S_{\rm G}(t') = 0$!

- Clearly, the observer is, in fact, not external, but lives inside the system.

- As he/she/it does their measurement, not just the entropy of the object of measurement (a subsystem) but also of the observer and their apparatus changes. The argument above implies that a very precise measurement leading to a decrease in the entropy of the measured subsystem must massively increase the entropy of the observer and his kit, to compensate and ensure that the total entropy increases [Eq. (12.49)].‡

We will return to these arguments in a slightly more quantitative (or, at any rate, more quantum) manner in §§13.4–13.5.

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13. P.P.S. Density Matrix and Entropy in Quantum Mechanics

[Literature: Binney & Skinner (2013), §6.3, 6.5]

So far the only way in which the quantum-mechanical nature of the world has figured in our discussion is via the sums of states being discrete and also in the interpretation of the indistinguishability of particles. Now I want to show you how one introduces the uncertainty about the quantum state of the system into the general quantum mechanical formalism.

13.1. *Expectation Values*

Suppose we are uncertain about the quantum state of our system but think that it is in one of a set of *orthogonal quantum states* $\{|\alpha\rangle\}$ ($\alpha = 1, \ldots, \Omega$) and our uncertainty about which one it is is expressed by *a priori* probabilities $\{p_\alpha\}$, as usual (assigned via the entropy-maximising procedure whose quantum further particulars I am about to explain). For any *observable* $\hat{O}$ (which is an operator, e.g., the Hamiltonian $\hat{H}$), its expectation value is

$$\bar{O} = \sum_\alpha p_\alpha \langle\alpha|\hat{O}|\alpha\rangle, \tag{13.1}$$

p_α is the *a priori* probability that the system is in the state $|\alpha\rangle$ and $\langle\alpha|\hat{O}|\alpha\rangle$ is the expectation value of $\hat{O}$ *if* the system is in the state $|\alpha\rangle$ (e.g., E_α if $\hat{O} = \hat{H}$). The states $\{|\alpha\rangle\}$ are *not* necessarily eigenstates of $\hat{O}$. Since, written in terms of its eigenstates and eignevalues, this operator is

$$\hat{O} = \sum_\mu O_\mu |O_\mu\rangle\langle O_\mu|, \tag{13.2}$$

† In a classical world, this would not be a problem because you can make measurements without altering the system, but in Quantum Mechanics, you cannot.

‡ This sort of argument was the basis of the exorcism of Maxwell's Demon by Szilard (1929).

its expectation value (13.1) can be written as

$$\bar{O} = \sum_{\alpha\mu} p_\alpha O_\mu \langle\alpha|O_\mu\rangle\langle O_\mu|\alpha\rangle = \sum_\mu O_\mu \underbrace{\sum_\alpha p_\alpha |\langle O_\mu|\alpha\rangle|^2}_{\substack{\text{total}\\ \text{probability to}\\ \text{measure } O_\mu}}. \tag{13.3}$$

This formula underscores the fact that the expected outcome of a measurement is subject to two types of uncertainty:

—our uncertainty as to what state the system is in (quantified by the probability p_α to be in the state $|\alpha\rangle$),

—quantum uncertainty as to the outcome of a measurement, given a definite quantum state (this uncertainty is quantified by $|\langle O_\mu|\alpha\rangle|^2$, the probability to measure O_μ if the system is in the state $|\alpha\rangle$).

13.2. *Density Matrix*

This construction motivates us to introduce the *density operator*

$$\boxed{\hat{\rho} = \sum_\alpha p_\alpha |\alpha\rangle\langle\alpha|}\,. \tag{13.4}$$

This looks analogous to Eq. (13.2), except note that $\hat{\rho}$ is *not* an observable because p_α's are subjective. In the context of this definition, one refers to the system being in a *pure state* if for some α, $p_\alpha = 1$ and so $\hat{\rho} = |\alpha\rangle\langle\alpha|$, or an *impure state* if all $p_\alpha < 1$.

The density operator is useful because, knowing $\hat{\rho}$, we can express expectation values of observables as

$$\boxed{\bar{O} = \mathrm{Tr}\big(\hat{\rho}\,\hat{O}\big)}. \tag{13.5}$$

Indeed, the above expression reduces to Eq. (13.1):

$$\mathrm{Tr}\big(\hat{\rho}\,\hat{O}\big) = \sum_{\alpha'}\langle\alpha'|\hat{\rho}\hat{O}|\alpha'\rangle = \sum_{\alpha'\alpha}\underbrace{\langle\alpha'|\alpha\rangle}_{\delta_{\alpha'\alpha}}\langle\alpha|\hat{O}|\alpha'\rangle = \sum_\alpha p_\alpha\langle\alpha|\hat{O}|\alpha\rangle = \bar{O}, \quad \text{q.e.d.} \tag{13.6}$$

It is useful to look at the density operator in the $\{|O_\mu\rangle\}$ representation: since

$$|\alpha\rangle = \sum_\mu \langle O_\mu|\alpha\rangle|O_\mu\rangle, \tag{13.7}$$

we have

$$\hat{\rho} = \sum_\alpha p_\alpha \sum_{\mu\nu}\langle O_\mu|\alpha\rangle\langle\alpha|O_\nu\rangle|O_\mu\rangle\langle O_\nu| \equiv \sum_{\mu\nu} p_{\mu\nu}|O_\mu\rangle\langle O_\nu|, \tag{13.8}$$

where we have introduced the *density matrix:*

$$\boxed{p_{\mu\nu} = \sum_\alpha p_\alpha \langle O_\mu|\alpha\rangle\langle\alpha|O_\nu\rangle}\,. \tag{13.9}$$

Thus, whereas $\hat{\rho}$ is diagonal in the "information basis" $\{|\alpha\rangle\}$, it is, in general, *not* diagonal in any given basis associated with the eigenstates of an observable, $\{|O_\mu\rangle\}$—in other words, the states to which we assign *a priori* probabilities are not necessarily the eigenstates of the observable that we then wish to calculate.

Let us express the expectation value of $\hat{O}$ in terms of the density matrix: using Eq. (13.8),

$$\bar{O} = \mathrm{Tr}\big(\hat{\rho}\,\hat{O}\big) = \sum_{\mu'}\langle O_{\mu'}|\hat{\rho}\hat{O}|O_{\mu'}\rangle = \sum_{\mu'\mu\nu} O_{\mu'}p_{\mu\nu}\underbrace{\langle O_{\mu'}|O_\mu\rangle}_{\delta_{\mu'\mu}}\underbrace{\langle O_\nu|O_{\mu'}\rangle}_{\delta_{\nu\mu'}} = \sum_\mu O_\mu p_{\mu\mu}, \tag{13.10}$$

the same expression as Eq. (13.3), seeing that

$$p_{\mu\mu} = \sum_\alpha p_\alpha |\langle O_\mu|\alpha\rangle|^2. \tag{13.11}$$

Thus, the diagonal elements of the density matrix in the $\hat{O}$ representation are the combined quantum and *a priori* probabilities of the observable giving eigenvalues O_μ as measurement outcomes.

The off-diagonal elements have no classical interpretation. They measure quantum correlations and come into play when, e.g., we want the expectation value of an observable other than that in whose representation we chose to write $\hat{\rho}$: for an observable $\hat{P}$, the expectation value is

$$\bar{P} = \operatorname{Tr}\bigl(\hat{\rho}\,\hat{P}\bigr) = \sum_{\mu'} \langle O_{\mu'}| \underbrace{\sum_{\mu\nu} p_{\mu\nu} |O_\mu\rangle\langle O_\nu|}_{\hat{\rho}} \hat{P} |O_{\mu'}\rangle = \sum_{\mu\nu} p_{\mu\nu} \langle O_\nu|\hat{P}|O_\mu\rangle. \tag{13.12}$$

13.3. *Quantum Entropy*

The generalisation of the Gibbs–Shannon entropy in this formalism is the *von Neumann entropy*:

$$\boxed{S_{\text{vN}} = -\operatorname{Tr}\bigl(\hat{\rho}\,\ln\hat{\rho}\bigr)}\,, \tag{13.13}$$

which is, in fact, the same as S_{G}:

$$\begin{aligned} S_{\text{vN}} &= -\sum_\alpha \langle\alpha| \underbrace{\left(\sum_\mu p_\mu |\mu\rangle\langle\mu|\right)}_{\hat{\rho}} \underbrace{\left(\sum_\nu \ln p_\nu |\nu\rangle\langle\nu|\right)}_{\substack{\ln\hat{\rho},\\ \text{by definition}}} |\alpha\rangle \\ &= \sum_{\alpha\mu\nu} p_\mu \ln p_\nu \underbrace{\langle\alpha|\mu\rangle}_{\delta_{\alpha\mu}} \underbrace{\langle\mu|\nu\rangle}_{\delta_{\mu\nu}} \underbrace{\langle\nu|\alpha\rangle}_{\delta_{\nu\alpha}} = -\sum_\alpha p_\alpha \ln p_\alpha. \end{aligned} \tag{13.14}$$

As always, we find p_α's by maximising S_{vN} subject to constraints imposed by the information we possess, often in the form of expectation values like $\bar{O}$.

The canonical distribution (9.8) has the following density matrix, *in energy representation*:

$$\hat{\rho} = \sum_\alpha \frac{e^{-\beta E_\alpha}}{Z} |\alpha\rangle\langle\alpha| = \frac{e^{-\beta\hat{H}}}{Z}, \tag{13.15}$$

where $\{|\alpha\rangle\}$ are understood to be eigenstates of $\hat{H}$ and the partition function is

$$Z = \sum_\alpha e^{-\beta E_\alpha} = \operatorname{Tr} e^{-\beta\hat{H}} \tag{13.16}$$

13.4. *Time Evolution*

If we know $\hat{\rho}$ at some time, we can easily find it at any later time:

$$\frac{\mathrm{d}\hat{\rho}}{\mathrm{d}t} = \sum_\alpha p_\alpha \left(\frac{\partial|\alpha\rangle}{\partial t}\langle\alpha| + |\alpha\rangle\frac{\partial\langle\alpha|}{\partial t}\right) = \frac{1}{i\hbar}\sum_\alpha p_\alpha \bigl(\hat{H}|\alpha\rangle\langle\alpha| - |\alpha\rangle\langle\alpha|\hat{H}\bigr) = \frac{\hat{H}\hat{\rho} - \hat{\rho}\hat{H}}{i\hbar} \tag{13.17}$$

is the *time-dependent Schrödinger equation*. In a more standard form:

$$\boxed{i\hbar\frac{\mathrm{d}\hat{\rho}}{\mathrm{d}t} = \bigl[\hat{H},\hat{\rho}\bigr]}\,. \tag{13.18}$$

Note that the probabilities p_α do not change with time: if the system was in a state $|\alpha(0)\rangle$ initially, it will be in its descendant state $|\alpha(t)\rangle$ at a later time t.

So, we may envision a situation in which *we are uncertain about a system's initial conditions*, work out $\hat{\rho}(t=0)$ via the maximum-entropy principle, constrained by some measurements, and then evolve $\hat{\rho}(t)$ forever *if* we know the Hamiltonian precisely. Since p_α's do not change, *the Gibbs–Shannon–von Neumann entropy of the system stays the same during this time evolution*—the only uncertainty was in the initial conditions.

What if we do not know the Hamiltonian (or choose to forget)? This was discussed in §12.5: then, at a later time, we may make another measurement and construct the new density matrix

$\hat{\rho}_{\rm new}(t)$ via another application of the maximum-entropy principle. Both $\hat{\rho}_{\rm new}(t)$ and $\hat{\rho}_{\rm old}(t)$—which is our $\hat{\rho}(0)$ evolved via Eq. (13.18) with the (inknown to us) precise $\hat{H}$—are consistent with the new measurement. But $\hat{\rho}_{\rm new}(t)$ corresponds to the maximum possible value of the entropy consistent with this measurement while $\hat{\rho}_{\rm old}(t)$ has the same entropy as $\hat{\rho}(0)$ did at $t = 0$. Therefore,

$$S_{\rm new}(t) > S_{\rm old}(0). \tag{13.19}$$

This is the Second Law and the argument above is the same as the argument already given in §12.5.

13.5. *How Information Is Lost*

When we discuss predictions or outcomes of physical measurements, we can think of the world as consisting of two parts:

—the system to be measured,
—the rest of the world: the environment, including the measuremwnt apparatus (sometimes only it if the experiment is "isolated").

The observables that we have and so the information that we will use for statistical inference will pertain to the system, while the environment will remain mysterious.

For example, imagine we measured the energy of the system at soem initial time. For lack of better knowledge, it is natural to make statistical inference about the microstates of the world in the following form:

$$|\alpha\alpha', 0\rangle = |E_\alpha^{(\rm sys)}(0)\rangle |E_{\alpha'}^{(\rm env)}(0)\rangle, \tag{13.20}$$

where 0 stands for $t = 0$, $|E_\alpha^{(\rm sys)}(0)\rangle$ are the states of the system in the energy representation and $|E_{\alpha'}^{(\rm env)}(0)\rangle$ are the states of the environment (unknown). Then

$$\hat{\rho}(0) = \sum_{\alpha\alpha'} p_{\alpha\alpha'} |\alpha\alpha', 0\rangle\langle\alpha\alpha', 0|, \tag{13.21}$$

where $p_{\alpha\alpha'}$ are the probabilities of $|E_\alpha^{(\rm sys)}(0)\rangle$, indifferent to $|E_{\alpha'}^{(\rm env)}(0)\rangle$.

Now evolve this density matrix according to the time-dependent Schrödinger equation (13.18): $p_{\alpha\alpha'}$'s will stay the same, while the states will evolve:

$$|\alpha\alpha', 0\rangle \to |\alpha\alpha', t\rangle \neq |E_\alpha^{(\rm sys)}(t)\rangle |E_{\alpha'}^{(\rm env)}(t)\rangle. \tag{13.22}$$

The descendants of the initial states (13.20) will *not* in general be superpositions of the energy states of the system and the environment. This is because the system and the environment get *entangled*. Formally speaking,

$|E_\alpha^{(\rm sys)}\rangle$ are eigenstates of $\hat{H}^{(\rm sys)}$, the Hamiltonian of the system,
$|E_{\alpha'}^{(\rm env)}\rangle$ are eigenstates of $\hat{H}^{(\rm env)}$, the Hamiltonian of the environment,

but

$|E_\alpha^{(\rm sys)}\rangle |E_{\alpha'}^{(\rm env)}\rangle$ are *not* eigenstates of the world's Hamiltonian:

$$\hat{H} = \hat{H}^{(\rm sys)} + \hat{H}^{(\rm env)} + \hat{H}^{(\rm int)} \tag{13.23}$$

because of the interaction Hamiltonian $\hat{H}^{(\rm int)}$.

If now, at time t, we measure the energy of the system again, we will have to make statistical inference about superposed eigenstates:

$$|\alpha\alpha', {\rm new}\rangle = |E_\alpha^{(\rm sys)}(t)\rangle |E_{\alpha'}^{(\rm env)}(t)\rangle \neq |\alpha\alpha, t\rangle. \tag{13.24}$$

In this representation, our old density matrix is *diagonal*:

$$\begin{aligned}\hat{\rho}^{(\mathrm{old})}(t) &= \sum_{\alpha\alpha'} p^{(\mathrm{old})}_{\alpha\alpha'} |\alpha\alpha', t\rangle\langle\alpha\alpha', t| \\ &= \sum_{\alpha\alpha'} p^{(\mathrm{old})}_{\alpha\alpha'} \sum_{\mu\mu'}\sum_{\nu\nu'} |\mu\mu', \mathrm{new}\rangle\langle\mu\mu', \mathrm{new}|\alpha\alpha, t\rangle\langle\alpha\alpha', t|\nu\nu', \mathrm{new}\rangle\langle\nu\nu', \mathrm{new}| \\ &= \sum_{\mu\mu'}\sum_{\nu\nu'} p^{(\mathrm{old})}_{\mu\mu'\nu\nu'}(t) |\mu\mu', \mathrm{new}\rangle\langle\nu\nu', \mathrm{new}|, \end{aligned} \tag{13.25}$$

where the old density matrix in the new representation is [cf. Eq. (13.9)]:

$$p^{(\mathrm{old})}_{\mu\mu'\nu\nu'}(t) = \sum_{\alpha\alpha'} p^{(\mathrm{old})}_{\alpha\alpha'} \langle\mu\mu', \mathrm{new}|\alpha\alpha, t\rangle\langle\alpha\alpha', t|\nu\nu', \mathrm{new}\rangle. \tag{13.26}$$

However, the measured energy of the system at time t only depends on the diagonal elements $p^{(\mathrm{old})}_{\alpha\alpha'\alpha\alpha'}(t)$ of this matrix:

$$\begin{aligned} U &= \mathrm{Tr}\big[\hat{\rho}^{(\mathrm{old})}(t)\hat{H}^{(\mathrm{sys})}(t)\big] \\ &= \sum_{\alpha\alpha'}\sum_{\mu\mu'}\sum_{\nu\nu'} p^{(\mathrm{old})}_{\mu\mu'\nu\nu'}(t) E^{(\mathrm{sys})}_{\nu}(t) \langle\alpha\alpha', \mathrm{new}|\mu\mu', \mathrm{new}\rangle\langle\nu\nu', \mathrm{new}|\alpha\alpha', \mathrm{new}\rangle \\ &= \sum_{\alpha\alpha'} p^{(\mathrm{old})}_{\alpha\alpha'\alpha\alpha'}(t) E^{(\mathrm{sys})}_{\alpha}(t). \end{aligned} \tag{13.27}$$

All information about correlations between the system and the environment is lost in this measurement.

When we maximise entropy and thus make a new statistical inference about the system, the new entropy will be higher than the old for two reasons:

1) all off-diagonal elements from the old density matrix are lost,

2) the diagonal elements $p^{(\mathrm{old})}_{\alpha\alpha'\alpha\alpha'}(t)$ are in general *not* the ones that maximise entropy (see the argument in §12.5):

$$p^{(\mathrm{new})}_{\alpha\alpha'} \neq p^{(\mathrm{old})}_{\alpha\alpha'\alpha\alpha'}(t) \tag{13.28}$$

Thus, the new density matrix

$$\hat{\rho}^{(\mathrm{new})} = \sum_{\alpha\alpha'} p^{(\mathrm{new})}_{\alpha\alpha'} |\alpha\alpha', \mathrm{new}\rangle\langle\alpha\alpha', \mathrm{new}|, \tag{13.29}$$

being the maximiser of entropy at time t, will have

$$\begin{aligned} S^{(\mathrm{new})}_{\mathrm{vN}} = -\mathrm{Tr}\big[\hat{\rho}^{(\mathrm{new})}\ln\hat{\rho}^{(\mathrm{new})}\big] \geqslant S^{(\mathrm{old})}_{\mathrm{vN}}(t) &= -\mathrm{Tr}\big[\hat{\rho}^{(\mathrm{old})}(t)\ln\hat{\rho}^{(\mathrm{old})}(t)\big] \\ &= \underbrace{S^{(\mathrm{old})}_{\mathrm{vN}}(0) = -\mathrm{Tr}\big[\hat{\rho}(0)\ln\hat{\rho}(0)\big]}_{\substack{\text{old entropy did not change} \\ \text{because } p_{\alpha\alpha'}\text{'s did not change}}} . \end{aligned} \tag{13.30}$$

So information is lost and we move forward to an ever more boring world... (which is a very interesting fact, so don't despair!)

You might think of what has happened as *our total ignorance about the environment having polluted our knowledge about the system as a result of the latter getting entangled with the former.*

PART IV
Statistical Mechanics of Simple Systems

This part of the course was taught by Professor Julien Devriendt.

PART V

Open Systems

14. Grand Canonical Ensemble

So you know what to do if you are interested in a system whose quantum states you know and whose probabilities for being in any one of these states you have to guess based on (the expectation of) the knowledge of some measurable mean quantities associated with the system. So far (except in §10.4) the measurable quantity has always been mean energy—and the resulting canonical distribution gave a good statistical description of a physical system in contact with a heat bath at some fixed temperature.

Besides the measurable mean energy U, our system depended on a number of *exactly fixed external parameters:* the volume V, the number of particles N—these were not constraints, they did not need to be measured, they were just there, set in stone (a box of definite volume, with impenetrable walls, containing a definite number of particles). Mathematically speaking, the microstates of the system† and their energies depended parametrically on V and N:

$$E_\alpha = E_\alpha(V, N). \tag{14.1}$$

There are good reasons to recast N as a measurable mean quantity rather than a fixed parameter. This will allow us to treat systems that are not entirely closed and so can exchange particles with other systems. For example:

—inhomogeneous systems in some external potential (gravity, electric field, rotation, etc.), in which parts of the system can be thought of as exchanging particles with other parts where the external potential has a different value (§14.5);

—multiphase systems, where different phases (e.g., gaseous, liquid, solid) can exchange particles via evaporation, condensation, sublimation, solidification, etc. (§15.2, Part VII);

—systems containing different substances that can react with each other and turn into each other (§15), e.g., chemical reacting mixtures (§§15.3 and 15.4), partially ionised plasmas subject to ionisation/recombination (Exercise 15.2);

—systems in which the number of particles is not fixed at all and is determined by the requirements of thermodynamical equilibrium, e.g., pair production/annihilation (Exercise 16.3), thermal radiation (§19), etc.;

—systems where N might be fixed, but, for greater ease of counting microstates, it is convenient formally to allow it to vary (Fermi and Bose statistics for quantum gases, §16).

† E.g., for ideal gas, α depended on V via the set of possible values of particles' momenta, Eq. (11.3), and on N via the fixed sum of the occupation numbers, Eq. (11.18).

14.1. *Grand Canonical Distribution*

We now declare that each microstate α has a certain energy and a certain number of particles associated with it,‡

$$\alpha \leftrightarrow E_\alpha, N_\alpha, \tag{14.2}$$

and there are two constraints:

$$\sum_\alpha p_\alpha E_\alpha = U \quad \text{mean energy}, \tag{14.3}$$

$$\sum_\alpha p_\alpha N_\alpha = \bar{N} \quad \text{mean number of particles}. \tag{14.4}$$

Both U and $\bar{N}$ are measurable; measuring $\bar{N}$ is equivalent to measuring the *mean density*

$$n = \frac{\bar{N}}{V} \tag{14.5}$$

(note that V remains an exactly fixed external parameter).

We know the routine: maximise entropy subject to these two constraints:

$$S_\mathrm{G} - \lambda\left(\sum_\alpha p_\alpha - 1\right) - \beta\left(\sum_\alpha p_\alpha E_\alpha - U\right) + \beta\mu\left(\sum_\alpha p_\alpha N_\alpha - \bar{N}\right) \to \max, \tag{14.6}$$

where $-\beta\mu$ is the new Lagrange multiplier responsible for enforcing the new constraint (14.4); the factor of $-\beta$ is introduced to follow the conventional definition of μ, whose physical meaning will shortly emerge. Carrying out the maximisation in the same manner as in §9.1, we find

$$\ln p_\alpha + 1 + \lambda + \beta E_\alpha - \beta\mu N_\alpha = 0. \tag{14.7}$$

This gives the *grand canonical distribution:*

$$\boxed{p_\alpha = \frac{e^{-\beta(E_\alpha - \mu N_\alpha)}}{\mathcal{Z}(\beta,\mu)}}, \tag{14.8}$$

where the normalisation factor (arising from the Lagrange multiplier λ) is the *grand partition function:*

$$\boxed{\mathcal{Z}(\beta,\mu) = \sum_\alpha e^{-\beta(E_\alpha - \mu N_\alpha)}}. \tag{14.9}$$

It remains to calculate β and μ. Since

$$\frac{\partial \ln \mathcal{Z}}{\partial \beta} = \frac{1}{\mathcal{Z}}\sum_\alpha (-E_\alpha + \mu N_\alpha) e^{-\beta(E_\alpha - \mu N_\alpha)} = -U + \mu\bar{N}, \tag{14.10}$$

the first implicit equation for β and μ in terms of U and $\bar{N}$ (or, if convenient, the expression for U) is

$$U(\beta,\mu) = -\frac{\partial \ln \mathcal{Z}}{\partial \beta} + \mu\bar{N}(\beta,\mu). \tag{14.11}$$

‡ There is no sacral meaning to N_α: we simply allow states with 1 particle, states with 2 particles, states with 3 particles, ..., states with N particles, ..., all to be part of our enlarged set of allowed microstates. Volume is still a parameter, on which both E_α and other quantum numbers may depend, but N_α is not a function of V: we always consider states with different values of N_α as different states. Obviously, different states will have different probabilities and so certain values of N_α will be more probable than others, so the *mean* number of particles *will* depend on V.

The second equation arises from noticing that

$$\frac{\partial \ln \mathcal{Z}}{\partial \mu} = \frac{1}{\mathcal{Z}} \sum_\alpha \beta N_\alpha e^{-\beta(E_\alpha - \mu N_\alpha)} = \beta \bar{N}, \tag{14.12}$$

and so

$$\bar{N}(\beta, \mu) = \frac{1}{\beta} \frac{\partial \ln \mathcal{Z}}{\partial \mu}. \tag{14.13}$$

Note that the canonical distribution and the canonical partition fuction (§9.1) can be recovered as a special case of our new theory: suppose that for all α, the number of particles is the same,

$$N_\alpha = N \quad \text{for all } \alpha. \tag{14.14}$$

Then Eq. (14.8) becomes

$$p_\alpha = e^{-\beta E_\alpha} \underbrace{\frac{e^{\beta \mu N}}{\mathcal{Z}}}_{=1/Z} = \frac{e^{-\beta E_\alpha}}{Z}, \tag{14.15}$$

which is our old canonical distribution [Eq. (9.8)], where, using Eq. (14.9),

$$Z = e^{-\beta \mu N} \mathcal{Z} = e^{-\cancel{\beta \mu N}} \sum_\alpha e^{-\beta(E_\alpha - \cancel{\mu N})} = \sum_\alpha e^{-\beta E_\alpha} \tag{14.16}$$

is the familiar non-grand partition function [Eq. (9.7)]. The relationship for the grand and non-grand partition functions, when written in the form

$$\mathcal{Z} = (e^{\beta \mu})^N Z(\beta), \tag{14.17}$$

highlights a quantity sometimes referred to as "fugacity" $= e^{\beta \mu}$.

14.2. *Thermodynamics of Open Systems and the Meaning of Chemical Potential*

We are now ready to generalise the construction of thermodynamics from §9.2 to the case of open systems.

The Gibbs entropy in the grand canonical equilibrium is

$$S_\mathrm{G} = -\sum_\alpha p_\alpha \ln p_\alpha = -\sum_\alpha p_\alpha \left[-\beta(E_\alpha - \mu N_\alpha) - \ln \mathcal{Z}\right] = \beta(U - \mu \bar{N}) + \ln \mathcal{Z}. \tag{14.18}$$

Its differential is

$$\begin{aligned}
\mathrm{d}S_\mathrm{G} &= \beta(\mathrm{d}U - \bar{N}\mathrm{d}\mu - \mu \mathrm{d}\bar{N}) + (U - \mu\bar{N})\mathrm{d}\beta + \frac{\mathrm{d}\mathcal{Z}}{\mathcal{Z}} \\
&= \beta(\mathrm{d}U - \cancel{\bar{N}\mathrm{d}\mu} - \mu \mathrm{d}\bar{N}) + \cancel{(U - \mu\bar{N})\mathrm{d}\beta} \\
&\quad + \sum_\alpha \underbrace{\frac{e^{-\beta(E_\alpha - \mu N_\alpha)}}{\mathcal{Z}}}_{=p_\alpha} \Big[-\beta(\mathrm{d}E_\alpha - \cancel{N_\alpha \mathrm{d}\mu}) - \cancel{(E_\alpha - \mu N_\alpha)\mathrm{d}\beta}\Big] \\
&= \beta\Bigg(\mathrm{d}U - \mu \mathrm{d}\bar{N} - \underbrace{\sum_\alpha p_\alpha \mathrm{d}E_\alpha}_{=\left\langle \frac{\partial E_\alpha}{\partial V} \right\rangle \mathrm{d}V}\Bigg) \\
&= \beta\left(\mathrm{d}U + P\mathrm{d}V - \mu \mathrm{d}\bar{N}\right) = \beta \mathrm{d}Q_\mathrm{rev}.
\end{aligned} \tag{14.19}$$

We have taken $E_\alpha = E_\alpha(V)$ (energy levels are a function of the single remaining external parameter V, the volume of the system) but $\mathrm{d}N_\alpha = 0$ (N_α is not a function of V; see footnote ‡ in §14.1); we have also used our standard definition of pressure (7.13).

It is clear that the right-hand side of Eq. (14.19) had to be identified as $\beta \mathrm{d}Q_{\mathrm{rev}}$ because we would like to keep the correspondence between S_{G} and the thermodynamical entropy [Eq. (9.14)] and between β and the thermodynamical temperature [Eq. (9.13)]:

$$S_{\mathrm{G}} = \frac{S}{k_{\mathrm{B}}}, \quad \beta = \frac{1}{k_{\mathrm{B}}T}. \tag{14.20}$$

This implies the physical interpretation of μ: *in a reversible process where U and V stay the same but $\bar{N}$ changes, adding each particle generates $-\mu$ amount of heat.* In other words,

$$\boxed{\mu = -T\left(\frac{\partial S}{\partial \bar{N}}\right)_{U,V}}. \tag{14.21}$$

Intuitively, adding particles should increase entropy (systems with more particles usually have a larger number of microstates available to them, so the uncertainty as to which of these microstates they are in is likely to be greater)—therefore, we expect μ to be a negative quantity, under normal circumstances. Equivalently, one might argue that a positive value of μ would imply that entropy increased with diminishing $\bar{N}$ and so, in its quest to maximise entropy, a system with positive μ would be motivated to lose all its particles and thus cease to be a system. This argument is mostly correct, although we will encounter an interesting exception in the case of degenerate Fermi gas (§17).

We are now ready to write the *fundamental equation of thermodynamics of open systems* [generalising Eq. (9.15)]:

$$\boxed{\mathrm{d}U = T\mathrm{d}S - P\mathrm{d}V + \mu \mathrm{d}\bar{N}}. \tag{14.22}$$

Writing it in this form highlights another interpretation of the chemical potential:

$$\mu = \left(\frac{\partial U}{\partial \bar{N}}\right)_{S,V}, \tag{14.23}$$

the energy cost of a particle to a system at constant volume and entropy.

To complete our new thermodynamics, let us generalise the concept of free energy: using Eqs. (14.18) and (14.20), we introduce a new thermodynamical quantity

$$\boxed{\Phi = -k_{\mathrm{B}}T \ln \mathcal{Z} = U - TS - \mu\bar{N} = F - \mu\bar{N}}, \tag{14.24}$$

called the *grand potential* (its physical meaning will become clear in §14.6.3). The usefulness of this quantity for open systems is the same as the usefulness of F for closed ones: it is the function by differentiating which one gets all the relevant thermodynamical quantities and equations. Indeed, using Eq. (14.22), we get

$$\boxed{\mathrm{d}\Phi = -S\mathrm{d}T - P\mathrm{d}V - \bar{N}\mathrm{d}\mu}, \tag{14.25}$$

and so,

$$S = -\left(\frac{\partial\Phi}{\partial T}\right)_{V,\mu}, \tag{14.26}$$

$$\bar{N} = -\left(\frac{\partial\Phi}{\partial\mu}\right)_{T,V} \quad \left(\text{equivalently, equation for density } n = \frac{\bar{N}}{V}\right), \tag{14.27}$$

$$U = \Phi + TS + \mu\bar{N}, \tag{14.28}$$

$$P = -\left(\frac{\partial\Phi}{\partial V}\right)_{T,\mu}, \quad \text{equation of state} \tag{14.29}$$

(note that the equation of state will, in fact, turn out to be obtainable in an even simpler way than this: see §14.6.3).

Similarly to the case of fixed number of particles, we have found that all we need to do, pragmatically, is calculate the (grand) partition function $\mathcal{Z}(\beta,\mu)$, which incorporates all the microphysics relevant to the thermodynamical description, infer from it the grand potential Φ, and then take derivatives of it—and we know everything we care about.

What is the role that μ plays in all this? Eq. (14.22) suggests that $-\mu$ to $\bar{N}$ is what P is to V or $-1/T$ to U, i.e., it regulates the way in which some form of equilibrium is achieved across a system.

14.3. *Particle Equilibrium*

Similarly to what we did in §10.2, consider two systems in thermal and particle contact (i.e., capable of exchanging energy and matter), but otherwise isolated. The name of the game is to maximise entropy subject to conserved total (mean) energy and particle number:

$$U = U_1 + U_2 = \text{const}, \tag{14.30}$$

$$\bar{N} = \bar{N}_1 + \bar{N}_2 = \text{const} = \text{const}, \tag{14.31}$$

$$S = S_1 + S_2 \to \max. \tag{14.32}$$

Taking differentials,

$$\begin{aligned}
\mathrm{d}S &= \left(\frac{\partial S_1}{\partial U_1}\right)_{\bar{N}_1,V_1}\mathrm{d}U_1 + \left(\frac{\partial S_1}{\partial \bar{N}_1}\right)_{U_1,V_1}\mathrm{d}\bar{N}_1 + \left(\frac{\partial S_2}{\partial U_2}\right)_{\bar{N}_2,V_2}\underbrace{\mathrm{d}U_2}_{=-\mathrm{d}U_1} + \left(\frac{\partial S_2}{\partial \bar{N}_2}\right)_{U_2,V_2}\underbrace{\mathrm{d}\bar{N}_2}_{=-\mathrm{d}\bar{N}_1} \\
&= \underbrace{\left[\left(\frac{\partial S_1}{\partial U_1}\right)_{\bar{N}_1,V_1} - \left(\frac{\partial S_2}{\partial U_2}\right)_{\bar{N}_2,V_2}\right]}_{=\frac{1}{T_1}-\frac{1}{T_2}}\mathrm{d}U_1 + \underbrace{\left[\left(\frac{\partial S_1}{\partial \bar{N}_1}\right)_{U_1,V_1} - \left(\frac{\partial S_2}{\partial \bar{N}_2}\right)_{U_2,V_2}\right]}_{=-\frac{\mu_1}{T_1}+\frac{\mu_2}{T_2}}\mathrm{d}\bar{N}_1 = 0,
\end{aligned} \tag{14.33}$$

where we have used Eq. (14.21) to identify the derivatives in the second term. Setting the first term to zero gives $T_1 = T_2 = T$ (thermal equilibrium). Then setting the second term to zero implies that

$$\boxed{\mu_1 = \mu_2}, \tag{14.34}$$

i.e., $\mu = \text{const}$ *across a system in equilibrium*. We also see that out of equilibrium, $\mathrm{d}S > 0$ implies that $\mathrm{d}\bar{N}_1 > 0 \Leftrightarrow \mu_2 > \mu_1$ and vice versa, so *particles flow from larger to smaller* μ.

Thus, if we figure out how to calculate μ, we should be able to predict equilibrium

states: how many particles, on average, there will be in each part of a system in equlibrium.

Exercise 14.1. Microcanonical Ensemble Revisited. Derive the grand canonical distribution starting from the microcanonical distribution (i.e., by considering a small subsystem exchanging particles and energy with a large, otherwise isolated system).

Hint. This is a generalisation of the derivation in §12.2.2.

14.4. *Grand Partition Function and Chemical Potential of Classical Ideal Gas*

So let us then learn how to calculate μ for our favorite special case of a classical monatomic ideal gas.

As always, the key question is what are the microstates? The answer is that they are the same as before [Eq. (11.18)], except now we can have an arbitrary number of particles, so

$$\alpha = (\alpha_N, N), \tag{14.35}$$

where

$$\alpha_N = \{n_{\boldsymbol{k}_1}, n_{\boldsymbol{k}_2}, \dots\}, \quad \sum_{\boldsymbol{k}} n_{\boldsymbol{k}} = N, \tag{14.36}$$

are the microstates of a gas of N particles and $n_{\boldsymbol{k}}$ are occupation numbers of the single-particle states designated by the wave vectors $\boldsymbol{k}$.

The grand partition function is, therefore,

$$\mathcal{Z} = \sum_{\alpha} e^{-\beta(E_\alpha - \mu N_\alpha)} = \sum_N e^{\beta\mu N} \sum_{\alpha_N} e^{-\beta E_{\alpha_N}} = \sum_N e^{\beta\mu N} Z_N, \tag{14.37}$$

where Z_N is the standard partition function of a gas of N particles, for which, neglecting quantum correlations, we may use Eq. (11.22):

$$\mathcal{Z} \approx \sum_N e^{\beta\mu N} \frac{Z_1^N}{N!} = \sum_N \frac{(e^{\beta\mu} Z_1)^N}{N!} = e^{Z_1 e^{\beta\mu}}. \tag{14.38}$$

The grand potential (14.24) is, therefore,

$$\Phi = -k_{\mathrm{B}} T \ln \mathcal{Z} = -k_{\mathrm{B}} T Z_1 e^{\mu/k_{\mathrm{B}}T}. \tag{14.39}$$

Using Eq. (14.27),

$$\bar{N} = -\left(\frac{\partial \Phi}{\partial \mu}\right)_{T,V} = Z_1 e^{\mu/k_{\mathrm{B}}T}, \tag{14.40}$$

whence

$$\boxed{\mu = -k_{\mathrm{B}} T \ln \frac{Z_1}{\bar{N}}}. \tag{14.41}$$

Now recall that the single-particle partition function is

$$Z_1 = \frac{V}{\lambda_{\mathrm{th}}^3} Z_1^{(\mathrm{internal})}, \quad \lambda_{\mathrm{th}} = \hbar \sqrt{\frac{2\pi}{m k_{\mathrm{B}} T}}, \tag{14.42}$$

where the first factor is the single-particle partition function associated with the particles' translational degrees of freedom and $Z_1^{(\mathrm{internal})}$ is the partition function associated with whatever internal degrees of freedom the particles have, e.g., for a diatomic gas,

$$Z_1^{(\mathrm{internal})} = Z_1^{(\mathrm{rotational})} Z_1^{(\mathrm{vibrational})}. \tag{14.43}$$

The chemical potential (14.41) is then

$$\boxed{\mu = k_{\rm B}T \ln \frac{n\lambda_{\rm th}^3}{Z_1^{(\rm internal)}}}, \tag{14.44}$$

where $n = \bar{N}/V$ is the (mean) number density of the gas. Note that, as $n\lambda_{\rm th}^3 \ll 1$ in the classical limit [Eq. (11.26)] and $Z_1^{(\rm internal)} \geqslant 1$ (because the number of internal states is at least 1), the above formula gives $\mu < 0$, as anticipated in §14.2.

Finally, using Eqs. (14.38–14.40), we get two remarkably simple formulae:

$$\mathcal{Z} = e^{\bar{N}}, \tag{14.45}$$

$$\Phi = -k_{\rm B}T\bar{N}, \tag{14.46}$$

whence the free energy is [see Eq. (14.24)]

$$F = \Phi + \mu\bar{N} = -k_{\rm B}T\bar{N}\left(1 - \ln \frac{n\lambda_{\rm th}^3}{Z_1^{(\rm internal)}}\right). \tag{14.47}$$

Comparing this with the expression that we previously had for the free energy of ideal gas, Eq. (11.28), we see that it has not changed, except for N having been replaced by $\bar{N}$ (and the appearance of $Z_1^{(\rm internal)}$, which we did not yet know about in §11.9). This means that *all our results previously derived for the case of a fixed number of particles survive, with* $N \to \bar{N}$.

Exercise 14.2. Particle Number Distribution. Consider a volume V of classical ideal gas with mean number density $n = \bar{N}/V$, where $\bar{N}$ is the mean number of particles in this volume. Starting from the grand canonical distribution, show that the probability to find exactly N particles in this volume is a Poisson distribution (thus, you will have recovered by a different method the result of Exercise 1.2a).

14.5. *Equilibria of Inhomogeneous Systems*

Let us now learn how to calculate the equilibrium states of a class of systems embedded in some external space-dependent potential $\varphi(\boldsymbol{r})$. For example, consider a classical ideal gas in a uniform gravitational field (an *atmosphere*), i.e., in a potential

$$\varphi(z) = gz. \tag{14.48}$$

If this system is in equilibrium, we know (§14.3) that both T and μ must be the same everywhere in it (think of subdividing the atmosphere into thin layers of constant z and requiring them all to be equilibrated with each other). So, we have

$$T(z) = \text{const} \tag{14.49}$$

("isothermal atmosphere"—obviously not a great model of the real thing, but this is not the point right now) and, using Eq. (14.41),

$$\mu(z) = -k_{\rm B}T \ln \frac{Z_1}{\bar{N}} = \text{const}. \tag{14.50}$$

The single-particle energy levels are the same as before plus potential energy per particle, $m\varphi = mgz$. In other words, $Z_1^{(\rm internal)}$ in Eq. (14.42) contains a factor corresponding the gravitational energy level (there is only one):

$$Z_1^{(\rm grav)} = e^{-\beta mgz}. \tag{14.51}$$

Therefore,

$$Z_1 = Z_1(g=0)Z_1^{(\mathrm{grav})} = Z_1(g=0)e^{-\beta mgz}, \tag{14.52}$$

where $Z_1(g=0)$ is the single-particle partition function for a gas with no gravity. Eq. (14.50) becomes

$$\mu = \mu(g=0) + mgz = k_\mathrm{B}T \ln \frac{n(z)\lambda_\mathrm{th}^3}{Z_1^{(\mathrm{internal})}} + mgz = \mathrm{const}, \tag{14.53}$$

where we have used Eq. (14.44) for $\mu(g=0)$, the chemical potential of ideal gas with zero gravity. The only way for z dependence to enter in the first term in Eq. (14.53) is via the particle density n because $T = \mathrm{const}$ in equilibrium. We find, therefore, a density profile known as the *Boltzmann distribution* (or *Boltzmann response*):

$$\boxed{n(z) = n(0)e^{-mgz/k_\mathrm{B}T}}\,, \tag{14.54}$$

where $n(0)$ is the density at $z = 0$.

The obvious generalisation of this result for a system in a general potential is

$$n(\boldsymbol{r}) \propto e^{-w(\boldsymbol{r})/k_\mathrm{B}T}, \tag{14.55}$$

where $w(\boldsymbol{r})$ is the potential energy per particle at location $\boldsymbol{r}$ (obviously, it must be assumed that that $w(\boldsymbol{r})$ varies at characteristic distances long enough for the quantum uncertanity as to the particle positions not to matter).

Remark. An interesting and useful formal lesson from Eq. (14.53) is that if we know the single-particle behaviour of some system and wish to adapt this knowledge to the same system but with energy levels shifted by some amount w, all we need to do is replace

$$\mu \to \mu + w \tag{14.56}$$

everywhere (this trick can be used, e.g., in the treatment of magnetisation; see Exercises 17.4 and 18.3). This makes sense: μ is the energy cost of adding a particle to the system [Eq. (14.23)] and so has to be larger by w if particles have additional energy w.

Exercise 14.3. Rotating Gas. A cylindrical container of radius R is filled with ideal gas at temperature T and rotating around the axis with angualar velocity Ω. The molecular mass is m. The mean density of the gas without rotation is $\bar{n}$. Assuming the gas is in isothermal equilibrium, what is the gas density at the edge of the cylinder, $n(R)$? Discuss the high and low temperature limits of your result.

Putting a mixture of two gases with different particle masses into a rotating container (a *centrifuge*) is a way to separate heavier from lighter particles (e.g., separate isotopes). Another method of doing this was via effusion (see comment at the end of §3). Making a set of sensible assumptions about all the parameters you need, assess the relative merits of the two methods.

14.6. *Chemical Potential and Thermodynamic Potentials*

Finally, we derive a few important general results concerning the relationship between μ and various thermodynamical quantities.

14.6.1. *Free Energy*

Using Eqs. (14.24) and (14.25), we find that

$$\boxed{\mathrm{d}F = -S\mathrm{d}T - P\mathrm{d}V + \mu\mathrm{d}\bar{N}}\,, \tag{14.57}$$

a generalisation of Eq. (7.2) to open systems. Hence the chemical potential is

$$\mu = \left(\frac{\partial F}{\partial \bar{N}}\right)_{T,V}, \tag{14.58}$$

free energy per particle in systems with fixed temperature and volume.

14.6.2. *Gibbs Free Energy*

By the same token, Gibbs free energy, $G = U - TS + PV = F + PV$, satisfies

$$\boxed{\mathrm{d}G = -S\mathrm{d}T + V\mathrm{d}P + \mu\mathrm{d}\bar{N}}, \tag{14.59}$$

and so the chemical potential is

$$\mu = \left(\frac{\partial G}{\partial \bar{N}}\right)_{T,P}, \tag{14.60}$$

Gibbs free energy per particle in systems with fixed temperature and pressure.

This result leads to a remarkable further simplification. Since $G = G(P, T, \bar{N})$ is an extensive quantity, P and T are intensive and $\bar{N}$ extensive, if we change $\bar{N}$ by a factor of λ, G must change by the same factor while P and T stay the same:

$$G(P, T, \lambda\bar{N}) = \lambda G(P, T, \bar{N}). \tag{14.61}$$

Differentiate this with respect to λ, then set $\lambda = 1$:

$$\left(\frac{\partial G}{\partial(\lambda\bar{N})}\right)_{P,T}\bar{N} = G, \quad \lambda = 1 \quad \Rightarrow \quad \underbrace{\left(\frac{\partial G}{\partial \bar{N}}\right)_{P,T}}_{\substack{=\mu,\\ \text{Eq. (14.60)}}}\bar{N} = G. \tag{14.62}$$

We have discovered that

$$\boxed{\mu = \frac{G}{\bar{N}}}, \tag{14.63}$$

i.e., *chemical potential is simply Gibbs-free-energy density!*

Exercise 14.4. Calculate $G = U - TS + PV$ for the ideal gas using the results of §11.9 and compare the outcome with Eq. (14.44).

Eq. (14.63) implies that μ *is an intensive quantity* (this was, of course, already obvious) and so

$$\forall\lambda, \quad \mu(P, T, \lambda\bar{N}) = \mu(P, T, \bar{N}) \quad \Rightarrow \quad \mu = \mu(P, T), \tag{14.64}$$

chemical potential is a function of pressure and temperature only. Indeed, for the ideal gas [Eq. (14.44)], using $P = nk_{\mathrm{B}}T$,

$$\mu = k_{\mathrm{B}}T\ln\frac{n\lambda_{\mathrm{th}}^3}{Z_1^{(\mathrm{internal})}} = k_{\mathrm{B}}T\ln P + \underbrace{k_{\mathrm{B}}T\ln\frac{\lambda_{\mathrm{th}}^3}{k_{\mathrm{B}}TZ_1^{(\mathrm{internal})}}}_{\text{function of } T \text{ only}}. \tag{14.65}$$

14.6.3. *Meaning of Grand Potential*

Using Eq. (14.63),

$$\Phi = F - \mu\bar{N} = F - G = -PV. \tag{14.66}$$

This implies that knowing Φ instantly gives us the equation of state:

$$\boxed{P = -\frac{\Phi}{V}}\,, \tag{14.67}$$

a simpler formula than Eq. (14.29), promised at the end of §14.2. Eq. (14.67) tells us that *pressure is minus the grand-potential density*, a way to give physical meaning to the thus far formal quantity Φ (it is minus "the total amount of pressure in the whole volume of the system").

Exercise 14.5. Pressure Ensemble. Throughout this course, we have repeatedly discussed systems whose volume is not fixed, but allowed to come to some equilibrium value under pressure. Yet, in both canonical (§9) and grand canonical (§14) ensembles, we treated volume as an external parameter, not as a quantity only measurable in the mean. In this Exercise, your objective is to construct an ensemble in which the volume is not fixed.

a) Consider a system with (discrete) microstates α to each of which corresponds some energy E_α and some volume V_α. Maximise the Gibbs entropy subject to the measured mean energy being U and the mean volume $\bar{V}$, with the number of particles N exactly fixed, and find the probabilities p_α. Show that the ("grandish") partition function for this ensemble can be defined as

$$\mathcal{Z} = \sum_\alpha e^{-\beta E_\alpha - \sigma V_\alpha}, \tag{14.68}$$

where β and σ are Lagrange multipliers. How are β and σ calculated?

b) Show that if we demand that the Gibbs entropy S_G for those probabilities be equal to S/k_B, where S is the thermodynamic entropy, then the Lagrange multiplier arising from the mean-volume constraint is

$$\sigma = \beta P = \frac{P}{k_\mathrm{B}T}, \tag{14.69}$$

where P is pressure. Thus, this ensemble describes a system under pressure set by the environment.

c) Prove that

$$\mathrm{d}U = T\mathrm{d}S - P\mathrm{d}\bar{V}. \tag{14.70}$$

d) Show that

$$-k_B T \ln \mathcal{Z} = G, \tag{14.71}$$

where G is the Gibbs free energy defined in the usual way. How does one calculate the equation of state for this ensemble?

e) Calculate the partition function $\mathcal{Z}$ for classical monatomic ideal gas in a container of changeable volume but impermeable to particles (e.g., a balloon made of inelastic material). You will find it useful to consider microstates of an ideal gas at fixed volume V and then sum up over all possible values of V. This sum (assumed discrete) can be converted to an integral via $\sum_V = \int_0^\infty \mathrm{d}V/\Delta V$, where ΔV is the "quantum of volume" (an artificial quantity shortly to be eliminated from the theory; how small must ΔV be in order for the sum and the integral to be good approximations of each other?).

Hint. You will need to use the formula $\int_0^\infty \mathrm{d}x\, x^N e^{-x} = N!$

f) Calculate G and find what conditions ΔV must satisfy in order for the resulting expression to coincide with the standard formula for the ideal gas (derived in Exercise 14.4) and be independent of ΔV (assume $N \gg 1$). If you can argue that the unphysical quantity ΔV does not affect any physically testable results, then your theory is sensible.

g) Show that the equation of state is

$$P = nk_\mathrm{B}T, \quad n = \frac{N}{\bar{V}}. \tag{14.72}$$

[cf. Lewis & Siegert 1956]

Exercise 14.6. Try constructing the "grandiose" ensemble, where all three of mean energy,

mean volume and mean number of particles are treated as measurable constraints. Why is such a theory impossible/meaningless?

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15. Multi-Species (Multi-Component) Systems

We now wish to consider systems containing several different "components": several species of molecules or particles, e.g.,

—solutions,

—mixtures of (reacting) chemicals,

—plasmas (ions + electrons + also neutral atoms if partially ionised).

15.1. *Generalisation of the Grand Canonical Formalism to Many Species*

We will characterise the thermodynamic state of a multi-species system by a mean energy, U, and the mean mumber of particles of each species s, $\bar{N}_s$. Completely analogously to what we did in §14.1, we maximise $S_{\rm G}$ subject all these quantities being fixed (by measurement)—there will be a Lagrange multiplier for each s, so each species will have its own chemical potential μ_s. Leaving the algebra to you as an **Exercise**, here are the results (where the equations that these results are generalisations of are indicated):

$$\text{Eq. (14.8)} \to \quad p_\alpha = \frac{e^{-\beta(E_\alpha - \sum_s \mu_s N_{s\alpha})}}{\mathcal{Z}}, \tag{15.1}$$

$$\text{Eq. (14.9)} \to \quad \mathcal{Z} = \sum_\alpha e^{-\beta(E_\alpha - \sum_s \mu_s N_{s\alpha})}, \tag{15.2}$$

where $N_{s\alpha}$ is the number of particles of species s corresponding to the microstate α of the whole system, and, given $s = 1, \ldots, m$ species, the $m+1$ Lagrange multipliers β, $\mu_1, \ldots, \mu_m$ are determined from the following $m+1$ equations:

$$\text{Eq. (14.11)} \to \quad U = -\frac{\partial \ln \mathcal{Z}}{\partial \beta} + \sum_s \mu_s \bar{N}_s, \tag{15.3}$$

$$\text{Eq. (14.13)} \to \quad \bar{N}_s = \frac{1}{\beta}\frac{\partial \ln \mathcal{Z}}{\partial \mu_s}. \tag{15.4}$$

The grand potential is

$$\text{Eq. (14.24)} \to \quad \Phi = -k_{\rm B} T \ln \mathcal{Z} = U - TS - \sum_s \mu_s \bar{N}_s, \tag{15.5}$$

$$\text{Eq. (14.25)} \to \quad \mathrm{d}\Phi = -S\mathrm{d}T - P\mathrm{d}V - \sum_s \bar{N}_s \mathrm{d}\mu_s, \tag{15.6}$$

and the multispecies thermodynamics, i.e., the expressions for S, $\bar{N}_s$, U and P, can be read off from this in the same manner as Eqs. (14.26–14.29) were. The differentials of the thermodynamic potentials (defined in the usual way) are, therefore

$$\text{Eq. (14.22)} \to \quad \mathrm{d}U = T\mathrm{d}S - P\mathrm{d}V + \sum_s \mu_s \mathrm{d}\bar{N}_s, \tag{15.7}$$

$$\text{Eq. (14.57)} \to \quad \mathrm{d}F = -S\mathrm{d}T - P\mathrm{d}V + \sum_s \mu_s \mathrm{d}\bar{N}_s, \tag{15.8}$$

$$\text{Eq. (14.59)} \to \quad \mathrm{d}G = -S\mathrm{d}T + V\mathrm{d}P + \sum_s \mu_s \mathrm{d}\bar{N}_s, \tag{15.9}$$

whence follow the expressions for the chemical potential of species s, analogous to Eqs. (14.21), (14.23), (14.58) and (14.63):

$$\mu_s = -T\left(\frac{\partial S}{\partial \bar{N}_s}\right)_{U,V,\bar{N}_{s'\neq s}} = \left(\frac{\partial U}{\partial \bar{N}_s}\right)_{S,V,\bar{N}_{s'\neq s}} = \left(\frac{\partial F}{\partial \bar{N}_s}\right)_{T,V,\bar{N}_{s'\neq s}} = \left(\frac{\partial G}{\partial \bar{N}_s}\right)_{T,P,\bar{N}_{s'\neq s}}, \tag{15.10}$$

where all these derivatives are taken at constant $\bar{N}_{s'}$, where $s' = 1, \ldots, s-1, s+1, \ldots, m$.

15.1.1. *Gibbs Free Energy vs. μ_s*

Similarly to the case of one species, there is a special relationship between the chemical potentials and Gibbs free energy (cf. §14.6.2). Indeed, since $G = U - TS + PV$ is extensive and so are all the particle numbers $\bar{N}_1, \ldots, \bar{N}_m$, scaling the system by λ gives

$$G(P, T, \lambda\bar{N}_1, \ldots, \lambda\bar{N}_m) = \lambda G(P, T, \bar{N}_1, \ldots, \bar{N}_m), \tag{15.11}$$

which, upon differentiation with respect to λ and then setting $\lambda = 1$, gives us [cf. Eq. (14.62)]

$$\sum_s \underbrace{\left(\frac{\partial G}{\partial \bar{N}_s}\right)_{T,P,\bar{N}_{s'\neq s}}}_{\substack{=\mu_s,\\ \text{Eq. (15.10)}}} \bar{N}_s = G, \tag{15.12}$$

whence it follows that the Gibbs free energy of a multispecies system is "the total amount of chemical potential" amongst all species:

$$\boxed{G = \sum_s \mu_s \bar{N}_s}\,. \tag{15.13}$$

Note that this implies, via Eq. (15.5), that

$$\Phi = U - TS - G = -PV, \tag{15.14}$$

and the equation of state can again be obtained from this [Eq. (14.67)].

15.1.2. *Fractional Concentrations*

Since μ_s are all intensive [follows from Eq. (15.10)], they do not depend on the total number of particles, but only on other intensive quantities, viz., pressure, temperature and the *fractional concentrations* of all the species:

$$\mu_s = \mu_s(P, T, c_1, \ldots, c_{m-1}), \tag{15.15}$$

where

$$c_s = \frac{\bar{N}_s}{\bar{N}}, \quad \bar{N} = \sum_s \bar{N}_s. \tag{15.16}$$

There are only $m - 1$ independent fractional concentrations as obviously $\sum_s c_s = 1$.

15.2. *Particle Equilibrium and Gibbs Phase Rule*

Arguing exactly like we did in §14.3, one can prove that *across a system in equilibrium,* $\mu_s = \text{const}$ *for each species* (**Exercise**). Note that chemical potentials of *different* species do *not* need to be equal even if they are in contact within a system.

There is a useful immediate consequence of this. Consider a system of m species, each

of which can be in r phases. Then, in equilibrium,

$$\begin{aligned}&\mu_1^{(\text{phase } 1)} = \mu_1^{(\text{phase } 2)} = \cdots = \mu_1^{(\text{phase } r)},\\ &\cdots\\ &\mu_m^{(\text{phase } 1)} = \mu_m^{(\text{phase } 2)} = \cdots = \mu_m^{(\text{phase } r)},\end{aligned} \tag{15.17}$$

where each $\mu_s^{(\text{phase } p)}$ is a function of P, T and the fractional concentrations of all species, each in r phases:

$$\mu_s^{(\text{phase } p)} = \mu_s^{(\text{phase } p)}\left(P, T, c_1^{(\text{phase } 1)}, \ldots, c_{m-1}^{(\text{phase } 1)}, \ldots, c_1^{(\text{phase } r)}, \ldots, c_{m-1}^{(\text{phase } r)}\right). \tag{15.18}$$

Thus, we have $m(r-1)$ equations for $2 + r(m-1)$ unknowns. In order for them to have solutions (not necessarily unique), the number of equations must not exceed the number of unknowns, viz.,

$$\boxed{r \leqslant m + 2}\,. \tag{15.19}$$

This is called the *Gibbs phase rule*. It implies, for example, that a single species ($m = 1$) can only support an equilibrium state with $r \leqslant 3$ coexisting phases (e.g., gas, liquid, solid).

Eqs. (15.17) are the starting point for the *theory of phase transitions*, of which more will be said in Part VII.

15.3. *Chemical Equilibrium*

Now let us work out how μ_s for *different* species are related to each other in equilibrium. Obviously, they need to be related at all only if these different species can transmute into each other and so the system can adjust their fractional concentrations in its quest for an optimal (maximum-entropy) equilibrium state—i.e., if these species are subject to *chemical (or atomic) reactions.* These reactions can usually be expressed in the form

$$\sum_s \nu_s A_s = 0, \tag{15.20}$$

where A_s designate the species and ν_s are integers encoding their relative amounts participating in a reaction. For example,

$$2\mathrm{H}_2 + \mathrm{O}_2 = 2\mathrm{H}_2\mathrm{O} \tag{15.21}$$

is encoded by $A_1 = \mathrm{H}_2$, $A_2 = \mathrm{O}_2$, $A_3 = \mathrm{H}_2\mathrm{O}$ and $\nu_1 = 2$, $\nu_2 = 1$, $\nu_3 = -2$;

$$e^+ + e^- = \gamma \quad \Leftrightarrow \quad \nu_1 = 1, \nu_2 = 2, \nu_3 = -1 \tag{15.22}$$

(pair production/annihilation),

$$p^+ + e^- = \mathrm{H} \quad \Leftrightarrow \quad \nu_1 = 1, \nu_2 = 2, \nu_3 = -1 \tag{15.23}$$

(ionisation/recombination of atomic hydrogen), etc. The set of numbers $\{\nu_s\}$ fully specifies a reaction, as far as Statistical Mechanics is concerned (as we are about to see). In general, *reactions can go both ways*, until there is a stable soup where the fractional concentration of each species has assumed its equilibrium value.

How do we find these equilibrium values?

At constant T and P (which is the usual set up in a chemistry lab), in order to find the equilibrium, one must maximise Gibbs free energy, viz., from Eq. (15.9),

$$\mathrm{d}G = \sum_s \mu_s \mathrm{d}\bar{N}_s = 0. \tag{15.24}$$

The proof of this is the standard so-called "availability" argument, which is as follows. Consider a system in contact with environment. As it equilibrates, the total energy is conserved,

$$\mathrm{d}(U + U_{\mathrm{env}}) = 0, \tag{15.25}$$

whereas the total entropy must grow,

$$\mathrm{d}(S + S_{\mathrm{env}}) \geqslant 0. \tag{15.26}$$

From Eq. (15.25),

$$\mathrm{d}U = -\mathrm{d}U_{\mathrm{env}} = -T_{\mathrm{env}}\mathrm{d}S_{\mathrm{env}} + P_{\mathrm{env}}\mathrm{d}V_{\mathrm{env}}. \tag{15.27}$$

Note that the number of particles in the environment does not change: we assume that all the exchanges/transmutations of matter occur within the system. Since $\mathrm{d}V_{\mathrm{env}} = -\mathrm{d}V$ (the volume of the world is constant), this gives

$$T_{\mathrm{env}}\mathrm{d}S_{\mathrm{env}} = -\mathrm{d}U - P_{\mathrm{env}}\mathrm{d}V. \tag{15.28}$$

Now, from this and Eq. (15.26),

$$0 \leqslant T_{\mathrm{env}}(\mathrm{d}S + \mathrm{d}S_{\mathrm{env}}) = T_{\mathrm{env}}\mathrm{d}S - \mathrm{d}U - P_{\mathrm{env}}\mathrm{d}V = -\mathrm{d}(U - T_{\mathrm{env}}S + P_{\mathrm{env}}V) = -\mathrm{d}G. \tag{15.29}$$

Thus, $\mathrm{d}G \leqslant 0$, so the final equilibrium is achieved at the minimum value of G (the same argument mandates $\mathrm{d}F \leqslant 0$ when $V = \text{const}$ and, unsurprisingly, $\mathrm{d}S \geqslant 0$ when also $U = \text{const}$, i.e., when the system is isolated).

Eq. (15.20) implies that, as the reaction occurs,

$$\mathrm{d}\bar{N}_1 : \mathrm{d}\bar{N}_2 : \cdots : \mathrm{d}\bar{N}_m = \nu_1 : \nu_2 : \cdots : \nu_m. \tag{15.30}$$

Therefore, Eq. (15.24) becomes

$$\boxed{\sum_s \nu_s \mu_s = 0}\,. \tag{15.31}$$

This is the *equation of chemical equilibrium*. There will be an equation like this for each reaction that the system is capable of (each specified by a set of numbers $\{\nu_s\}$). All these equations together give a set of constraints on fractional concentrations $c_1, \ldots, c_{m-1}$ because these are the only variables μ_s depends on, at constant P and T [Eq. (15.15)]. Note that the number of equations (15.31) is not necessarily equal to the number of unknowns and so solutions are not necessarily unique (or exist).

15.4. *Chemical Equilibrium in a Mixture of Classical Ideal Gases: Law of Mass Action*

In order to apply Eq. (15.31), we need explicit expressions for $\mu_s(P, T, c_1, \ldots, c_{m-1})$. We can get them from, e.g., Eq. (15.4), which we can rewrite so:

$$c_s = \frac{k_{\mathrm{B}}T}{\bar{N}} \frac{\partial \ln \mathcal{Z}}{\partial \mu_s} \tag{15.32}$$

(a system of m equations for $s = 1, \ldots, m$). This means we need the grand partition function for our mixture—if the mixture is of classical ideal gases, we can calculate it by direct generalisation of the relevant results of §14.4. Since the gases are ideal, there are no interactions between particles and so each species within a gas behaves as a separate subsystem,† in equilibrium with the rest. Therefore,

$$\mathcal{Z} = \prod_s \mathcal{Z}_s = \exp\left(\sum_s Z_{1s} e^{\beta\mu_s}\right), \tag{15.33}$$

† This is not true in general for multicomponent chemical systems as they can, in principle, interpenetrate, be strongly interacting and have collective energy levels not simply equal to sums of the energy levels of individual components.

where $\mathcal{Z}_s$ is the grand partition function of the species s, we have used Eq. (14.38), and

$$Z_{1s} = \frac{V}{\lambda_{\mathrm{th}s}^3} Z_{1s}^{(\mathrm{internal})}, \quad \lambda_{\mathrm{th}s} = \hbar\sqrt{\frac{2\pi}{m_s k_{\mathrm{B}} T}}, \tag{15.34}$$

is the single-particle partition function of species s.

Exercise 15.1. Derive Eq. (15.33) directly, by constructing the microstates of a mixture of ideal gases and then summing over all these microstates to get $\mathcal{Z}$.

Using Eq. (15.33) in Eq. (15.32), we find

$$c_s \bar{N} = k_{\mathrm{B}} T \frac{\partial}{\mu_s} \sum_{s'} Z_{1s'} e^{\beta \mu_{s'}} = Z_{1s} e^{\beta \mu_s} \tag{15.35}$$

and, after using Eq. (15.34), we get [cf. Eqs. (14.41) and (14.44)],

$$\mu_s = -k_{\mathrm{B}} T \ln \frac{Z_{1s}}{c_s \bar{N}} = k_{\mathrm{B}} T \ln \left[\frac{c_s n \lambda_{\mathrm{th}s}^3}{Z_{1s}^{(\mathrm{internal})}} \right] = k_{\mathrm{B}} T \ln \left[\frac{c_s}{Z_{1s}^{(\mathrm{internal})}} \frac{P \lambda_{\mathrm{th}s}^3}{k_{\mathrm{B}} T} \right], \tag{15.36}$$

where $n = \bar{N}/V$ is the overall number density of the mixture and we have used *Dalton's law:* total pressure is the sum of the pressures of individual species,

$$P = \sum_s n_s k_{\mathrm{B}} T = n k_{\mathrm{B}} T \tag{15.37}$$

(see Exercise 4.3b or convince yourself, starting from Eq. (15.14), that this is true).

Finally, inserting Eq. (15.36) into Eq. (15.31),

$$k_{\mathrm{B}} T \sum_s \nu_s \ln \left[\frac{c_s}{Z_{1s}^{(\mathrm{internal})}} \frac{P \lambda_{\mathrm{th}s}^3}{k_{\mathrm{B}} T} \right] = 0, \tag{15.38}$$

so the fractional concentrations must obey

$$\sum_s \nu_s \ln c_s = -\sum_s \nu_s \ln \left[\frac{P \lambda_{\mathrm{th}s}^3}{Z_{1s}^{(\mathrm{internal})} k_{\mathrm{B}} T} \right], \tag{15.39}$$

or, to write this in the commonly used form highlighting pressure and temperature dependence,

$$\boxed{\prod_s c_s^{\nu_s} = P^{-\sum_s \nu_s} \underbrace{\prod_s \left[\frac{k_{\mathrm{B}} T}{\lambda_{\mathrm{th}s}^3} Z_{1s}^{(\mathrm{internal})} \right]^{\nu_s}}_{\text{function of } T \text{ only}} \equiv K(P,T)}. \tag{15.40}$$

The right-hand side of this equation is called the *chemical equilibrium constant*, which, for any given reaction (defined by ν_s's), is a known function of P, T and the microphysics of the participating particles. The equation itself is known as the *Law of Mass Action* (because of the particle masses m_s entering $K(P,T)$ via $\lambda_{\mathrm{th}s}$).

Eq. (15.40), together with the requirement that $\sum_s c_s = 1$, constrains fractional concentrations in chemical equilibrium. It also allows one to determine in which direction the reaction will go from some initial non-equlibrium state:

—if $\prod_s c_s^{\nu_s} > K(P,T)$, the reaction is *direct*, i.e., the concentrations c_s of the species with $\nu_s > 0$, which are on the left-hand side of Eq. (15.20), will go down, while those of

the species with $\nu_s < 0$, on the right-hand side of Eq. (15.20), will go up;

—if $\prod_s c_s^{\nu_s} < K(P,T)$, the reaction is *reverse*.

This is all the chemistry you need to know! (At least in this course.)

Exercise 15.2. Partially Ionised Plasma. Consider atomic-hydrogen gas at high enough temperature that ionisation and recombination are occurring. The reaction is given by Eq. (15.23). Our goal is to find, as a function of density and temperature (or pressure and temperature), the degree of ionisation $\chi = n_p/n$, where n_p is the proton number density, $n = n_{\rm H} + n_p$ is the total number density of hydrogen, ionised or not, and $n_{\rm H}$ is the number density of the un-ionised H atoms. Note that n is fixed (conservation of nucleons). Assume overall charge neutrality of the system.

a) What is the relation between chemical potentials of the H, p and e gases if the system is in chemical equlibrium?

b) Treating all three species as classical ideal gases, show that in equlibrium,

$$\boxed{\frac{n_e n_p}{n_{\rm H}} = \left(\frac{m_e k_{\rm B} T}{2\pi\hbar^2}\right)^{3/2} e^{-R/k_{\rm B}T}}\,, \tag{15.41}$$

where $R = 13.6$ eV (1 Rydberg) is the ionisation energy of hydrogen. This formula is known as the *Saha equation*.

Hint. Remember that you have to include the internal energy levels into the partition function for the hydrogen atom. You may assume that only the ground state energy level $-R$ matters (i.e., neglect all excited states).

c) Find the degree of ionisation $\chi = n_p/n$ as a function of n and T. Does χ go up or down as density is decreased? Why? Consider a cloud of hydrogen with $n \sim 1$ cm^{-3}. Roughly at what temperature would most of it be ionised? These are approximately the conditions in the so called "warm" phase of the interstellar medium—the stuff that much of the Galaxy is filled with (although, admittedly, the Law of Mass Action is not thought to be a very good approximation for interstellar medium, because it is not exactly in equilibrium).

d) Now find an expression for χ as a function of total gas pressure P and temperature T. Sketch χ as a function of T at several constant values of P.

PART VI
Quantum Gases

16. Quantum Ideal Gases

So far, in all our calculations of partition functions for gases, we have stayed within the classical limit, where the key assumption was that the number of single-particle states available to particles was much greater than the number of these particles, so the probability of any one particle occupying any given single-particle state was small and, therefore, the probability of more than one particle laying claim to the same state could be completely discounted. The time has now come to relax this assumption, but first, let me explain what are those *quantum correlations* dealing with which we have so far been so assiduous in avoiding.

16.1. *Fermions and Bosons*

- Consider a 2-particle wave function, $\psi(1,2)$, where the first argument corresponds to the first particle, the second ti the second and the notation means that the first is in state 1 and the second in state 2.
- Now swap the two particles: $\psi(1,2) \to \psi(2,1)$.

• If the particles are indistinguishable, this operation cannot change any observables, so the probability density cannot change under the swapping operation:

$$|\psi(2,1)|^2 = |\psi(1,2)|^2 \quad \Rightarrow \quad \psi(2,1) = e^{i\phi}\psi(1,2) \tag{16.1}$$

(swapping can only bring in a phase factor).

• Apply the swapping operation twice:

$$\psi(2,1) = e^{i\phi}\psi(1,2) = e^{2i\phi}\psi(2,1) \quad \Rightarrow \quad e^{2i\phi} = 1 \quad \Rightarrow \quad e^{i\phi} = \pm 1. \tag{16.2}$$

This argument tells us that there can be (and, as it turns out, there are) two types of particles, corresponding to two possible exchange symmetries:†

$$1) \quad \boxed{\psi(2,1) = \psi(1,2)}\,, \tag{16.3}$$

called *bosons*—they can be proven to be particles with *integer spin*, e.g., photons (spin 1), ^{4}He atoms (spin 0);

$$2) \quad \boxed{\psi(2,1) = -\psi(1,2)}\,, \tag{16.4}$$

called *fermions*—these are particles with *half-integer spin*, e.g., e, n, p, ^{3}He (spin 1/2).

The fermions are subject to the *Pauli exclusion principle:* if the states 1 and 2 are the same, then

$$\psi(1,1) = -\psi(1,1) = 0, \tag{16.5}$$

so *no two fermions can be in the same state.* This is precisely an example of *quantum correlations*: even though the gas is ideal and so the fermions are non-interacting, the system as a whole "knows" which single-particle states are occupied and so unavailable to other particles.

What does all this mean for the statistical mechanics of systems composed of bosons or fermions? Recall that the microstates of a box of ideal gas were specified in terms of occupation numbers n_i of single-particle states i‡. What we have just inferred from the exchange symmetries determines what values these occupation numbers can take:

—for bosons, $n_i = 0, 1, 2, 3, \ldots$ (any integer),

—for fermions, $n_i = 0$ or 1 (no more than 1 particle in each state).

Armed with this knowledge, we are ready to start computing.

16.2. *Partition Function*

The grand partition function is given by Eq. (14.9), where the microstates are $\alpha = \{n_i\}$ (sets of occupation numbers), the energy levels of the system are

$$E_\alpha = \sum_i n_i \varepsilon_i \tag{16.6}$$

(ε_i are the energies of the single-particle states), and the particle number in state α is

$$N_\alpha = \sum_i n_i. \tag{16.7}$$

† See Landau & Lifshitz (1981), §61–62 for a rigorous generalisation to N-particle wave functions and the derivation of the connection between a particle's spin and the exchange symmetry.

‡ In §11, i was $\boldsymbol{k}$, but in general, single-particle states will depend on other quantum numbers as well, e.g., spin, angular momentum, vibrational levels, etc.—but they are all still discrete, so we simply index them by i.

Then

$$\mathcal{Z} = \sum_{\alpha} e^{-\beta(E_\alpha - \mu N_\alpha)} = \sum_{\{n_i\}} e^{-\beta \sum_i n_i(\varepsilon_i - \mu)}$$
$$= \underbrace{\sum_{n_1}\sum_{n_2}\sum_{n_3}\cdots}_{\substack{\text{over all possible} \\ \text{values of } \{n_i\}}} \prod_i e^{-\beta n_i(\varepsilon_i-\mu)} = \prod_i \sum_{n_i} e^{-\beta n_i(\varepsilon_i-\mu)}. \tag{16.8}$$

For fermions, $n_i = 0$ or 1, so the sum $\sum_{n_i}$ has only two members and so

$$\mathcal{Z} = \prod_i \left[1 + e^{-\beta(\varepsilon_i-\mu)}\right]. \tag{16.9}$$

For bosons, $n_i = 0, 1, 2, 3, \ldots$, so

$$\mathcal{Z} = \prod_i \underbrace{\sum_{n_i=0}^{\infty} \left[e^{-\beta(\varepsilon_i-\mu)}\right]^{n_i}}_{\text{geometric series}} = \prod_i \frac{1}{1 - e^{-\beta(\varepsilon_i-\mu)}}. \tag{16.10}$$

Or, to write this compactly,

$$\boxed{\ln \mathcal{Z} = \pm \sum_i \ln\left[1 \pm e^{-\beta(\varepsilon_i-\mu)}\right]}, \tag{16.11}$$

where "+" corresponds to fermions and "−" to bosons.

16.3. *Occupation Number Statistics and Thermodynamics*

The probability for a give set of occupation numbers to occur is given by the grand canonical distribution (14.8):

$$p_\alpha \equiv p(n_1, n_2, n_3, \ldots) = \frac{1}{\mathcal{Z}}\, e^{-\beta \sum_i n_i(\varepsilon_i-\mu)}. \tag{16.12}$$

Therefore, the *mean occupation number* of a single-particle state j is

$$\bar{n}_j \equiv \langle n_j \rangle = \sum_{\{n_i\}} n_j\, p(n_1, n_2, n_3, \ldots) = \frac{1}{\mathcal{Z}} \sum_{\{n_i\}} n_j\, e^{-\beta \sum_i n_i(\varepsilon_i-\mu)} = -\frac{1}{\beta}\frac{\partial \ln \mathcal{Z}}{\partial \varepsilon_j}. \tag{16.13}$$

Using Eq. (16.11), we get

$$\boxed{\bar{n}_i = \frac{1}{e^{\beta(\varepsilon_i-\mu)} \pm 1}}. \tag{16.14}$$

Thus, we can predict how many particles will be in any given state on average (this is exactly the same as calculating the *distribution function*, which was our main vehicle in Kinetic Theory). The "+" sign in Eq. (16.14) gives us the *Fermi–Dirac statistics* and the "−" sign the *Bose–Einstein statistics*.

Exercise 16.1. Entropy of Fermi and Bose Gases out of Equilibrium. It is possible to construct the statistical mechanics of quantum ideal gases directly in terms of occupation numbers. In the spirit of Gibbs (§12.2.3), consider an ensemble of $\mathcal{N}$ copies of our system (gas in a box). Let $\mathcal{N}_i$ be the number of particles that are in the single-particle state i across this entire über-system. Then the average occupation number of the state i per copy is $\bar{n}_i = \mathcal{N}_i/\mathcal{N}$. If the number of ways in which a given assignment $\{\mathcal{N}_1, \mathcal{N}_2, \ldots, \mathcal{N}_i, \ldots\}$ of particles to single-particle

states can be achieved is $\Omega_{\mathcal{N}}(\mathcal{N}_1, \mathcal{N}_2, \dots)$, then the Gibbs entropy associated with the set of occupation numbers $(\bar{n}_1, \bar{n}_2, \dots, \bar{n}_i, \dots)$ will be

$$S_{\rm G}(\bar{n}_1, \bar{n}_2, \dots) = \frac{\ln \Omega_{\mathcal{N}}(\mathcal{N}_1, \mathcal{N}_2, \dots)}{\mathcal{N}} \tag{16.15}$$

in the limit $\mathcal{N} \to \infty$ and all $\mathcal{N}_i \to \infty$ while keeping $\bar{n}_i$ constant. This is very similar to the construction in §§8.1.3 or 12.2.3 of the Gibbs entropy of a set of probabilities of microstates, except we now have different rules about how many particles can be in any given microstate i:

— for fermions, each copy of the system in the ensemble can have only one or none of the $\mathcal{N}_i$ particles available for each state i;

— for bosons, the $\mathcal{N}_i$ particles in each state i can be distributed completely arbitrarily between the $\mathcal{N}$ copies.

a) Prove that the Gibbs entropy, as defined above, will be

$$S_{\rm G} = -\sum_i \left[\bar{n}_i \ln \bar{n}_i \pm (1 \mp \bar{n}_i) \ln(1 \mp \bar{n}_i)\right], \tag{16.16}$$

where the upper sign is for fermions and the lower for bosons.

Hint. Observe that $\Omega_{\mathcal{N}}(\mathcal{N}_1, \mathcal{N}_2, \dots) = \prod_i \Omega_i$, where Ω_i is the number of ways to asign the $\mathcal{N}_i$ particles available for the microstate i to the $\mathcal{N}$ copies in the ensemble.

Note that Eq. (16.16) certainly holds for Fermi and Bose gases *in equilibrium*, as derived above (convince yourself that this is the case), but you have shown now that it also holds *out of equilibrium*, i.e., for arbitrary sets of occupation numbers (arbitrary *particle distributions*).

b) Considering a system with fixed mean energy and number of particles and maximising $S_{\rm G}$, derive from Eq. (16.16) the Fermi–Dirac and Bose–Einstein formulae (16.14) for the mean occupation numbers in equilibrium.

c) Devise a way to treat a classical ideal gas by the same method.

Eq. (16.14) is useful provided we know

—the single-particle energy levels ε_i for a given system (which we get from Quantum Mechanics),

—the chemical potential $\mu(n, T)$ ($n = N/V$ is the overall particle density), the equation for which is simply

$$\boxed{N = \sum_i \bar{n}_i} \tag{16.17}$$

[equivalent to Eq. (14.13)]. From this point on, we will drop the bars on N as we really are interested in the case with a fixed number of particles again and the use of grand canonical ensemble was a matter of analytical convenience. As we explained at the end of §14.1, canonical results are recoverable from the grand canonical ones because they correspond to the special case of $N_\alpha = N$ for all α.

Exercise 16.2. Show that using Eq. (16.11) in Eq. (14.13) gives the same result as using Eq. (16.14) in Eq. (16.17).

To construct the thermodynamics of a quantum gas, we then need to calculate the mean energy

$$\boxed{U = \sum_i \varepsilon_i \bar{n}_i} \tag{16.18}$$

[equivalent to Eq. (14.11)], the grand potential and the equation of state [Eqs. (14.24) and (14.67)],

$$\boxed{\Phi = -k_{\rm B} T \ln \mathcal{Z} \quad \Rightarrow \quad P = -\frac{\Phi}{V}}, \tag{16.19}$$

and the entropy

$$\boxed{S = \frac{U - \Phi - \mu N}{T}} \tag{16.20}$$

[equivalent to Eq. (14.26)], whence we can get the heat capacities, etc.

16.4. *Preview of Further Developments*

16.4.1. *High Temperatures*

16.4.2. *Fermions at Low Temperature*

16.4.3. *Bosons at Low Temperature*

16.4.4. *Photons*

16.5. *Calculations in Continuum Limit*

[Literature: Landau & Lifshitz (1980), §56; Schrödinger (1990), Ch. VIII]

16.5.1. *From Sums to Integrals*

16.5.2. *Chemical Potential*

16.5.3. *Energy*

16.5.4. *Grand Potential and Equation of State*

16.5.5. *Entropy*

16.6. *Classical Limit (Sanity Check)*

16.7. *Degeneration*

Exercise 16.3. Pair Plasma.
Exercise 16.4. Creation/Annihilation of Matter.

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17. Degenerate Fermi Gas

[Literature: Landau & Lifshitz (1980), §§57–58; Schrödinger (1990), Ch. VIII(a)]

17.1. *Energy and Equation of State*

Exercise 17.1. Stability of Stars.

17.2. *Heat Capacity*

17.2.1. *Qualitative Calculation*

17.2.2. *Quantitative Calculation*

Exercise 17.2. Ratio of Heat Capacities for Fermi Gas.
Exercise 17.3. Ultrarelativistic Quantum Gas.
Exercise 17.4. Paramagnetism of a Degenerate Electron Gas (Pauli Magnetism).

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18. Degenerate Bose Gas

[Literature: Landau & Lifshitz (1980), §62; Schrödinger (1990), Ch. VIII(b)]

18.1. *Bose-Einstein Condensation*
18.2. *Thermodynamics of Degenerate Bose Gas*
18.2.1. *Energy*
18.2.2. *Equation of State*
18.2.3. *Heat Capacity*

Exercise 18.1. Degenerate Bose Gas in 2D.
Exercise 18.2. Low Energy Levels in Degenerate Bose Gas.
Exercise 18.3. Paramagnetism of Degenerate Bose Gas.

19. Thermal Radiation (Photon Gas)

This part of the course was taught by Professor Julien Devriendt.

Exercise 19.1. Work out the theory of thermal radiation using the results of Exercise 17.3.

Exercise 19.2. Heat Capacity of Metals.

PART VII
Thermodynamics of Real Gases

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REFERENCES

Binney, J. & Skinner, D. 2013 *The Physics of Quantum Mechanics*. Oxford University Press.
Blundell, S. J. & Blundell, K. M. 2009 *Concepts in Thermal Physics*. Oxford University Press, 2nd Edition.

Boltzmann, L. 1995 *Lectures on Gas Theory*. Dover.

Chandrasekhar, S. 2003 *Hydrodynamic and Hydromagnetic Stability*. Dover.

Chapman, S. & Cowling, T. G. 1991 *The Mathematical Theory of Non-uniform Gases*. Cambridge University Press, 3rd Edition.

Dellar, P. J. 2015 *Kinetic Theory of Gases*. Lecture Notes for the Oxford MMathPhys course on Kinetic Theory; URL: http://people.maths.ox.ac.uk/dellar/MMPkinetic.html.

Dunkel, J. & Hilbert, S. 2014 Consistent Thermostatistics Forbids Negative Absolute Temperatures. *Nature Physics* **10**, 67.

Ginzburg, V. L., Levin, L. M., Sivukhin, D. V. & Yakovlev, I. A. 2006 *Problems for a Course of General Physics II. Thermodynamics and Molecular Physics*. Moscow: Fizmatlit.

Jaynes, E. T. 1965 Gibbs vs. Boltzmann Entropies. *American Journal of Physics* **33**, 391.

Jaynes, E. T. 2003 *Probability Theory: The Logic of Science*. Cambridge: Cambridge University Press.

Landau, L. D. & Lifshitz, E. M. 1980 *Statistical Physics (L. D. Landau and E. M. Lifshitz's Course of Theoretical Physics, Volume 5)*. Butterworth–Heinemann.

Landau, L. D. & Lifshitz, E. M. 1981 *Quantum Mechanics: Non-Relativistic Teeory (L. D. Landau and E. M. Lifshitz's Course of Theoretical Physics, Volume 3)*. Elsevier.

Lewis, M. B. & Siegert, A. J. F. 1956 Extension of the Condensation Theory of Yang and Lee to the Pressure Ensemble. *Physical Review* **101**, 1227.

Lifshitz, E. M. & Pitaevskii, L. P. 1981 *Physical Kinetics (L. D. Landau and E. M. Lifshitz's Course of Theoretical Physics, Volume 10)*. Elsevier.

Maxwell, J. C. 1860 Illustrations of the dynamical theory of gases.—Part I. On the motions and collisions of perfectly elastic spheres. *The London, Edinburgh and Dublin Philosophical Magazine and Journal of Science* **19**, 19.

Pauli, W. 2003 *Thermodynamics and the Kinetic Theory of Gases (Pauli Lectures on Physics, Volume 3)*. Dover.

Schekochihin, A. A. 2015 *Kinetic Theory of Plasmas*. Lecture Notes for the Oxford MMathPhys course on Kinetic Theory; URL: http://www-thphys.physics.ox.ac.uk/people/AlexanderSchekochihin/KT/2015/KTLectureNotes.pdf.

Schrödinger, E. 1990 *Statistical Thermodynamics*. Dover.

Shannon, C. 1948 A Mathemnatical Theory of Communication. *The Bell System Technical Journal* **27**, 379.

Sinai, Ya. G. 1992 *Probability Theory: An Introductory Course*. Springer.

Szilard, L. 1929 On the decrease of entropy in a thermodynamic system by the intervention of intelligent beings. *Zeitschrift für Physik* **53**, 840, English translation in *Behavioral Science* **9**, 301 (1964).

CONTENTS

1. Statistical Description of a Gas 1
1.1. Introduction 1
1.2. Energy 4
1.3. Thermodynamic Limit 5
1.4. Kinetic Calculation of Pressure 7
1.5. Isotropic Distributions 9
2. Classical Ideal Gas in Equilibrium 11
2.1. Maxwell's Distribution 12
2.2. Equation of State and Temperature 15
2.3. Validity of the Classical Limit 16
3. Effusion 17
4. Collisions 20
4.1. Cross-section 21
4.2. Collision Time 21
4.3. Mean Free Path 21
4.4. Relative Speed 21
5. From Local to Global Equilibrium (Transport Equations) 23
5.1. Inhomogeneous Distributions 23
5.2. Local Maxwellian Equilibrium 24
5.3. Conservation Laws 26
5.4. Thermal Conductivity and Viscosity 30
5.5. Transport Equations 32
5.6. Relaxation to Global Equilibrium 32
5.7. Diffusion 38
6. Kinetic Calculation of Transport Coefficients 40
6.1. A Nice but Dodgy Derivation 40
6.2. Kinetic Expressions for Fluxes 43
6.3. Kinetic Equation 43
6.4. Local Conservation Equations Rederived 44
6.5. Collision Operator 47
6.6. Solution of the Kinetic Equation 48
6.7. Calculation of Fluxes 49
7. From Microphysics to Macrophysics 52
7.1. What Are We Trying to Do? 52
7.2. The System and Its States 53
7.3. Pressure 53
8. Principle of Maximum Entropy 54
8.1. Quantifying Ignorance 54
8.2. Method of Lagrange Multipliers 58
8.3. Test of the Method: Isolated System 59
9. Canonical Ensemble 60
9.1. Gibbs Distribution 60
9.2. Construction of Thermodynamics 61
9.3. Some Mathematical Niceties 62
9.4. Third Law 63
9.5. Part I Obviated, Road Ahead Clear 64
10.Thermodynamic Equilibria and Stability 65
10.1. Additivity of Entropy 66

10.2. Thermal Equilibrium 67
10.3. Physical Interpretation of the Canonical Ensemble 68
10.4. Mechanical and Dynamical Equilibria 68
10.5. Stability 71
11.Statistical Mechanics of Classical Monatomic Ideal Gas 73
11.1. Single-Particle States 73
11.2. Down the Garden Path... 74
11.3. Single-Particle Partition Function 74
11.4. Digression: Density of States 75
11.5. Disaster Strikes 75
11.6. Gibbs Paradox 76
11.7. Distinguishability 76
11.8. Correct Partition Function 77
11.9. Thermodynamics of Classical Ideal Gas 78
11.10Maxwell's Distribution 79
12.P.S. More About Entropy 80
12.1. Shannon's Theorem 80
12.2. Boltzmann Entropy and the Ensembles 83
12.3. Gibbs vs. Boltzmann and the Meaning of Probabilities 87
12.4. Whose Uncertainty? 89
12.5. Second Law 89
13.P.P.S. Density Matrix and Entropy in Quantum Mechanics 91
13.1. Expectation Values 91
13.2. Density Matrix 92
13.3. Quantum Entropy 93
13.4. Time Evolution 93
13.5. How Information Is Lost 94
14.Grand Canonical Ensemble 96
14.1. Grand Canonical Distribution 97
14.2. Thermodynamics of Open Systems and the Meaning of Chemical Potential 98
14.3. Particle Equilibrium 100
14.4. Grand Partition Function and Chemical Potential of Classical Ideal Gas 101
14.5. Equilibria of Inhomogeneous Systems 102
14.6. Chemical Potential and Thermodynamic Potentials 103
15.Multi-Species (Multi-Component) Systems 106
15.1. Generalisation of the Grand Canonical Formalism to Many Species 106
15.2. Particle Equilibrium and Gibbs Phase Rule 107
15.3. Chemical Equilibrium 108
15.4. Chemical Equilibrium in a Mixture of Classical Ideal Gases: Law of Mass Action 109
16.Quantum Ideal Gases 111
16.1. Fermions and Bosons 111
16.2. Partition Function 112
16.3. Occupation Number Statistics and Thermodynamics 113
16.4. Preview of Further Developments 115
16.5. Calculations in Continuum Limit 115
16.6. Classical Limit (Sanity Check) 115
16.7. Degeneration 115

17.Degenerate Fermi Gas 115
17.1. Energy and Equation of State 115
17.2. Heat Capacity 115
18.Degenerate Bose Gas 115
18.1. Bose-Einstein Condensation 116
18.2. Thermodynamics of Degenerate Bose Gas 116
19.Thermal Radiation (Photon Gas) 116